軟體工程

第四課：系統塑模與統一架構 (System Modeling & Unified Architecture)

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第四章：課程藍圖與核心架構

第一部分：塑模基礎與核心靜態模型

- **4.1 系統塑模基礎：** 塑模本質、四大核心視角、五大必備圖表與盲人摸象寓言。
- **4.2 統一塑模語言 (UML)：** 1990 年代方法論之戰、三巨頭三位一體分工與 OMG 標準化。
- **4.3 功能塑模與使用案例：** 參與者目標、系統邊界、include/extend 關聯與 AI 提示詞指引。
- **4.4 結構塑模與領域類別圖：** 領域實體、可見度符號、關聯重數、整體與部分及 AI 提示詞。
- **4.5 互動塑模與循序圖：** 訊息傳遞時序、生命線、啟動條、BCE 模式與 AI 提示詞指引。

第二部分：工作流程、行為狀態與文字塑模

- **4.6 流程塑模與活動圖：** 並行流程、決策分支、分岔/結合、泳道職責與 AI 提示詞。
- **4.7 行為塑模與有限狀態機：** 反應型系統、訂單生命週期狀態機、事件觸發、守衛條件與 AI 提示詞。
- **4.8 PlantUML 宣告式文字塑模：** 架構即代碼 (Code-as-Architecture)、語法速查與多圖管線。
- **4.9 概念複習與統整：** 核心工程原則回顧、填空小測驗與經典文獻。

4.1 系統塑模基礎

> *"A language that doesn't affect the way you think about programming is not worth knowing."*
> *— Alan Perlis*

什麼是系統塑模？ (What is System Modeling?)

> *"系統塑模是為系統開發抽象模型的過程，每個模型代表該系統不同的視角或觀點。"*
> *— Ian Sommerville, Software Engineering (10th ed.)*

- **系統塑模的核心工程原則：**
  - **抽象化 (Abstraction)：** 隱藏不必要的實作細節，突顯關鍵的架構結構、資料流動與相依關係。
  - **多重視角 (Multiple Perspectives)：** 沒有任何單一圖表能解釋整個複雜軟體；不同利害關係人需要不同的視角。
  - **溝通與驗證 (Communication & Verification)：** 作為產品經理、系統架構師、開發者與測試工程師之間的通用視覺語言。
  - **實作藍圖 (Blueprint for Implementation)：** 提供正式的結構與行為規格，作為後續編寫程式碼與資料庫結構的精確藍圖。

盲人摸象與多重視角模型 (The Parable of the Elephant & Multiple Perspectives)

🦏 寓言啟示：盲人摸象與系統塑模的「多重視角」必要性

① 摸到象鼻：「像一條巨蟒 / 水管！」

🎯 **互動視角 (Interaction Perspective)**

- 關注：訊息傳遞順序、呼叫流向與動態協同
- 盲點：忽視靜態資料結構與內部持久化

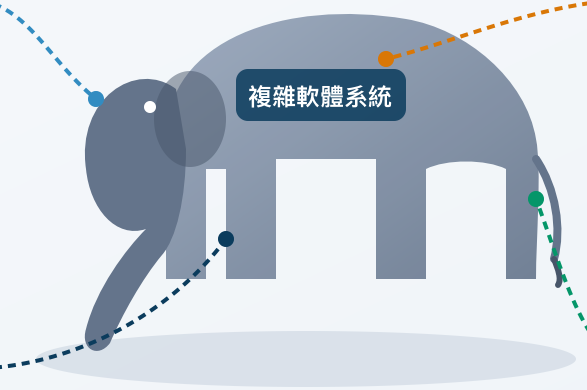
對應 UML：循序圖 (Sequence Diagram)

② 摸到象腿：「像一根堅硬巨柱！」

🏛️ **結構視角 (Structural Perspective)**

- 關注：實體資料屬性、關聯、組合與靜態骨幹
- 盲點：看不出系統隨時間演進的動態行為

對應 UML：類別圖 (Class Diagram)



③ 摸到象身：「像一面高大磚牆！」

🛡️ **外部視角 (External / Context)**

- 關注：系統邊界、運行環境與外部第三方 API
- 盲點：無法得知內部核心邏輯如何拆解實現

對應 UML：環境圖 (Context Diagram)

④ 摸到象尾：「像一條擺動細繩！」

⚡ **行為視角 (Behavioral Perspective)**

- 關注：業務流程流轉、外在事件刺激與狀態轉換
- 盲點：無法單獨反映靜態物件職責與系統邊界

對應 UML：活動圖 / 狀態圖 (Activity / State)

💡 結論：任何單一模型皆為「局部投影」；唯有綜合不同視角，才能完整還原軟體架構全貌！

系統塑模的 4 大核心視角 (4 Core Perspectives)

- **1. 外部視角 (External Perspective)：**
  - 塑模系統處在的運行環境、外部合作夥伴與邊界範圍。
  - 明確劃分系統邊界：哪些由內部團隊開發，哪些委託給外部第三方服務。
- **2. 互動視角 (Interaction Perspective)：**
  - 塑模外部參與者與系統之間，或是內部協同物件之間的動態溝通與訊息傳遞。
- **3. 結構視角 (Structural Perspective)：**
  - 塑模系統資料組織、類別、屬性、方法與關聯性的靜態架構，與執行時間先後無關。
- **4. 行為視角 (Behavioral Perspective)：**
  - 塑模系統的動態執行行為、業務流程步驟，以及系統針對外在事件所產生的反應式狀態轉換。

現代實務必備的 5 大 UML 核心模型

模型 / 圖表	對應章節	所屬視角	動態 / 靜態本質	主要軟體工程職責
1. 使用案例模型	4.3	互動視角	功能契約	界定系統邊界、參與者目標與系統功能範疇
2. 領域類別圖	4.4	結構視角	靜態骨幹	定義領域實體、屬性、方法與整體/部分關聯
3. 物件循序圖	4.5	互動視角	動態時序	沿時間軸追蹤訊息傳遞與 BCE 強韌性職責劃分
4. 業務活動圖	4.6	行為視角	動態流程	塑模業務流程、並行分岔/結合與跨角色泳道
5. 有限狀態機圖	4.7	行為視角	反應生命週期	捕捉事件驅動的離散狀態流轉、守衛與不變量

> *落實與加速： 透過 **4.8 PlantUML** (架構即代碼) 統一實踐，各章節深度整合專屬 **AI 提示詞指引**。*

觀念檢核測驗 1 (CCQ 1)

軟體架構師若欲定義「**領域實體資料的靜態組織方式，以及類別之間的繼承、關聯與包含關係**」（完全不隨執行時序變動），應採取哪一種塑模視角？

- **A.** 外部視角 (External Perspective)
- **B.** 互動視角 (Interaction Perspective)
- **C.** 結構視角 (Structural Perspective)
- **D.** 行為視角 (Behavioral Perspective)

4.2 統一塑模語言 (UML)

> *「標準化的視覺語法，將軟體工程從封閉的方法論孤島，淬煉為全球共通的工程學門。」*

標準化的迫切需求：1990 年代「方法論之戰」

- **物件導向程式設計的崛起 (1980 年代末至 1990 年代初)：**
  - 軟體產業從程序導向程式碼 (C, Pascal) 大規模轉移至物件導向典範 (C++, Smalltalk)。
  - 工程師迫切需要一套圖形符號來視覺化表達類別、物件與關聯。
- **百家爭鳴的「方法論之戰 (Method Wars)」時代：**
  - 超過 **50 種相互競爭的物件導向塑模記號**充斥商業市場。
  - 各派大師激烈爭論：類別究竟該畫成雲朵、矩形還是橢圓？繼承到底該用空心三角、實心箭頭還是虛線？
  - **嚴重的產業割裂碎片化：** 各公司之間無法交換架構模型，CASE 塑模軟體彼此互不相容，工程師換工作就得被迫重新學習一套新符號。

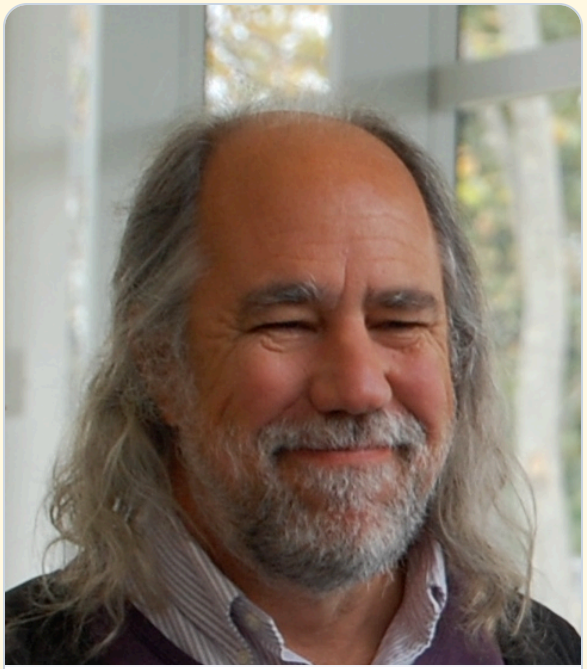
統一與標準化：從 Rational 到 OMG

- **1994 – Rational 展開歷史性大整合：**
  - Jim Rumbaugh 加入 Rational 與 Grady Booch 聯手，融合 Booch 方法與 OMT 為 Unified Method (v0.8)。
- **1995 – 三巨頭 (The Three Amigos) 合體：**
  - Ivar Jacobson 加入 Rational，帶入革命性的**使用案例 (Use Case)** 方法論 (OOSE)。
- **1997 – OMG 國際官方標準化：**
  - 正式提交至**物件管理組織 (OMG)**，於 1997 年 11 月一致通過採納為 **UML 1.1** 國際產業標準。
- **2005 – UML 2.0 重大架構改版：**
  - 擴充至 13 種（後續擴增至 14 種）正式圖表，具備形式化執行元模型 (Metamodels)。

UML 奠基先驅：Grady Booch

- **角色與學術榮譽：**
  - ACM Fellow、IEEE Fellow、IBM 院士、Rational 共同創辦人兼首席科學家。
- **開創經典方法論：**
  - **Booch 方法 (Booch Method)** 與重量級專書：《物件導向分析與設計》。
- **對 UML 的核心貢獻：**
  - 專精於**物件導向實作設計 (Design & Implementation)**，精確定義微觀物件協同與宏觀系統架構。
  - 賦予 UML 精準對應至 C++、Java 等 OOP 程式語言的實作直譯能力。
- **著名軟體工程箴言：**

> *"良好的架構，是讓未來的重大決策顯得容易且自然的藝術。"*

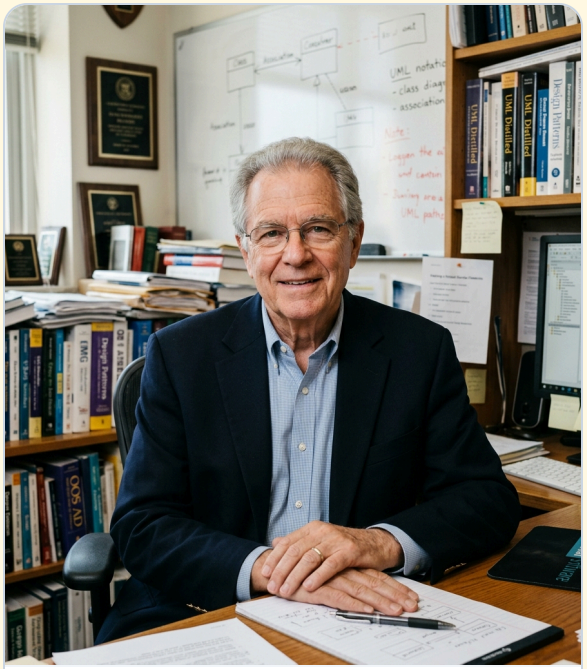


Grady Booch
Rational Software / IBM Fellow

UML 奠基先驅：James Rumbaugh

- **角色與學術榮譽：**
  - 奇異公司 (GE) 全球研發中心軟體主任研究員、Rational Software、IBM。
- **開創經典方法論：**
  - **物件塑模技術 (OMT, Object Modeling Technique)** 與專書：《物件導向塑模與設計》。
- **對 UML 的核心貢獻：**
  - 強調嚴謹的**領域分析 (Domain Analysis)**、語義資料塑模與實體關聯對應。
  - 首創將靜態物件結構與 David Harel 的**狀態圖 (Statecharts)** 深度結合，用以塑模動態反應式系統。
- **著名軟體工程箴言：**

> *"如果不透徹理解領域的真實本質，就絕不可能打造出卓越的軟體。"*



James Rumbaugh
GE Research / Rational Software

UML 奠基先驅：Ivar Jacobson

- **角色與學術榮譽：**
  - 愛立信 (Ericsson) 首席架構師、Objectory AB 創辦人、Rational Software、SEMAT 發起人。
- **開創經典方法論：**
  - **物件導向軟體工程 (OOSE)**。
- **對 UML 的核心貢獻：**
  - **使用案例 (Use Cases) 的發明者 (1986)：** 革命性地將系統架構錨定在可衡量的使用者目標之上。
  - 提出經典的**邊界–控制–實體 (BCE, Boundary–Control–Entity)** 強韌性分析模式與元件化架構。
- **著名軟體工程箴言：**

> *"一個沒有使用者的系統，就沒有任何存在的理由。塑模永遠要先從使用者的目標開始。"*

Ivar Jacobson
Ericsson / Objectory / Rational

UML 三巨頭：軟體塑模的三位一體 (The Trinity of Modeling)

Ivar Jacobson

核心回答「Why」(目標 / 使用者)

- **使用案例驅動：** 外部參與者想達成什麼目標？
- **BCE 強韌性分析：** 徹底分離 UI 邊界與核心資料實體。
- **元件契約規範：** 可重用子系統的架構邊界。

James Rumbaugh

核心回答「What」(領域與資料)

- **領域物件分析：** 現實世界存在哪些核心概念？
- **類別與資料綱要：** 結構化的實體關聯關係。
- **有限狀態機：** 反應式事件驅動的生命週期。

Grady Booch

核心回答「How」(設計與程式碼)

- **物件導向設計：** 類別之間如何精巧協同運作？
- **巨觀架構拆解：** 分層模組與相依性控制。
- **實作直譯對應：** 視覺模型與 OOP 程式碼直接對齊。

> 📌 *架構整合精髓：Jacobson 捕捉系統存在的初衷目的 **(Why)**；Rumbaugh 塑模領域存在的資料本質 **(What)**；Booch 設計程式執行的實作架構 **(How)**。*

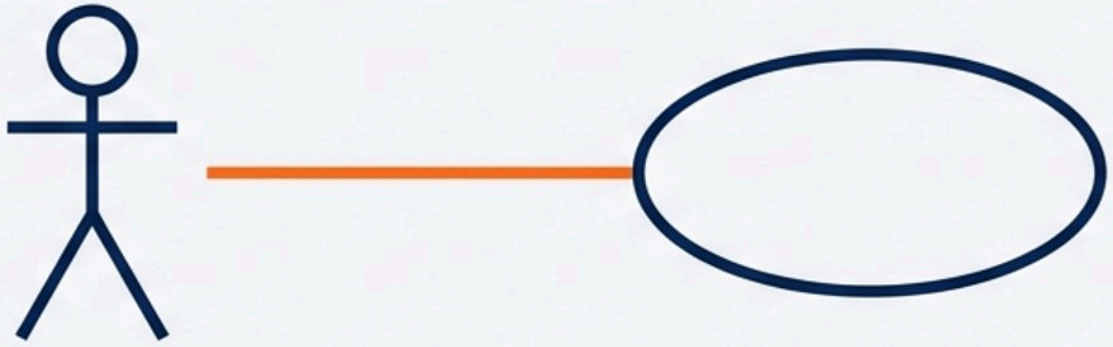
觀念檢核測驗 2 (CCQ 2)

在被尊稱為「UML 三巨頭」的軟體工程大師中，哪一位以於 1986 年發明**使用案例 (Use Cases)**、將軟體架構直接錨定於使用者具體目標而聞名？

- **A.** Grady Booch
- **B.** James Rumbaugh
- **C.** Ivar Jacobson
- **D.** Martin Fowler

4.3 功能與使用案例模型

> *"使用案例是系統與其參與者之間為了達成可衡量業務目標所簽署的行為契約。"*
> *— Alistair Cockburn*



Beginner's Guide to UML Use-Case Diagram Modeling

Transforming abstract requirements into clear system designs.

Focus: Business Value & User Goals

High-level visual representation of how users or external systems interact with the application.

💡 **The Golden Rule:**
Build use-case models from the project stake perspective, not the developer's.

Exclude: Technical Implementation

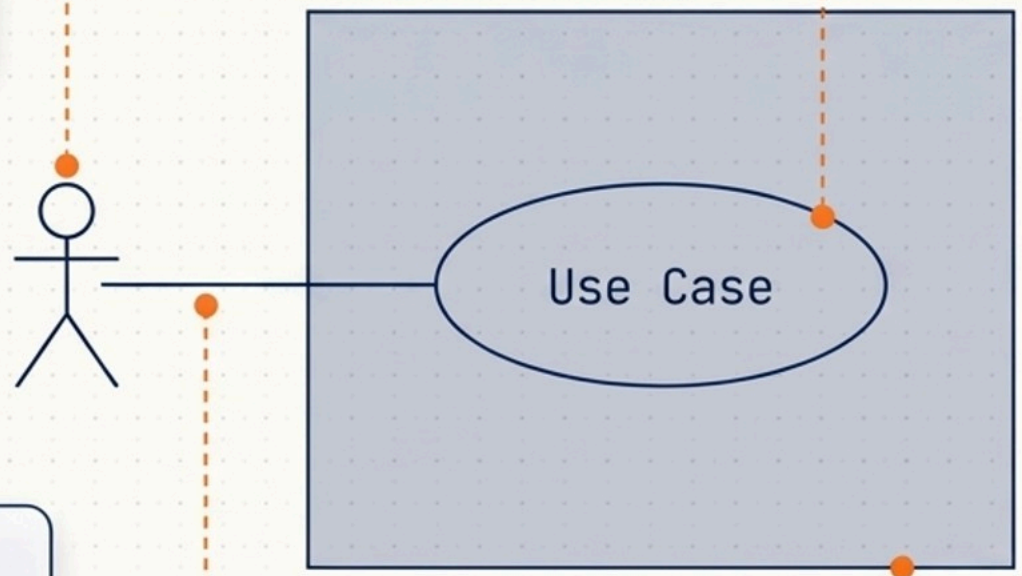
Do not map data flows, functional decomposition, or database processes.

Actor

Any person, organization, external system, or timer interacting with the system.

Use Case

A sequence of actions delivering measurable value to the actor.

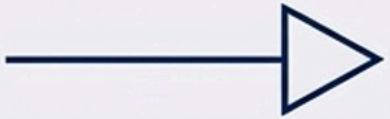


Relationship

Shows how actors and use cases connect, depend on, or generalize each other.

System Boundary

Defines what is inside your system vs. what is outside your control.

Relationship	UML Notation	Behavior	Real-World Analogy
Association		Actor participates in a use case.	A customer walks into a store.
<<include>>		Mandatory, reusable logic. Runs at a specific point.	Calling a function: `checkout()` → `calculateTax()`
<<extend>>		Optional/conditional logic. Might run during certain steps.	A pop-up coupon or conditional error check.
Generalization		Is a relationship. Child inherits logic from parent.	International Student is like a Student, but with extra steps.

清楚的使用案例命名語意 (Clear Use Case Naming)

Clear Semantics

Start with Strong Verbs:
`Withdraw Funds, Deliver Shipment`

Name singular, role-based nouns:
`Customer Support`

Apply the “Is Like” test for generalization: `[Child] is like [Parent]` must make logical sense.

Implementation Leaks

Avoid Weak Verbs:
`Process, Perform, Do`

Avoid Technical Jargon:
`Process_DB, Convey Package Via Vehicular Transportation`

Avoid Job Titles:
`Junior CSR, Manager` (Model roles, roles, not HR titles).

使用案例塑模核心法則 (Rules of Building Use Case Models)

NESTING LIMIT

Max 2 Levels of Nesting

Avoid functional decomposition (e.g., A includes B includes C).
Keep it requirements-focused, not design-focused.

ACTOR INTERACTIONS

No Actor-to-Actor Lines

Actors do not interact directly on the diagram.
Detail these interactions in your text specifications.

SYSTEM BOUNDARIES

Use Boundaries with Purpose

Only use the boundary box to define system scope or distinct release phases (Phase 1, Phase 2). Omit if redundant.

Note: Call out external systems with <<system>> and use a Time actor (clock icon) for scheduled triggers.

Case Study: University Enrollment System

Actors (The "Who")

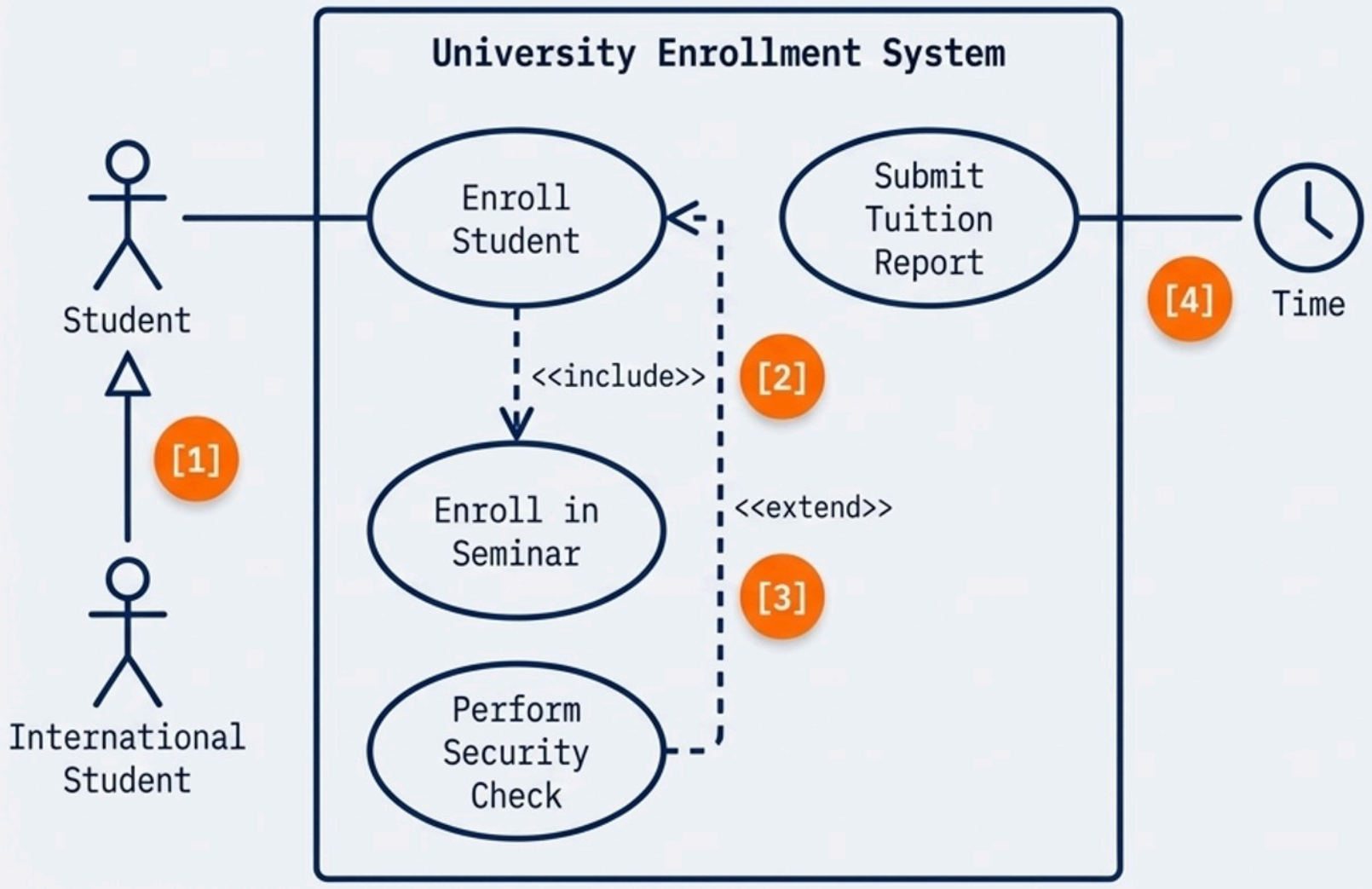
Student
Primary actor

International Student
Specialized actor

Time
Triggers periodic tasks

Core Use Cases (The "What")

- Enroll Student
- Enroll in Seminar
- Perform Security Check
- Submit Tuition Report (Monthly)



Blueprint Legend

[1] Generalization:
Int'l Student inherits base student logic.

[2] <<include>>:
Enrolling always requires a seminar (Mandatory).

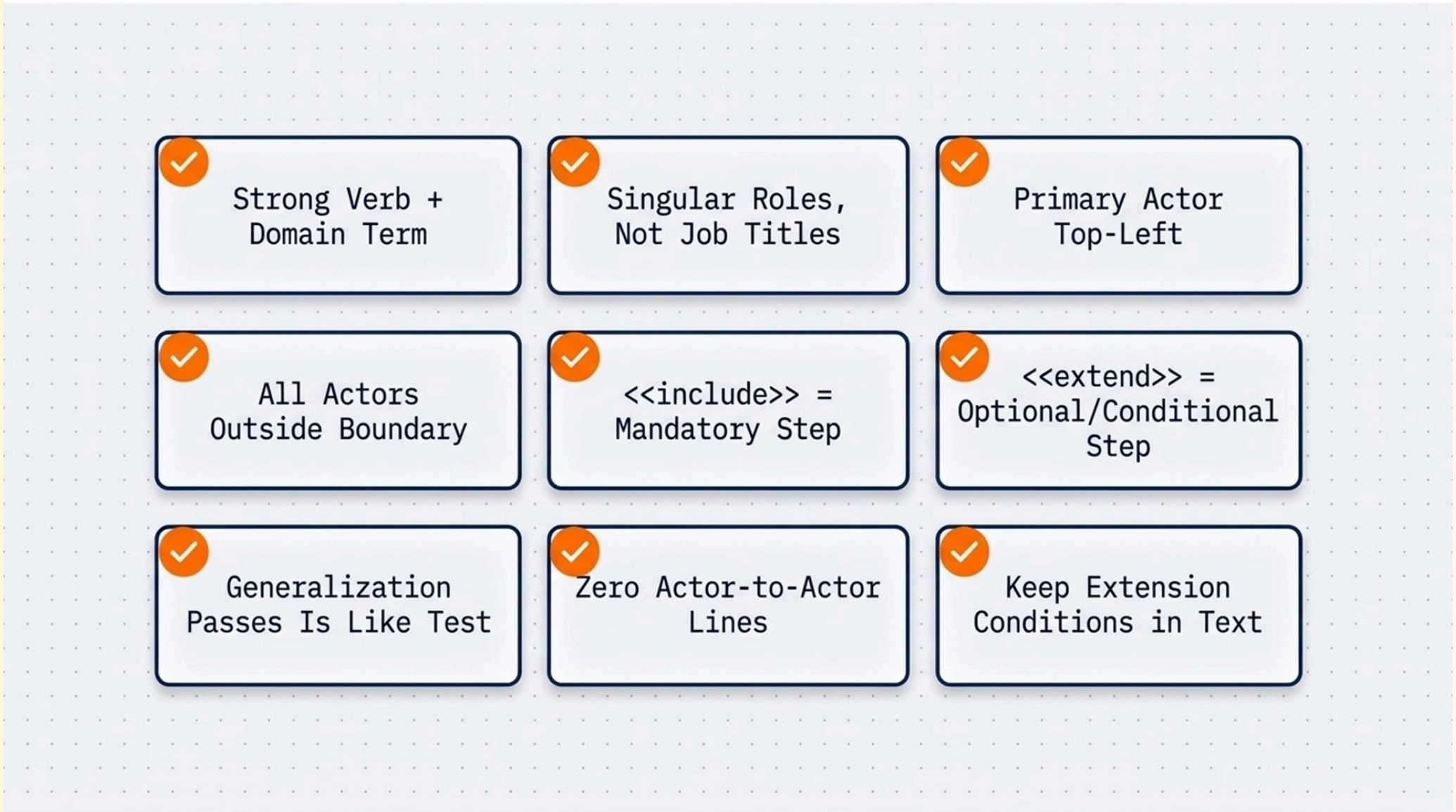
[3] <<extend>>:
Security check only runs for certain students (Conditional).

[4] Time Actor:
Triggers the monthly tuition report.

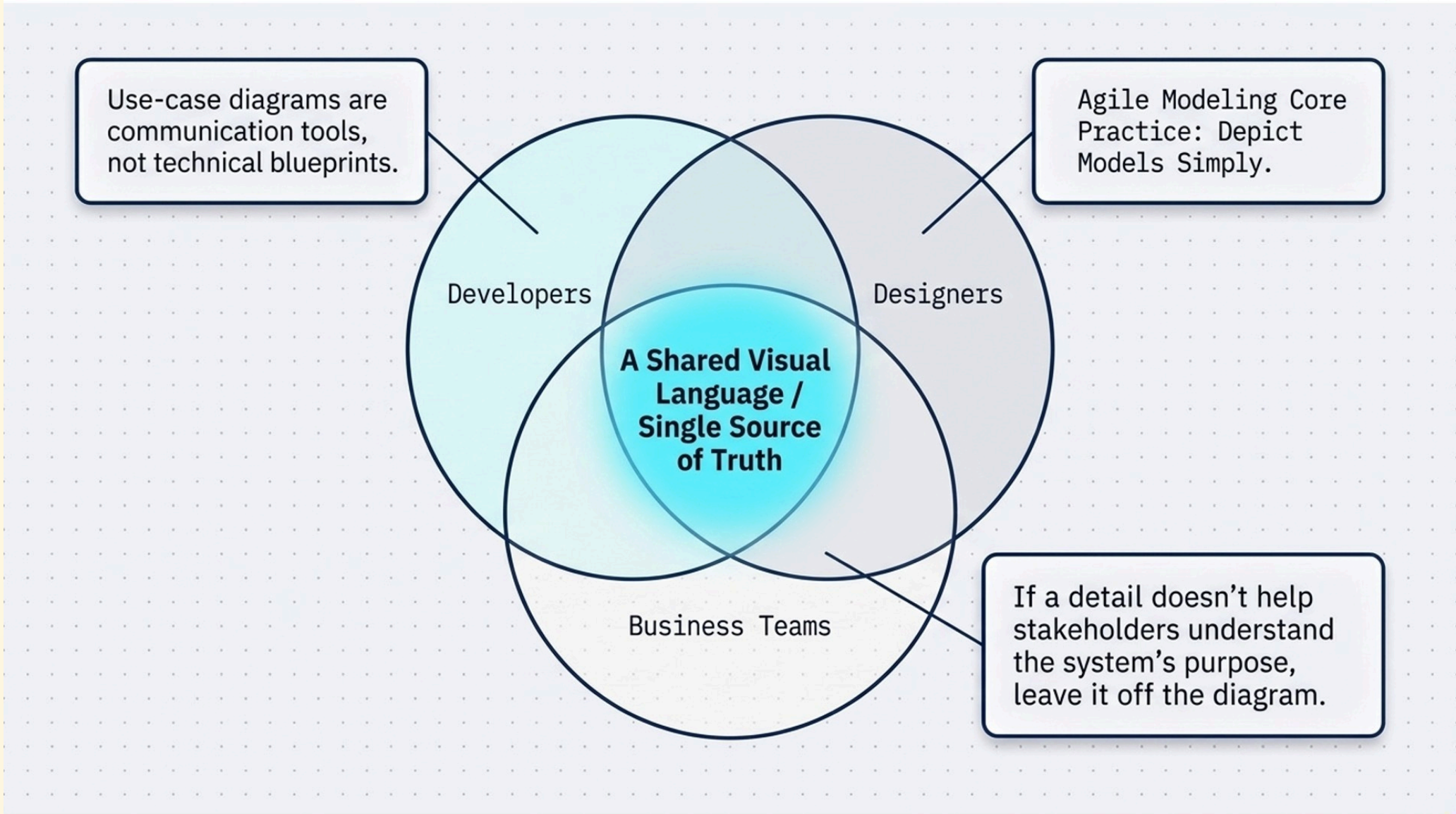
使用案例常見反模式 (Common Anti-Patterns of Use Case Models)

The Mistake (Symptom)	Why It Fails (Diagnosis)	The Fix (Cure)
✗ Technical names (`Process_DB_Transaction`)	Stakeholders don't understand implementation details.	✓ Use domain-friendly names (`Withdraw Funds`, `Generate Report`).
✗ Drawing arrows showing data flow	Associations show initiation, not data movement.	✓ Remove arrowheads unless showing passive actors.
✗ Overusing `<<extend>>` lines	Creates messy, unreadable diagrams.	✓ Move optional/conditional logic into textual use-case specs.
✗ Connecting Customer to Admin directly	Actors don't interact on use-case diagrams.	✓ Describe human interaction in the text scenario.

使用案例最佳實踐檢核清單 (Best Practices Checklists)



溝通優先思維 (Communication First Mindset)



使用案例規格書：UC-01 學生選課 (Enroll Student)

欄位名稱	規格細節說明
使用案例代號 / 名稱	**UC-01：學生選課 (Enroll Student in Course)**
主要參與者 (Actor)	學生 (透過學校單一登入 SSO 驗證身分)
前置條件 (Preconditions)	學生具備有效學籍在學狀態；當前處於選課開放時段內
後置條件 (Postconditions)	學生正式納入該課程名冊；該課程剩餘修課名額自動扣減 1
主要成功路徑 (Main Flow)	**1.** 學生依學系代碼或開課課號搜尋開放修習的課程清單。 **2.** 系統回傳並展示符合條件的班級資訊、上課時間表與剩餘名額。 **3.** 學生選定目標課程，向系統提出正式加選請求。 **4.** 系統執行 `<<include>> 檢查先修科目資格`，比對學生歷史歷年成績單。 **5.** 系統鎖定保留名額，寫入選課交易紀錄，並產生成員確認通知單。 **6.** 學生即時檢視個人更新後的課表，並收到選課成功通知信。
例外與擴充路徑 (Extensions)	**4a. 先修科目未通過：** 系統提示未修畢之必要擋修科目代碼，中斷選課流程。 **4b. 上課時間衝堂：** 系統發出警告，提示與已選課程時間重疊，放棄本次加選。 **5a. 課程名額已滿：** 系統觸發 `<<extend>> 加入候補等候名單` 流程。

AI 輔助指南：使用案例塑模 (Use Case Modeling)

- **1. 角色設定與系統邊界約束 (Persona & Boundary Priming)：**
  - 指示 AI 扮演「精通 UML 2.5 國際標準的資深需求分析師」。
  - **嚴格要求宣告邊界：** 必須先定義 `rectangle "System Name"`，禁止 AI 將外部雲端服務混入內部邊界。
- **2. 參與者與關聯性嚴格語義規範：**
  - **參與者分類：** 要求 AI 區分「主要參與者 (人)」與「次要外部支援服務 (金流 API、簡訊服務)」。
  - `<<include>>` **規則：** 僅用於「每次下單必執行的強制共用子流程」。
  - `<<extend>>` **規則：** 僅用於「具備特定觸發條件的可選附加行為」，並指明擴充點。
  - **反模式防護：** 嚴禁功能分解（禁止產生「點擊按鈕」、「輸入密碼」等瑣碎泡泡）。
- **3. 架構師人工審查清單 (Verification Checklist)：**
  - [ ] 每個使用案例是否皆以「動詞 + 受詞」命名（如 `下單訂購`、`追蹤外送`）？
  - [ ] 第三方基礎設施（金流閘道、簡訊）是否正確置於系統邊界外側？
  - [ ] include 箭頭是否正確指向被包含的共用功能？

AI 提示詞範例：外送平台使用案例模型

1. 角色設定與任務指令：

> 你是一位資深系統分析架構師。
> 請根據以下需求描述，產生合規且簡練的
> `PlantUML` 使用案例圖與使用案例規格書。

2. 塑模約束條件：

- 使用 `<<extends>>` 與 `<<includes>>` 建構模組化關係。
- 嚴禁功能分解（禁止出現「點擊按鈕」、「輸入密碼」等瑣碎用例）。
- 以 `rectangle "校園外送平台"` 劃定系統邊界。
- 外部金流閘道與簡訊服務置於邊界外作為次要參與者。

3. 輸入需求敘述 (Requirements Statement)：

> *「顧客可以在平台瀏覽餐廳菜單、將餐點加入購物車並進行結帳下單。*
> *顧客下單時，系統必須強制連線外部**金流閘道**授權扣款，並驗證外送地址有效性。*
> *結帳時，顧客可以自由選擇是否套用優惠券代碼（`套用折扣優惠`），或勾選放置門口的無接觸配送。*
> ***餐廳員工**可接收並審核進單、更新廚房備餐進度。*
> ***外送員**可承接外送任務、回報餐點取件與最終送達。*
> *當訂單狀態變更時，系統會自動呼叫外部**簡訊服務**向顧客發送即時通知。」*
>
> ”

AI 生成之 PlantUML 與架構圖：外送平台使用案例模型

AI 生成之 PlantUML 規格代碼：

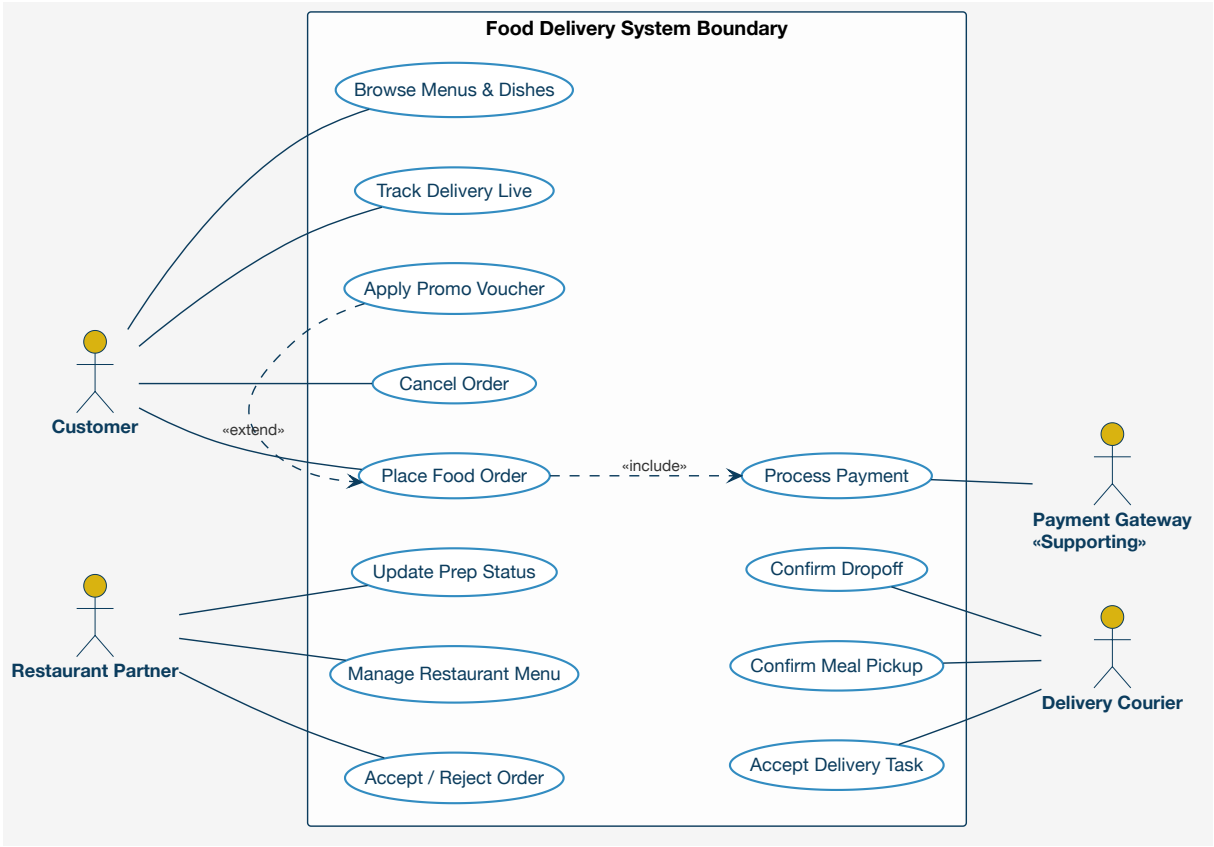
```
@startuml
left to right direction
actor Customer
actor "Restaurant Partner" as Restaurant
actor "Payment Gateway\n«Supporting»" as Payment
actor "Delivery Courier" as Courier

rectangle "Food Delivery System" {
  usecase "Browse Menus" as UC_Browse
  usecase "Place Food Order" as UC_PlaceOrder
  usecase "Apply Promo Voucher" as UC_Voucher
  usecase "Track Delivery Live" as UC_Track
  usecase "Manage Menu" as UC_Menu
  usecase "Accept / Reject Order" as UC_AcceptOrder
  usecase "Process Payment" as UC_Payment
  usecase "Accept Delivery Task" as UC_AcceptTask
}

Customer -- UC_Browse
Customer -- UC_PlaceOrder
Customer -- UC_Track
Restaurant -- UC_Menu
Restaurant -- UC_AcceptOrder

UC_PlaceOrder ..> UC_Payment : <<include>>
UC_Voucher .> UC_PlaceOrder : <<extend>>
UC_Payment -- Payment
```

渲染產出之架構圖：



課堂活動：使用 AI 演進使用案例 (雙人動手實作)

動手實作實驗：變更業務需求並審核 AI 產出 (4 分鐘)

- **1. 在使用案例提示詞中加入一項新需求功能：**
  - *預約定時下單：* 允許 `顧客` 預約未來特定時段的外送服務。
  - *線上即時客服：* 建立 `顧客` 與 `客服人員` 或 `外送員` 的即時對談。
  - *揪團共同訂餐：* 允許多位使用者將餐點加入同一筆匯總購物車。
- **2. 要求 AI 重新生成：** 提示 LLM 更新對應的 PlantUML 使用案例代碼。
- **3. 渲染並嚴格審查：** 貼入 PlantText 渲染並審查：
  - AI 是否將外部系統/金流閘道保持在系統邊界**矩形之外**？
  - AI 是否正確區分**核心必經步驟** (`<<include>>`) 與**可選延伸功能** (`<<extend>>`)？
  - AI 是否成功避開了點擊按鈕等微觀 UI 反模式（例如：避免出現「點擊確認」）？

觀念檢核測驗 3 (CCQ 3)

在 UML 使用案例模型中，若使用案例 A 與使用案例 B 之間存在 `<<include>>` 關聯（A 指向 B），這在軟體架構上代表什麼意義？

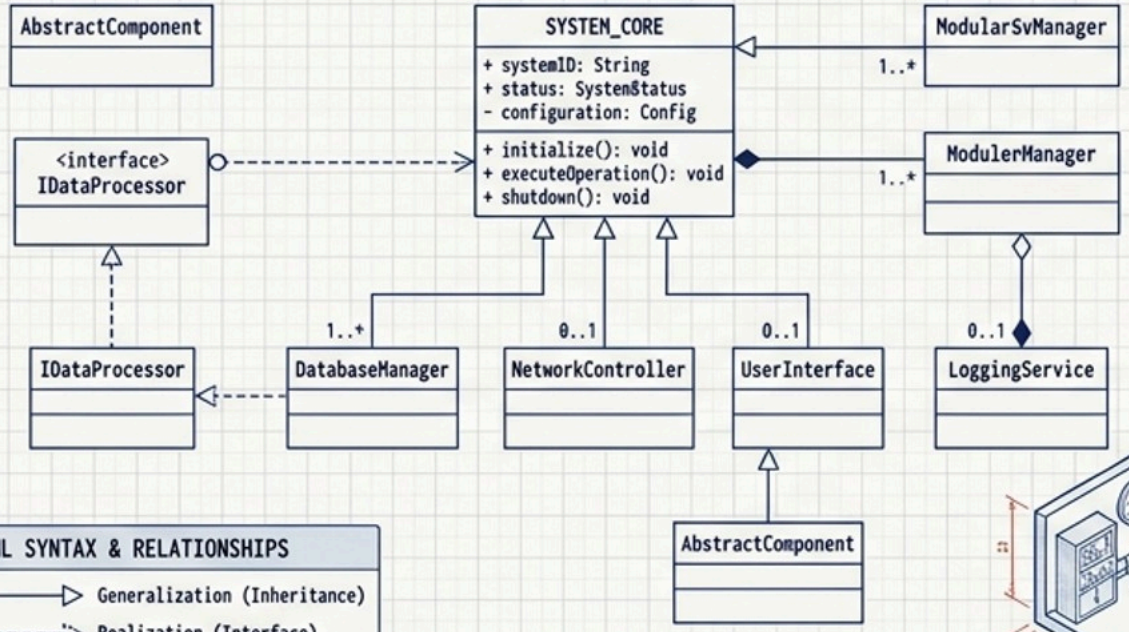
- **A.** 使用案例 B 是選擇性的，僅在特定異常狀況下才會由 A 觸發執行。
- **B.** 每當使用案例 A 執行時，使用案例 B 所定義的行為「必定且強制」會被納入執行。
- **C.** 使用案例 A 繼承了使用案例 B 的所有屬性與操作方法。
- **D.** 使用案例 B 是外部第三方系統，不屬於軟體邊界範疇。

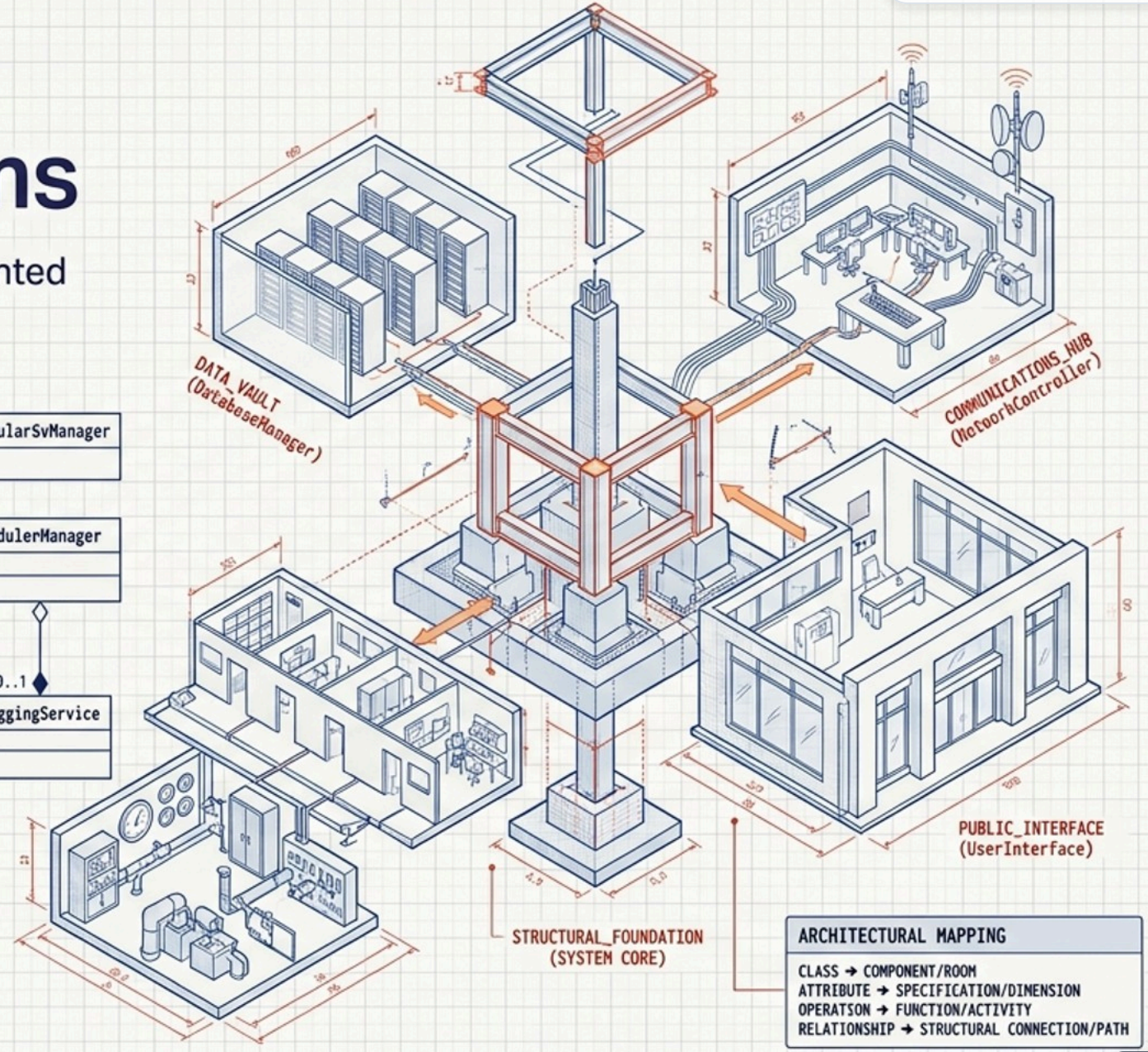
4.4 結構與領域類別模型

> *"類別圖是軟體系統的靜態建築骨幹；如果骨幹畸形，任何敏捷開發都拯救不了系統的崩潰。"*

The Anatomy of UML Class Diagrams

A visual reference guide to mapping object-oriented systems, syntax, and relationships.





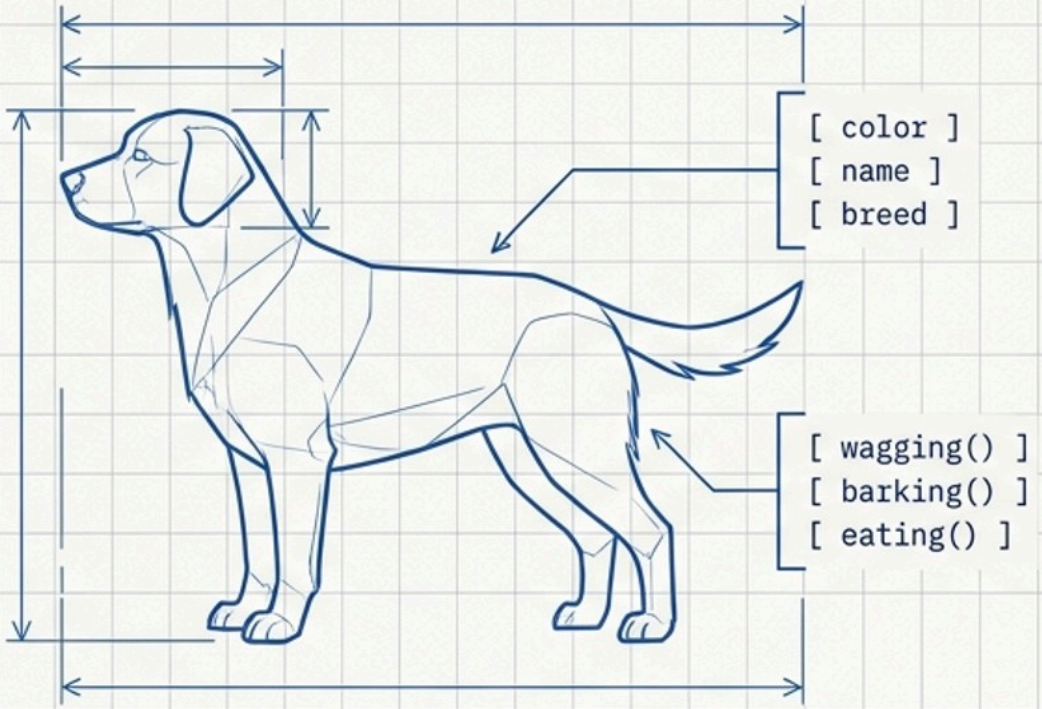
UML SYNTAX & RELATIONSHIPS

- Generalization (Inheritance)
- Realization (Interface)
- Aggregation (Shared) — Shared Ownership
- Composition (Composite) — Strong Ownership
- Association

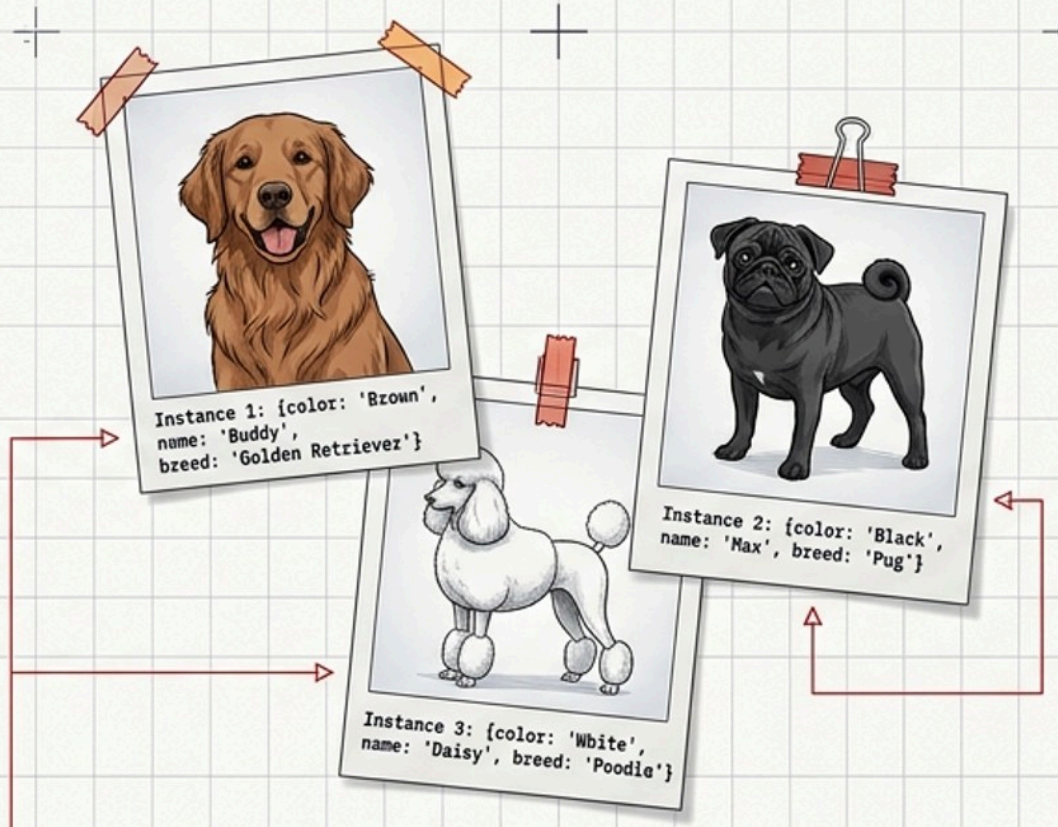
ARCHITECTURAL MAPPING

CLASS → COMPONENT/ROOM
ATTRIBUTE → SPECIFICATION/DIMENSION
OPERATSON → FUNCTION/ACTIVITY
RELATIONSHIP → STRUCTURAL CONNECTION/PATH

Blueprint vs. **Instance**: The Foundation of OOD



A **Class** is a blueprint. It describes the type of objects and encapsulates state (**attributes**) and behavior (**operations**). It is not the object itself.



Objects are usable instances of classes. Built from the same blueprint, they contain the same components but hold distinct, real-world states.

Instance

Define ar Instance's enzwy fzom cluapcst iantar

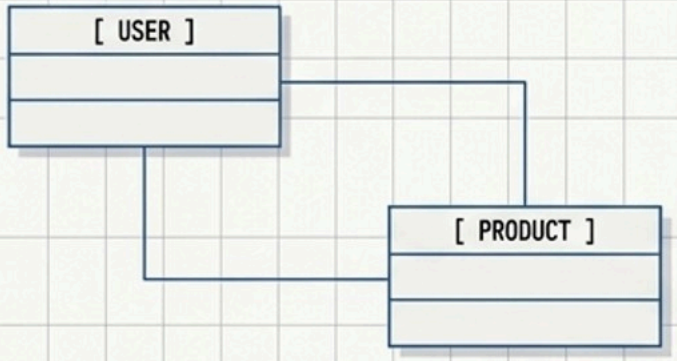
The **Three Perspectives** of Class Modeling

DEVELOPMENT TIME

Conceptual Perspective

Focus: Broad domain concepts.

Detail Level: Low. Seldom moves past class names and basic associations.

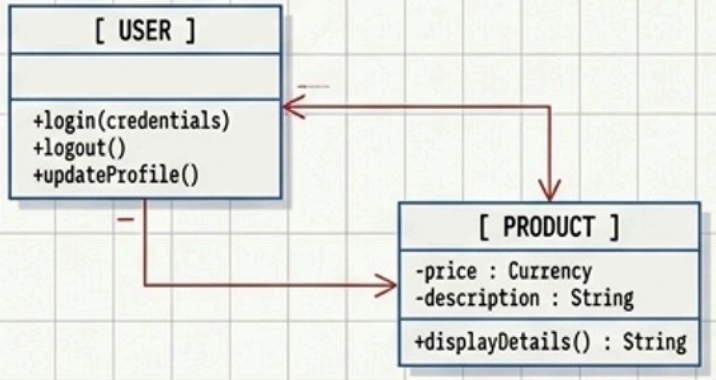


Simplified view for initial understanding.

Specification Perspective

Focus: Interfaces of Abstract Data Types (ADTs).

Detail Level: Medium. Emphasizes what the software does, independent of implementation.

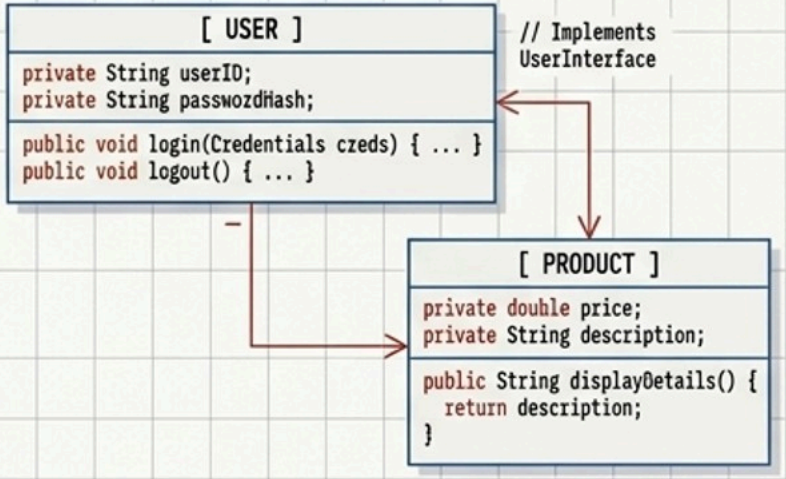


Defines the 'what' without the 'how'.

Implementation Perspective

Focus: Physical implementation of interfaces.

Detail Level: High. Exact replication of written code.



Represents the exact code structure.

Instance

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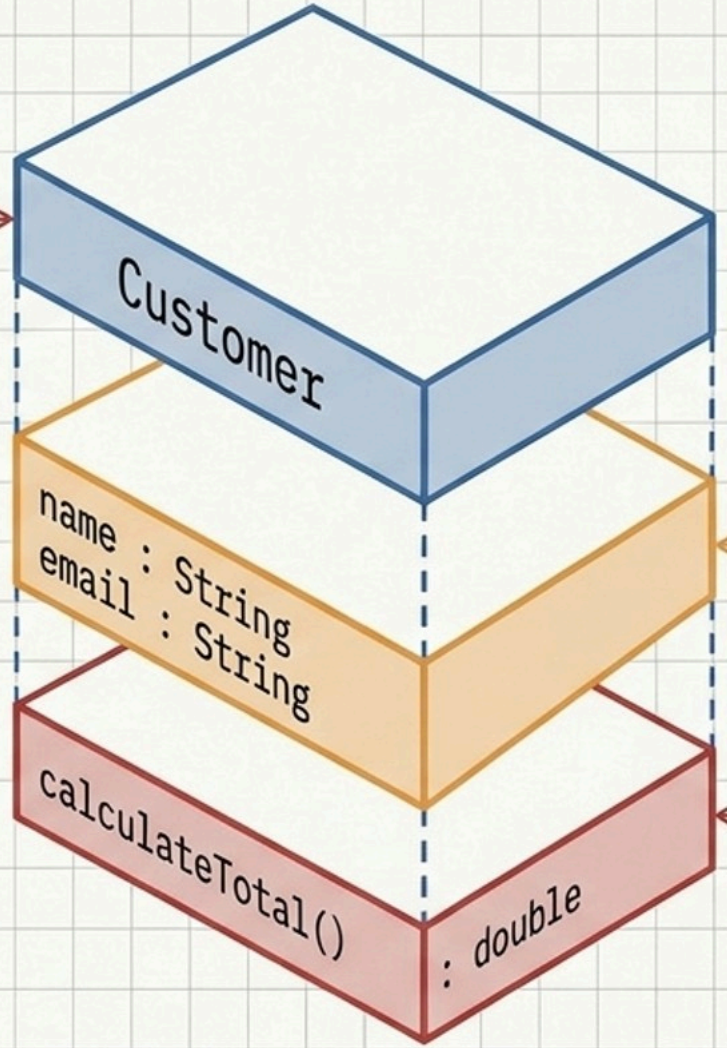
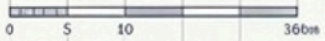
Anatomy of the Class Box

Class Name: The only mandatory information. Represents the core concept.



Attributes:
Encapsulates state.
Maps to member variables in code.
Type is shown after the colon.

Operations / Methods:
Encapsulates behavior.
Services the class provides.
Return type shown at the end.



Instance

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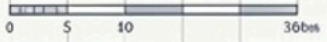
The **Visibility Matrix**

Symbol	Label	Meaning	Code Snippet
	Public	Accessible from anywhere.	`public String name;`
	Private	Accessible only within the class itself.	`private int id;`
	Protected	Accessible within the class and its subclasses.	`protected double balance;`



Instance

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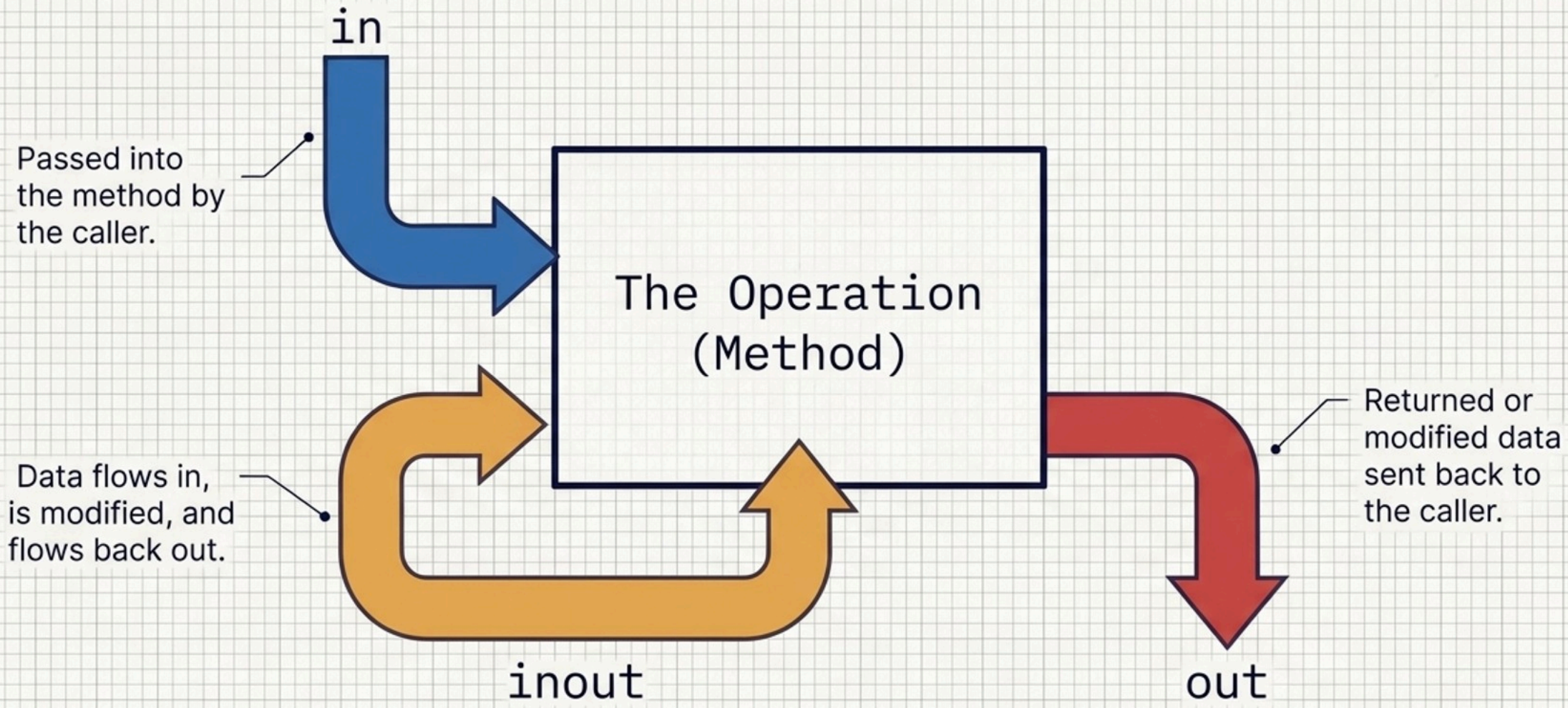
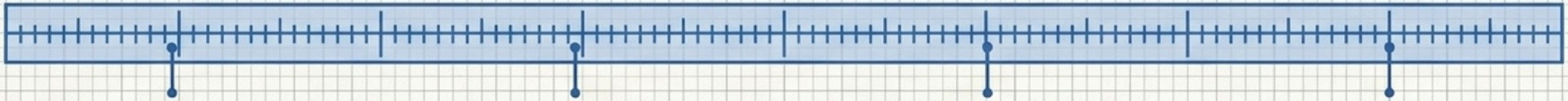
Parameter Directionality Dashboard

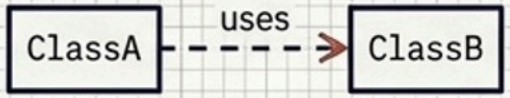


Taxonomy of Relationships

Loosest Connection — Tightest Structure

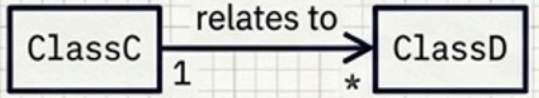


Dependency



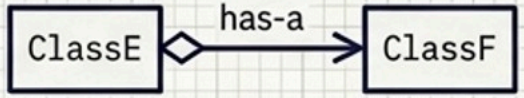
Class A uses Class B temporarily (e.g., as a parameter). No structural ownership.

Association (Simple)



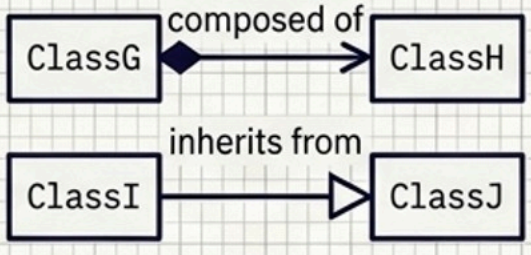
Classes know about each other. Can be unidirectional or bidirectional. Loose coupling.

Aggregation



A whole-part relationship, but parts can exist independently. E.g., a library has books.

Composition & Inheritance

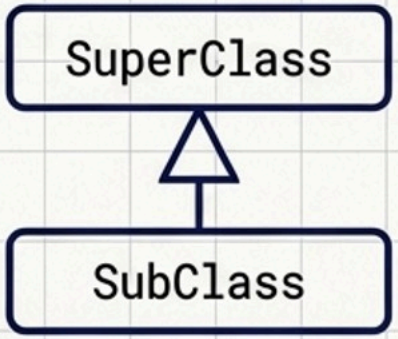


Composition: Strict whole-part; parts die with the whole.
Inheritance: Structural `is-a` hierarchy.

UML precisely conveys how code should be implemented. Connections range from temporary usage to strict structural life-and-death bindings.

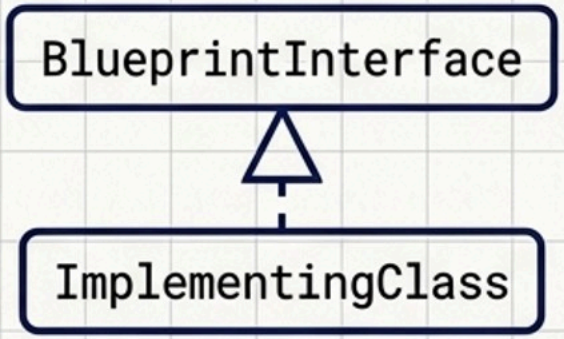
The Connector Cheat Sheet

Inheritance (Generalization)



Represents an “is-a” relationship. Subclass inherits features.

Realization



Relationship between interface and implementation.

Dependency

A “uses” relationship. Object used in a method, not stored in a field.

Simple Association

A structural link between two peer classes.

Lifecycle Diagnostic: Aggregation vs. Composition

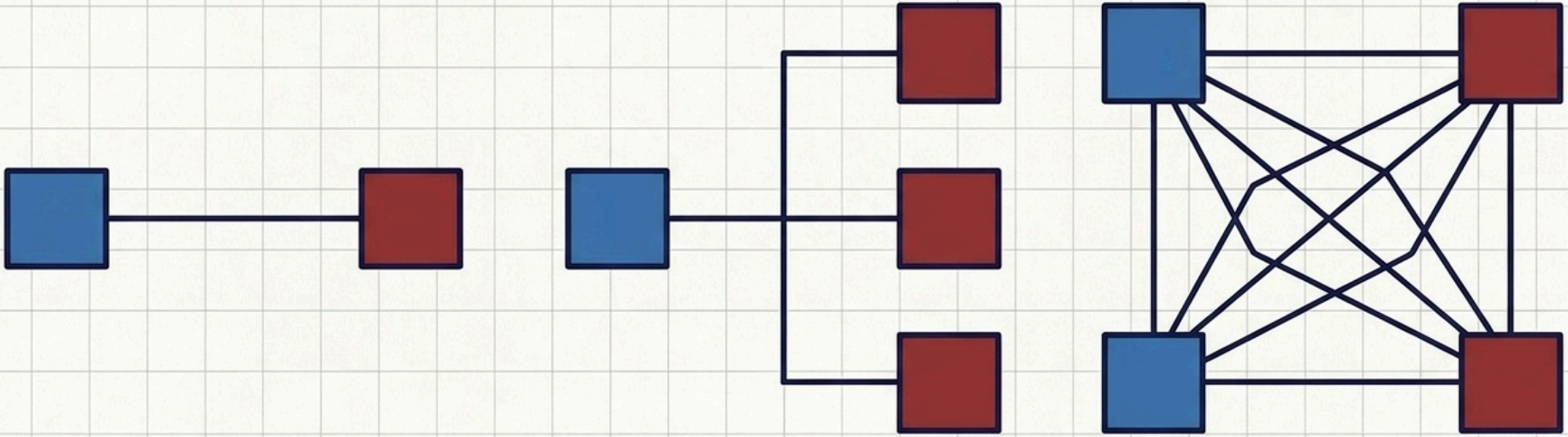
Aggregation	Composition
• **Concept:** A ‘part of’ relationship. • **Lifespan:** Separate lifetimes. • **Analogy:** Team ◇— Player (If the team disbands, players still exist)	• **Concept:** Strict containment. • **Lifespan:** Live and die together. Part cannot stand alone. • **Analogy:** House ◆— Room (If the house is demolished, rooms cease to exist)

Cardinality & Constraints

One to One (1 to 1)

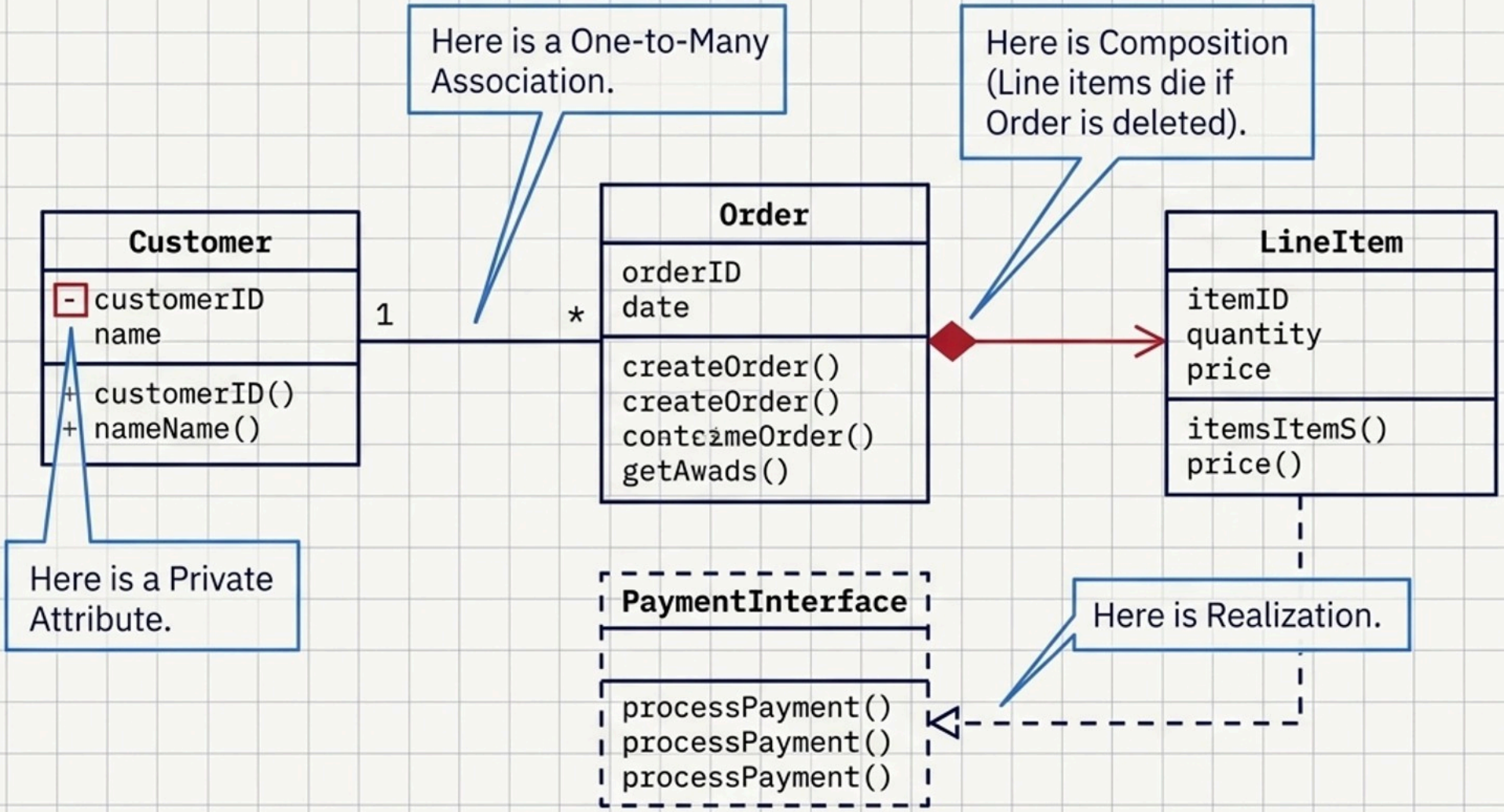
One to Many (1 to *)

Many to Many (* to *)



Cardinality defines the numerical limits of an association, dictating exactly how many instances of one class can connect to another.

Syntax to System: The Order Model



AI 輔助指南：類別塑模 (Class Modeling)

- **1. 角色設定與抽象層次界定 (Persona & Abstraction Level)：**
  - 指示 AI 扮演「物件導向領域模型專家 (Domain Architect)」，並明確要求產出「概念層領域模型」或「實作層設計模型」。
- **2. 整體與部分關聯性嚴格紀律 (Whole-Part Coupling)：**
  - **組合 (實心菱形, ◆)：** 強制 AI 辨識具備生命週期相依性的實體（整體銷毀時，部分連帶 cascade-delete）。
  - **聚合 (空心菱形, ◇)：** 要求部分具備獨立生命週期時使用空心菱形。
  - **禁止濫用泛型關聯：** 嚴禁 AI 全部使用普通實線 (實線) 敷衍了事，必須為每條線指明重數與關聯型態。
- **3. 封裝與雙向重數規範：**
  - 要求強制加上可見度符號（屬性私有 `-`、操作公開 `+`、型態宣告）。
  - 關聯線「兩端」皆必須明確標註重數（如 `1` 對 `1..*`）。
- **4. 架構師人工審查清單 (Verification Checklist)：**
  - [ ] AI 是否誤將繼承 (三角繼承箭頭) 當成組合 (實心菱形)？
  - [ ] 敏感屬性（密碼、金流 token、地址）是否皆設定為 `- private`？
  - [ ] 關聯重數在業務邏輯上是否合理（例如訂單能否包含 0 個項目）？

AI 提示詞範例：外送平台領域類別模型

1. 角色設定與任務指令：

> 你是一位資深物件導向軟體架構師。
> 請根據以下領域需求，產出符合 DDD 規範的
> 標準 PlantUML 類別圖 (Class Diagram)。

2. 塑模約束條件：

- `Order` 與 `OrderItem` 必須使用**組合關聯** (實心菱形，重數 `1` 對 `1..*`)。
- `Order` 與 `Courier` 必須使用**聚合關聯** (空心菱形，重數 `*` 對 `0..1`)。
- `OrderItem` 關聯至 `MenuItem` (`*` 對 `1`)。
- 所有屬性均須宣告 `- private` 與明確資料型態。
- 標註關鍵業務方法 (`+ calculateTotal()`, `+ assignCourier()`)。

3. 輸入需求敘述 (Requirements Statement)：

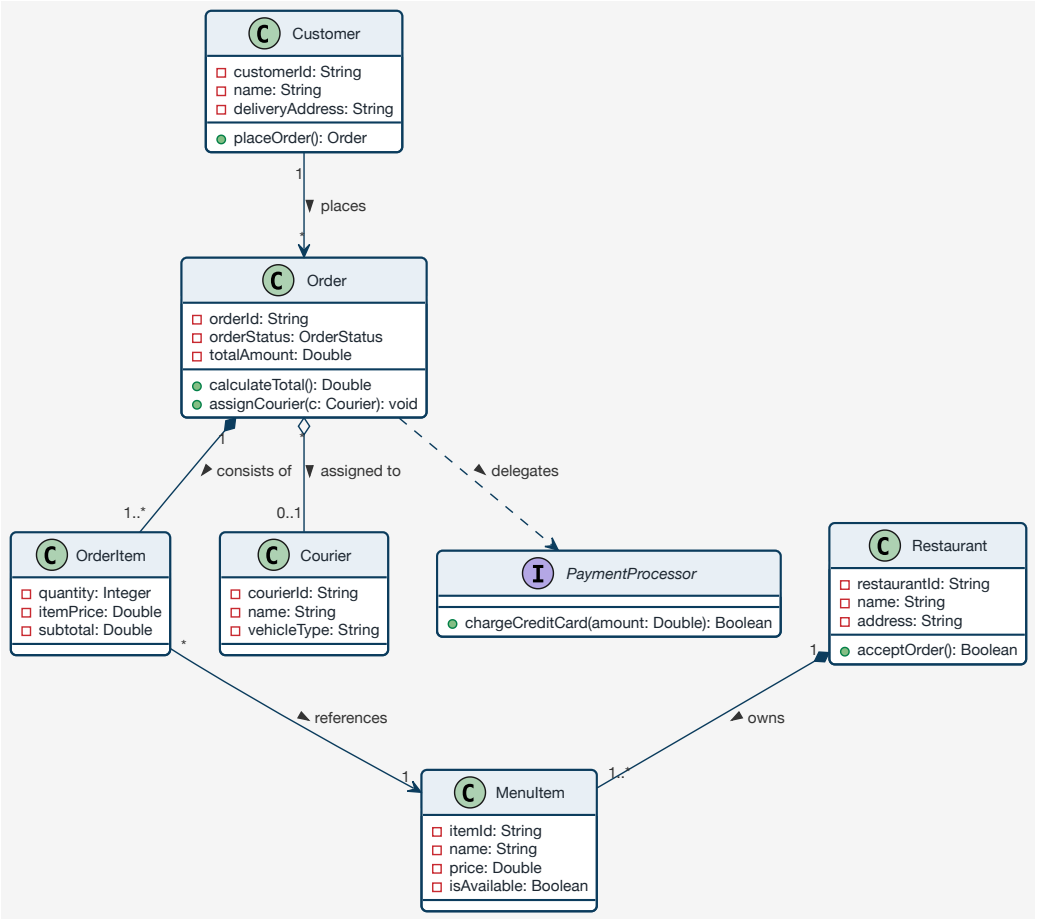
> *「系統包含 **Customer**（顧客 ID、姓名、Email、電話、外送地址）。*
> ***Restaurant**（餐廳 ID、店名、地址）擁有多項 **MenuItem**（餐點 ID、品名、單價、是否在庫）。*
> *顧客可建立多筆 **Order**。每筆訂單包含訂單編號、狀態列舉 (OrderStatus: 待處理、製作中、配送中、已送達)、下單時間與總金額。*
> *一筆訂單由一個或多個 **OrderItem**（購買數量、購買當下單價、小計）組成；若訂單被取消刪除，其明細項目必須連帶銷毀。*
> *平台可指派一位 **Courier**（外送員 ID、姓名、電話、交通工具）負責配送訂單。*
> *訂單透過 **PaymentProcessor** 介面委託第三方完成扣款。」*

AI 生成之 PlantUML 與架構圖：外送平台領域類別模型

AI 生成之 PlantUML 規格代碼：

```
@startuml
class Customer {
  - customerId: String
  - deliveryAddress: String
  + updateAddress(newAddress: String): void
}
class Restaurant {
  - restaurantId: String
  - cuisineType: String
}
class MenuItem {
  - itemId: String
  - price: Double
}
class Order {
  - orderId: String
  - status: OrderStatus
  + calculateTotal(): Double
  + assignCourier(c: Courier): void
}
class OrderItem {
  - quantity: Integer
  - subtotal: Double
}
class Courier {
  - courierId: String
  - vehicleType: String
}
interface PaymentProcessor {
  + chargeCreditCard(amount: Double): Boolean
}

Restaurant "1" *-- "1..*" MenuItem : owns >
```

渲染產出之架構圖：



課堂活動：使用 AI 演進領域類別架構 (雙人動手實作)

動手實作實驗：變更架構規則並審核 AI 產出 (4 分鐘)

- **1. 在類別模型提示詞中加入一項新架構規則：**
  - *多店合併結帳：* 允許單一筆訂單跨多家餐廳合併下單結帳。
  - *外送小費與評價：* 增加對外送員的直接打賞與雙向服務評分。
  - *VIP 會員訂閱制：* 導入每月固定扣款的會員免運機制。
- **2. 要求 AI 重新生成：** 提示 LLM 更新對應的 PlantUML 類別模型代碼。
- **3. 渲染並嚴格審查：** 貼入 PlantText 渲染並審查：
  - AI 是否正確維持**複合 (◆)** 與**聚合 (◇)** 的生命週期差異？
  - **重數 (Multiplicities)** 是否合乎邏輯（跨店訂單需不需要拆分子訂單）？
  - 關鍵欄位是否維持 `- private` 封裝，且未被設計模式過度複雜化？

觀念檢核測驗 4 (CCQ 4)

在 UML 類別圖中，若類別 A 與類別 B 之間繪製了一條連線，且在 A 端標示了**實心黑色菱形 (Filled Black Diamond, ◆)** 指向 B，這代表什麼架構意涵？

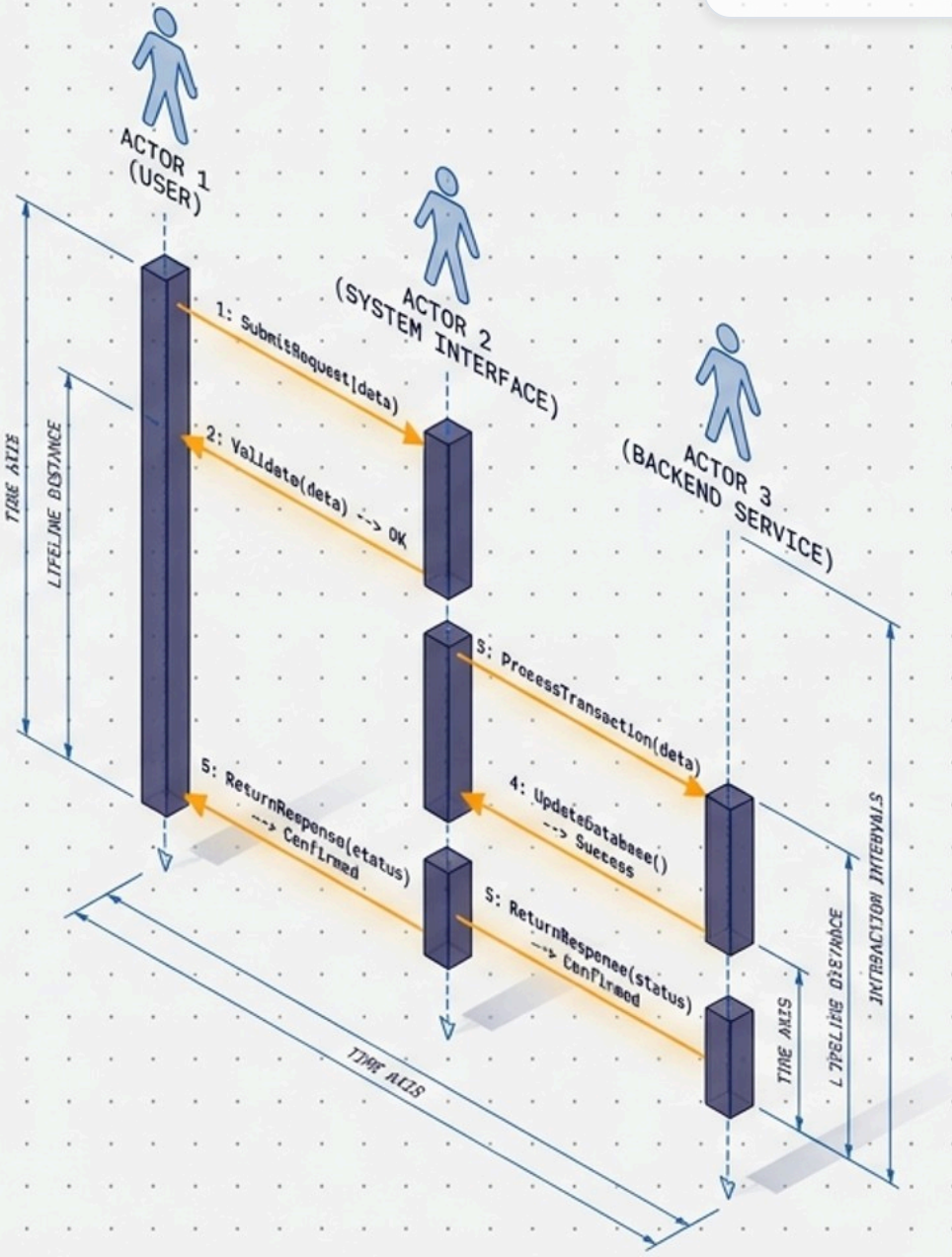
- **A.** 類別 B 繼承了類別 A 的所有公開方法與屬性。
- **B.** 類別 A 組合 (Composition) 了類別 B；當物件 A 被銷毀時，所屬的物件 B 也將連帶被銷毀。
- **C.** 類別 A 與類別 B 是鬆散聚合關係，B 擁有獨立生存的生命週期。
- **D.** 類別 A 與類別 B 之間僅存在執行期的暫時性依賴。

4.5 互動與循序圖模型

> *"時間是軟體架構中最嚴苛的考驗；循序圖將時間的流逝具象化為可驗證的訊息契約。"*

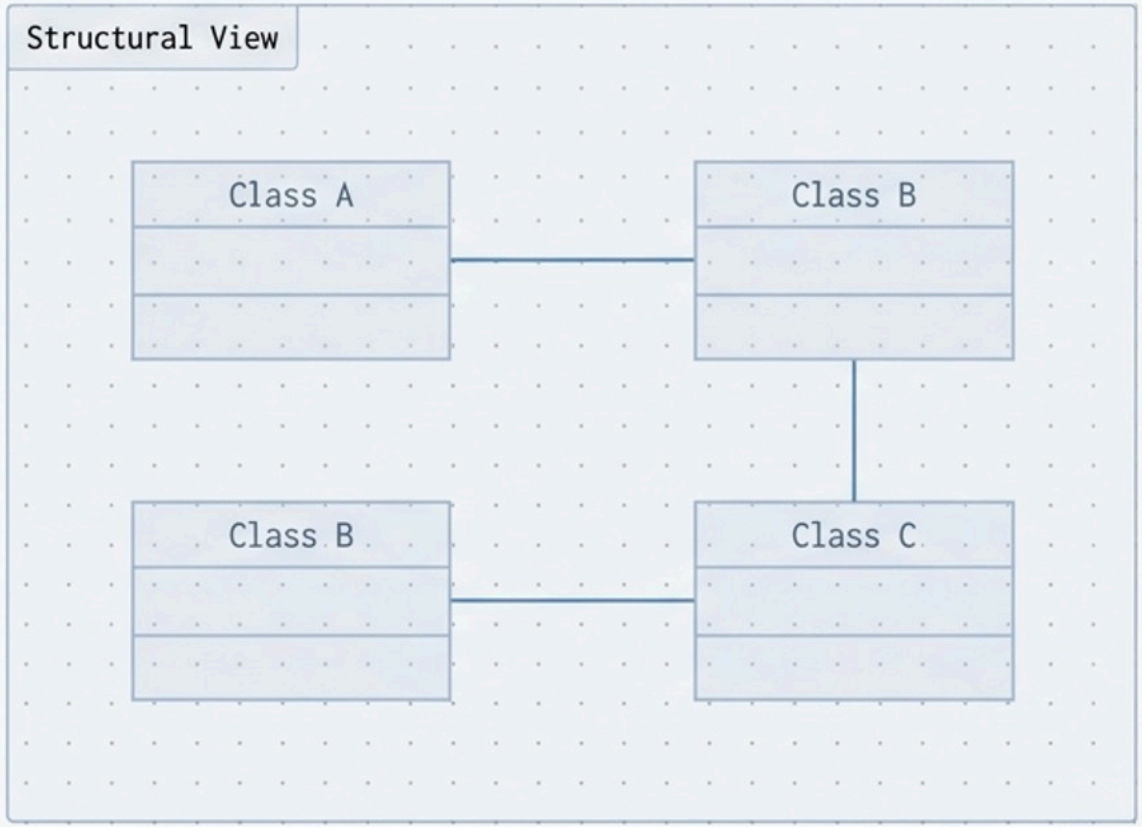
Sequence Diagrams: The Architectural Blueprint

A visual reference for modeling system interactions and logic.

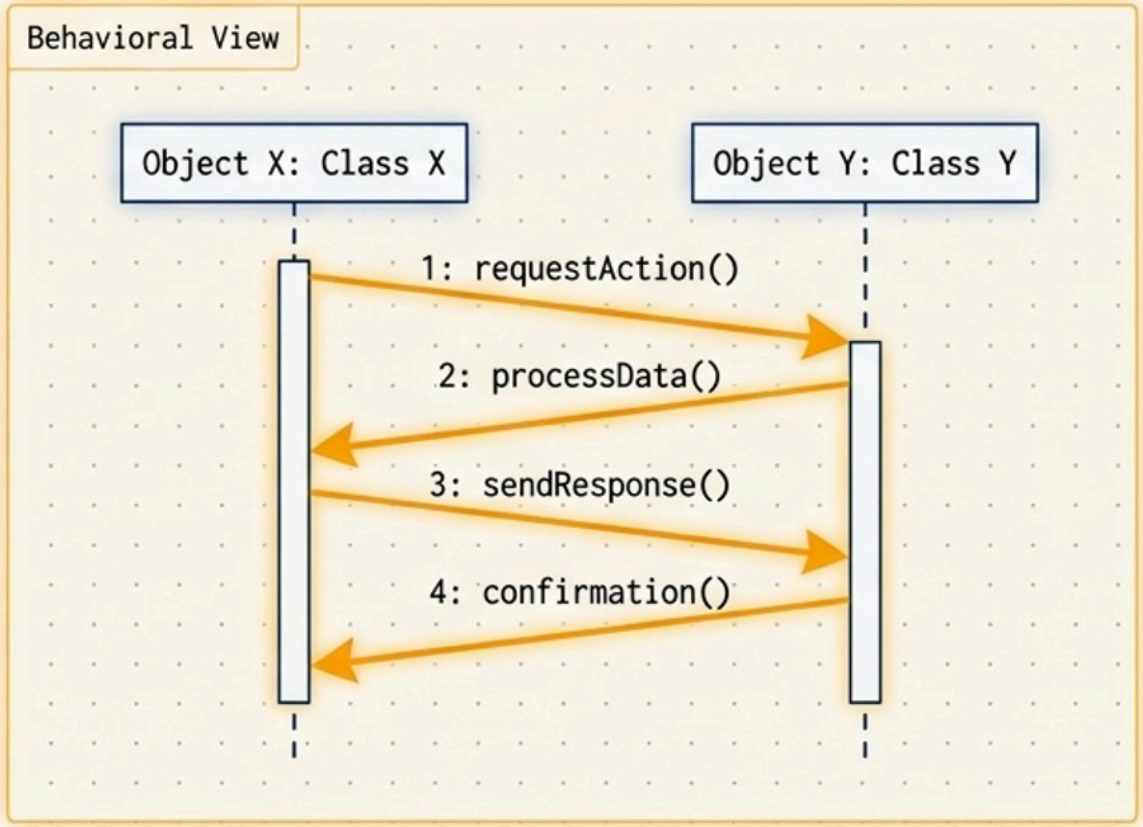


Mapping Dynamic Collaboration

STATIC MODEL

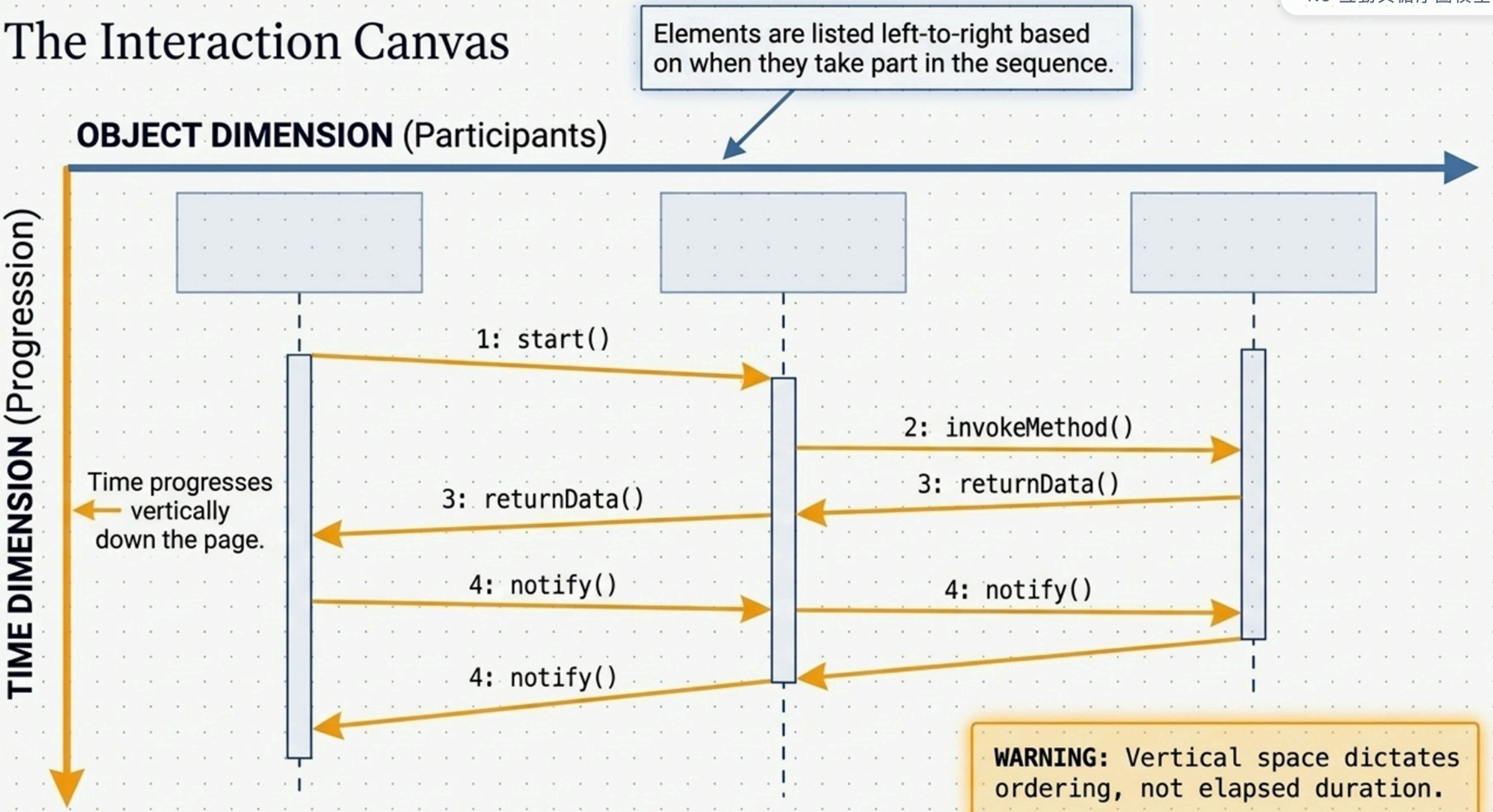


DYNAMIC MODEL

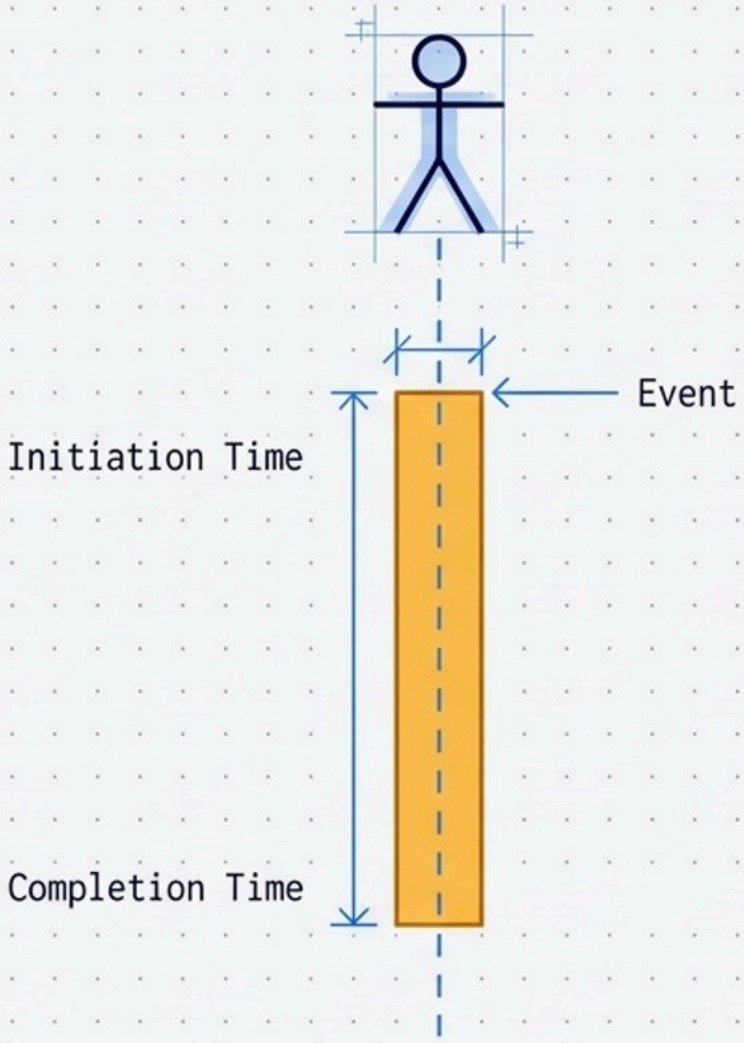


UML Sequence Diagrams detail how operations are carried out over time. Unlike class or object diagrams that provide static views, sequence diagrams capture the dynamic flow of messages sent and received within a collaboration.

The Interaction Canvas



Structural Anatomy



Actor

External entity interacting with the subject. A role, not necessarily a physical person.

Lifeline

An individual participant in the interaction.

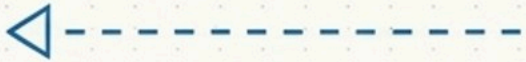
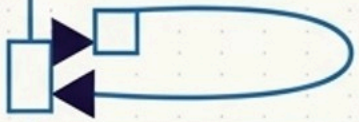
Focus of Control

The period during which an element performs an operation. Visualized by the activation rectangle.

Event

Any precise point where an occurrence happens (e.g., sending or receiving a message).

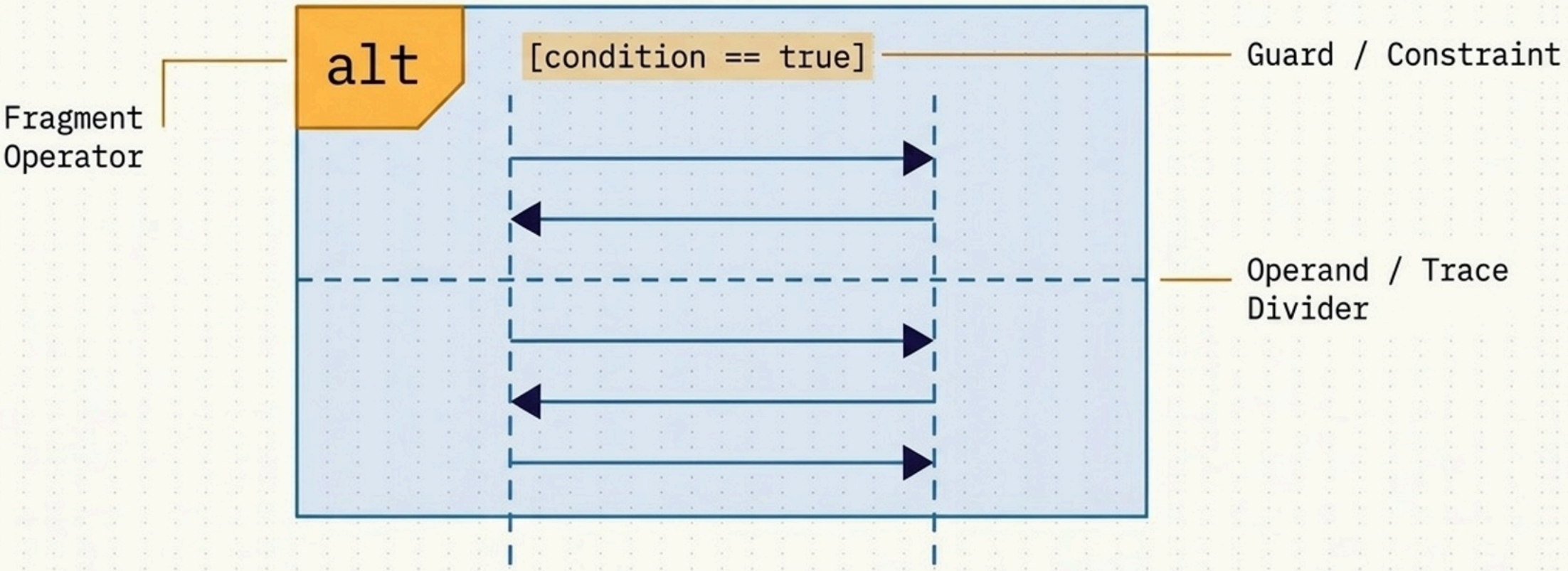
The Messaging Matrix

Message Type	Visual Notation	Semantic Meaning
`Call`		Invokes an operation on a target lifeline.
`Return`		Passes information back to the caller of a former message.
`Self`		Invokes a message on the same lifeline.
`Recursive`		Invokes a message targeting a new activation stacked on the caller.
`Create`		Instantiates a new target lifeline.
`Destroy`		Requests the destruction of the target lifeline's lifecycle.
`Duration`		Shows the distance between two time instants for an invocation.

Note: A comment carries no semantic force but attaches useful remarks.

UML 2.0 Sequence Fragments

Combined Fragments enclose a portion of the interaction to capture branching, loops, and parallel execution, moving beyond single linear paths.

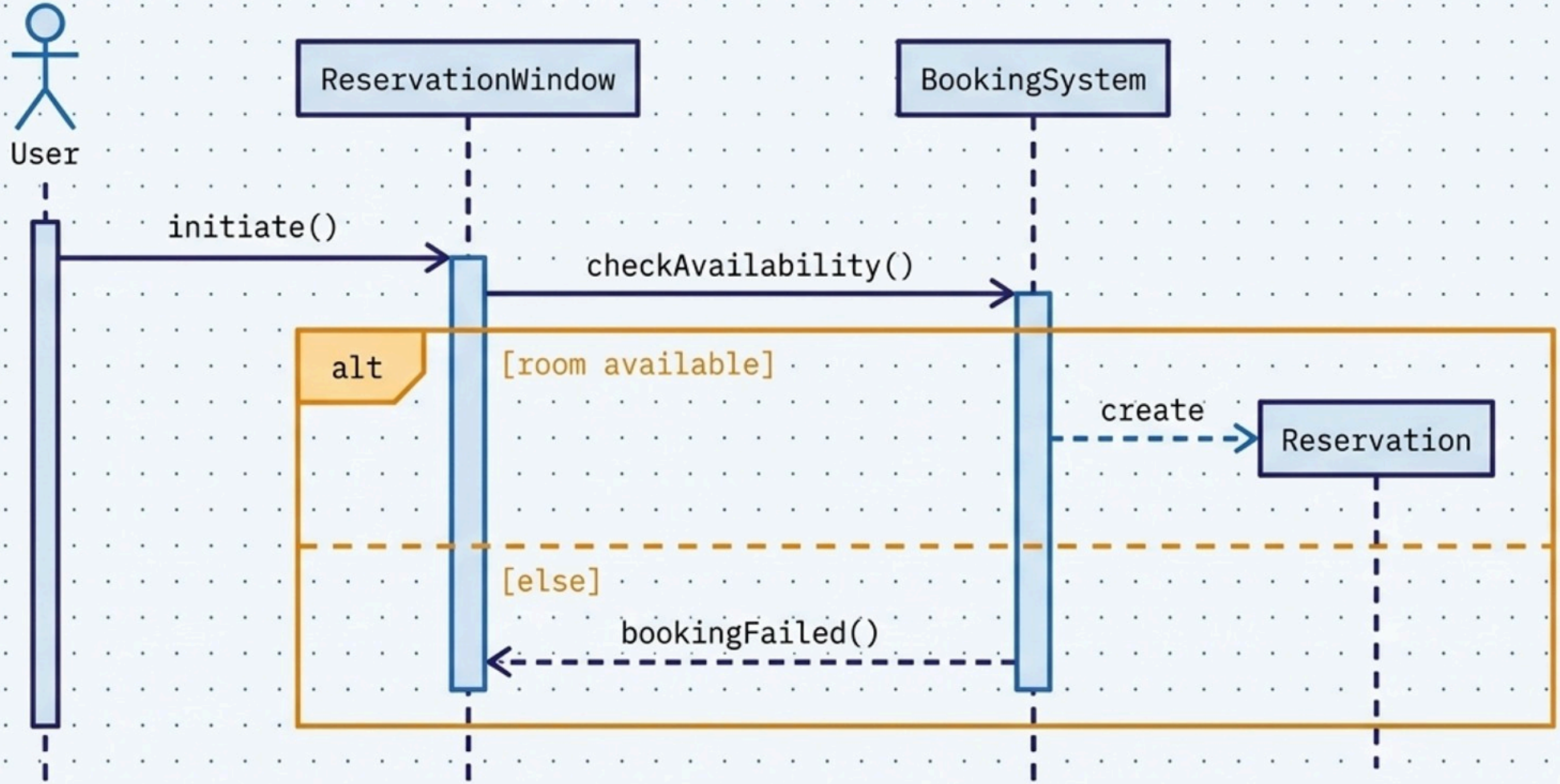


Fragment Operators to Code Logic

Operator	Definition	Code Equivalent
`alt`	**Alternative:** Multiple fragments; only the one with a true condition executes.	`if / else if / else`
`opt`	**Optional:** Executes only if the supplied condition is true. Single trace.	`if (without else)`
`loop`	**Loop:** Executes multiple times based on a guard condition.	`while / for loop`
`par`	**Parallel:** Each fragment runs concurrently.	`multi-threading / async`
`region`	**Critical Region:** Only one thread can execute this fragment at once.	`Mutex lock / sync`
`neg`	**Negative:** Shows an explicitly invalid interaction.	`Error path / Exception`
`ref`	**Reference:** Refers to an interaction defined on another diagram.	`Function / Module call`
`sd`	**Sequence Diagram:** Surrounds an entire root sequence diagram.	`Root program`

System Model: Hotel Reservation

Notice how the ReservationWindow initiates the sequence, progressing downward through an “alt” fragment to handle availability logic dynamically.



The Requirements-to-Code Pipeline

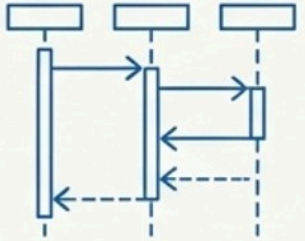
1. Use Case

The specification of a sequence of actions a system can perform with external actors.

2. Scenario

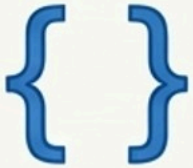
One specific path or flow through that high-level use case.

3. Sequence Diagram



The structural interaction modeled precisely over time.

4. Implementation



The final compiled algorithms and system code.

A scenario is a sequence of events occurring during one execution of a system—the Sequence Diagram is its exact architectural blueprint.

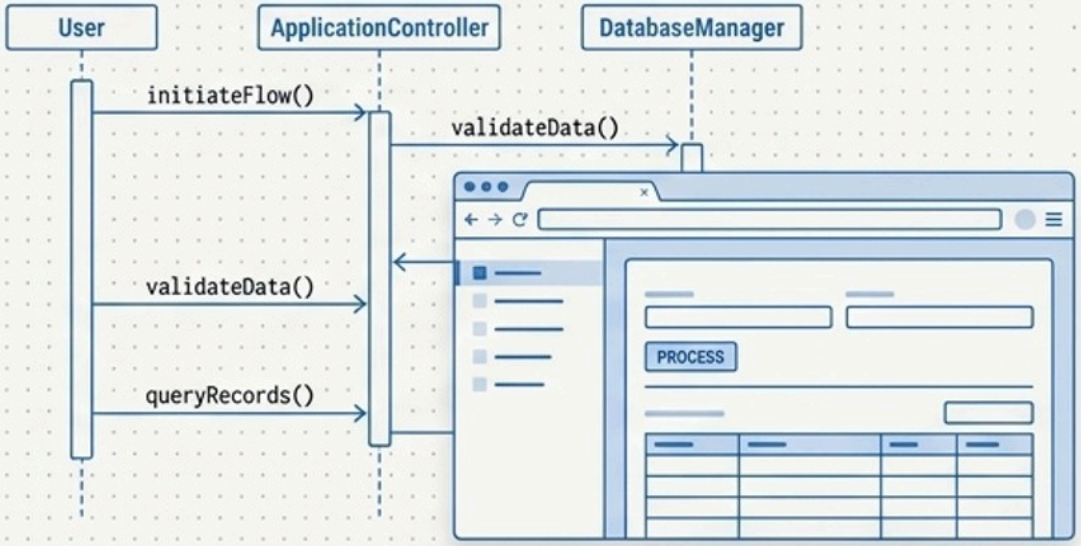
Model Before Code

CHAOTIC RAW CODE

```
if(data){
  return;
} else {
  ...
}
  function calculateRisk(input){
    //todo: implement
  }
       FOR user IN db.users
       SELECT * FROM...
   class SystemController {
     private int id; ...
   }
```

```
{
users {
  ...
  ...
}
{ }
ERROR: null pointer
 exception in module X
/> </
[ ]
[]
```

PRISTINE MODEL & BLUEPRINT



Language Neutral
Design systems without being locked into Java, Python, or C++ constraints.

Cross-Functional
Readable by non-coders; aligns business analysts and developers early.

Team Consensus
Easier to collaborate and debate a visual flow than to rewrite pull requests.

Test & UX Ready
Perfect foundational asset for writing accurate test cases and UI/UX.

Architecture is cheaper to change than code.
Draft the blueprint first.

AI 輔助指南：循序圖塑模 (Sequence Modeling)

- **1. 角色設定與架構模式約束 (Persona & Architecture Pattern)：**
  - 指示 AI 扮演「分散式系統與 API 整合架構師」。
  - **強制引入 BCE 模式：** 嚴格要求區分 Boundary (介面)、Control (業務邏輯)、Entity (資料庫實體)，嚴禁 UI 直接繞過控制器存取資料庫。
- **2. 訊息語法與複合片段嚴格規範：**
  - **同步呼叫 (`->`) 與返回 (`-->`)：** 強制要求對稱繪製返回虛線與回傳型態，避免啟動條無限懸空。
  - **非同步派發 (`->>`)：** 對於背景事件排隊、推播通知，強制使用開放式箭頭。
  - **複合片段 (`alt`)：** 強制要求 AI 必須針對業務失敗情境（如扣款失敗、超時）繪製 `alt [成功] else [失敗]` 區塊。
- **3. 架構師人工審查清單 (Verification Checklist)：**
  - [ ] 是否存在違背分層架構的「越級呼叫」（例如前端 UI 直接調用資料庫實體）？
  - [ ] `alt` 片段中是否具備互斥的守衛條件 `[guard]`？
  - [ ] 啟動條 (activate/deactivate) 的生命區間是否與方法呼叫週期完全吻合？

AI 提示詞範例：外送平台結帳循序模型

1. 角色設定與任務指令：

> 你是一位資深分散式系統後端架構師。
> 請根據以下結帳情境，採用 BCE 模式
> 產出標準 PlantUML 循序圖 (Sequence Diagram)。

2. 塑模約束條件：

- 宣告生命線職責：
  - `boundary CheckoutUI as ":結帳介面"`
  - `control OrderCtrl as ":訂單控制器"`
  - `boundary StripeAPI as ":金流服務"`
  - `entity OrderEntity as ":訂單實體"`
  - `queue KitchenQueue as ":廚房事件佇列"`
- 必須使用 `alt` 片段處理 `[扣款成功]` 與 `[扣款失敗]`。
- 廚房通知必須使用非同步箭頭 `->>` 發送至事件佇列。

3. 輸入需求敘述 (Requirements Statement)：

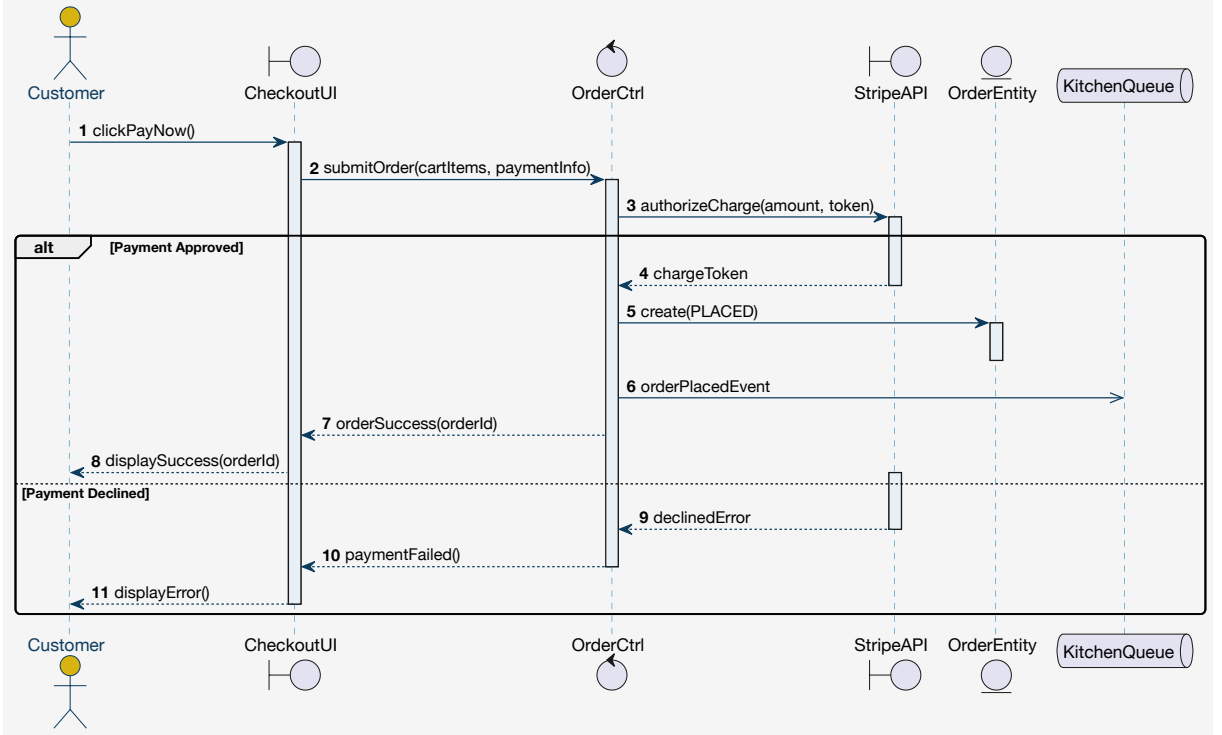
> *「顧客在* ***CheckoutUI*** *點擊『立即付款』按鈕。*
> ***CheckoutUI*** *轉發* `submitOrder(購物車清單, 支付憑據)` *至* ***OrderCtrl***。
> ***OrderCtrl*** *呼叫* ***StripeAPI*** *執行* `authorizeCharge(金額, 憑據)`。
> *若授權成功：*
> 1. *StripeAPI 回傳授權碼* `chargeToken`。
> 2. *OrderCtrl 呼叫* ***OrderEntity*** *建立持久化訂單紀錄（狀態設為* `PLACED`*）。*
> 3. *OrderCtrl 以非同步訊息發布* `orderPlacedEvent` *至* ***KitchenQueue***。
> 4. *OrderCtrl 回傳* `orderSuccess(訂單編號)` *給 CheckoutUI。*
>
> *若授權失敗：*
> 5. *StripeAPI 回傳* `declinedError`。
> 6. *OrderCtrl 記錄失敗日誌，不建立訂單，直接回傳* `paymentFailed()`。」

AI 生成之 PlantUML 與架構圖：外送平台結帳循序模型

AI 生成之 PlantUML 規格代碼：

```
@startuml
autonumber
actor Customer
boundary CheckoutUI
control OrderCtrl
boundary StripeAPI
entity OrderEntity
queue KitchenQueue

Customer -> CheckoutUI: clickPayNow()
activate CheckoutUI
CheckoutUI -> OrderCtrl: submitOrder(cart, payment)
activate OrderCtrl
OrderCtrl -> StripeAPI: authorizeCharge(amount, token)
activate StripeAPI

alt Payment Approved
  StripeAPI --> OrderCtrl: chargeToken
  deactivate StripeAPI
  OrderCtrl -> OrderEntity: create(PLACED)
  OrderCtrl ->> KitchenQueue: orderPlacedEvent
  OrderCtrl --> CheckoutUI: orderSuccess(orderId)
  CheckoutUI --> Customer: displaySuccess()
else Payment Declined
  StripeAPI --> OrderCtrl: declinedError
  OrderCtrl --> CheckoutUI: paymentFailed()
  deactivate OrderCtrl
```

渲染產出之架構圖：



課堂活動：使用 AI 演進循序模型 (雙人動手實作)

動手實作實驗：變更動態時序並審核 AI 產出 (4 分鐘)

- **1. 在結帳流程中加入一個極端邊界情境 (Edge Case)：**
  - *金流逾時 (Gateway Timeout)：* 外部付款閘道超過 5 秒未回應（觸發重試或補償）。
  - *拆分付款 (Split Payment)：* 允許顧客使用電子錢包點數折抵搭配信用卡差額支付。
  - *庫存售罄 (Out of Stock)：* 在付款授權過程中，餐廳剛好宣告餐點售罄。
- **2. 要求 AI 重新生成：** 提示 LLM 更新對應的 PlantUML 循序圖代碼。
- **3. 渲染並嚴格審查：** 貼入 PlantText 渲染並審查：
  - AI 是否維護**分層架構**（UI → 控制器），杜絕 UI 越權直連資料庫？
  - **成功與異常分支** (`alt`) 是否皆具備完整明確的回傳訊息？
  - 非同步後台通知（如通知廚房）是否確實與同步付款解耦？

觀念檢核測驗 5 (CCQ 5)

在 BCE (Boundary-Control-Entity) 架構循序圖中，當顧客透過結帳前端點擊『確認下單』按鈕時，該事件應先傳遞給哪一種類型的物件處理？

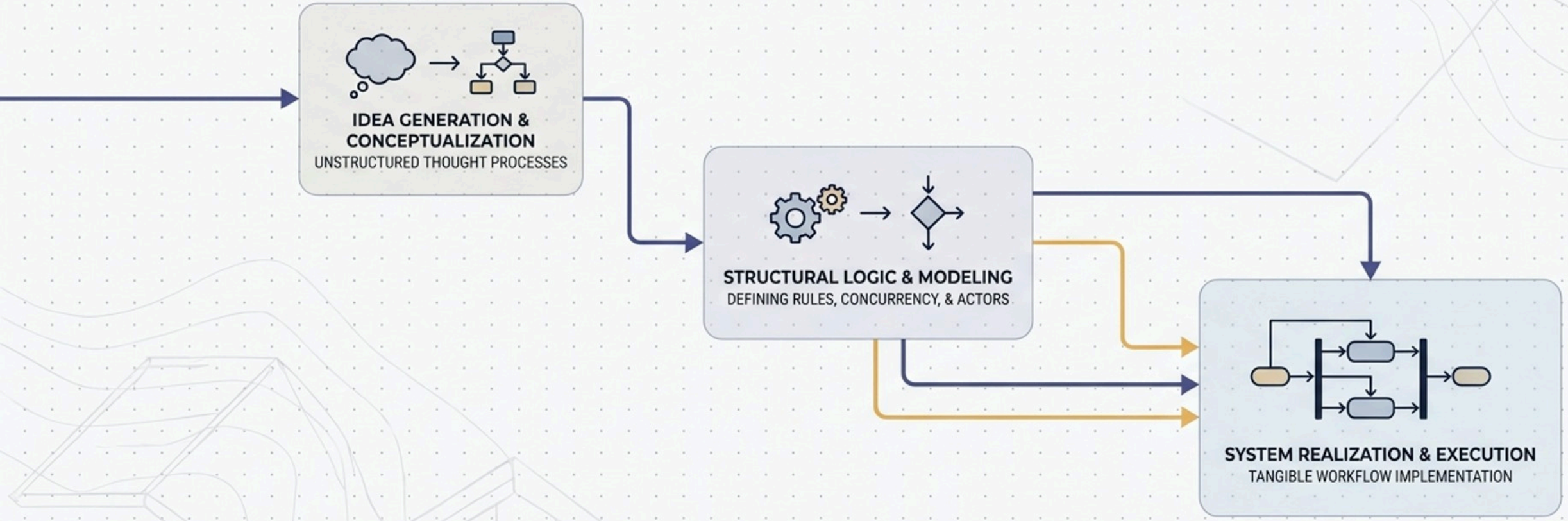
- **A.** 直接傳遞給 `Order` 實體物件立即寫入資料庫
- **B.** 直接呼叫 `PaymentGatewayAPI` 邊界物件立即刷卡扣款
- **C.** 傳遞給 `OrderController` 控制物件協調業務邏輯驗證與調用
- **D.** 直接傳遞給 `Restaurant` 實體物件確認餐廳接單

4.6 流程與活動圖模型

> *"複雜的業務流程就像多條河流匯聚；活動圖能讓我們精確掌控每一道支流的並行與匯合。"*

Mapping the Dynamic System

A guide to modeling complex, concurrent, and multi-actor workflows with UML Activity Diagrams.



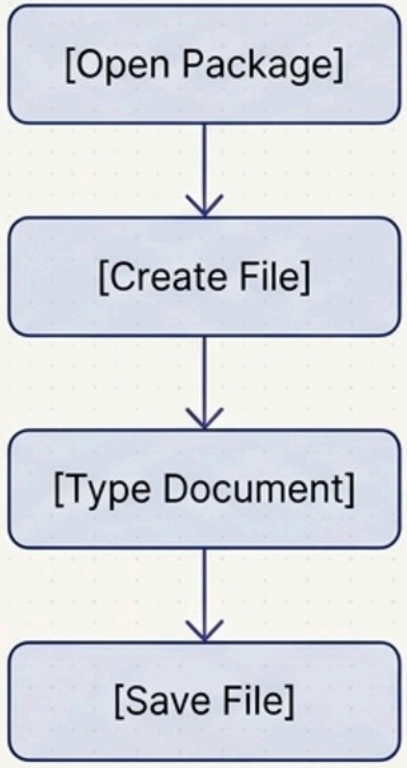
Moving beyond static structures to model dynamic behavior

An Activity Diagram is a crucial behavioral diagram in the Unified Modeling Language (UML). It is an advanced version of a flowchart designed to describe the dynamic aspects of a system, mapping the precise flow from one activity to the next.

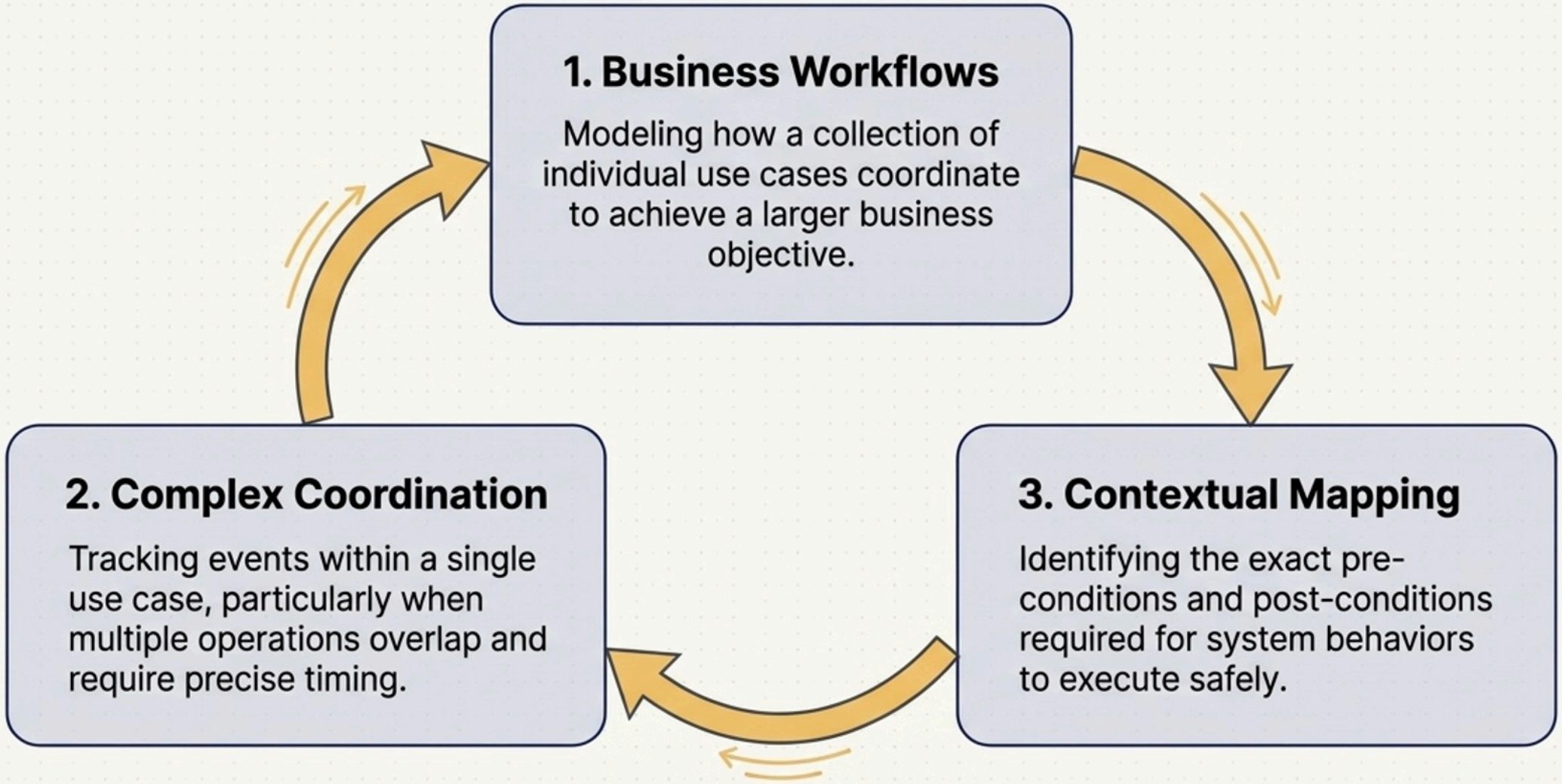
The Chaos: Text-Based Processes

Process starts when the user opens the package and then needs to initialize the system, following which they must create must create a new file and ensure all settings are correct, after which they proceed to type the document, remembering to format it according to the guidelines, and finally, after reviewing, they must save the file in the appropriate directory before closing the application and moving on to the next task.

The Clarity: Visual Logic



Three triggers for behavioral modeling



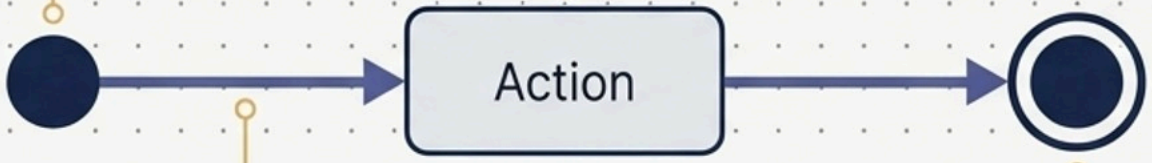
Standard flowcharts fail at systemic complexity

Capability	Standard Flowchart	UML Activity Diagram
Sequential Logic	✓	✓
Conditional Splits	✓	✓
Parallel / Concurrent Processes	✗	✓
Actor Accountability (Swimlanes)	✗	✓
Object States & Transformation	✗	✓

The core vocabulary of system flow

Initial Node
Portrays the spark—the beginning of a set of actions or activities.

Action
A specific, singular task to be performed by the system or actor.

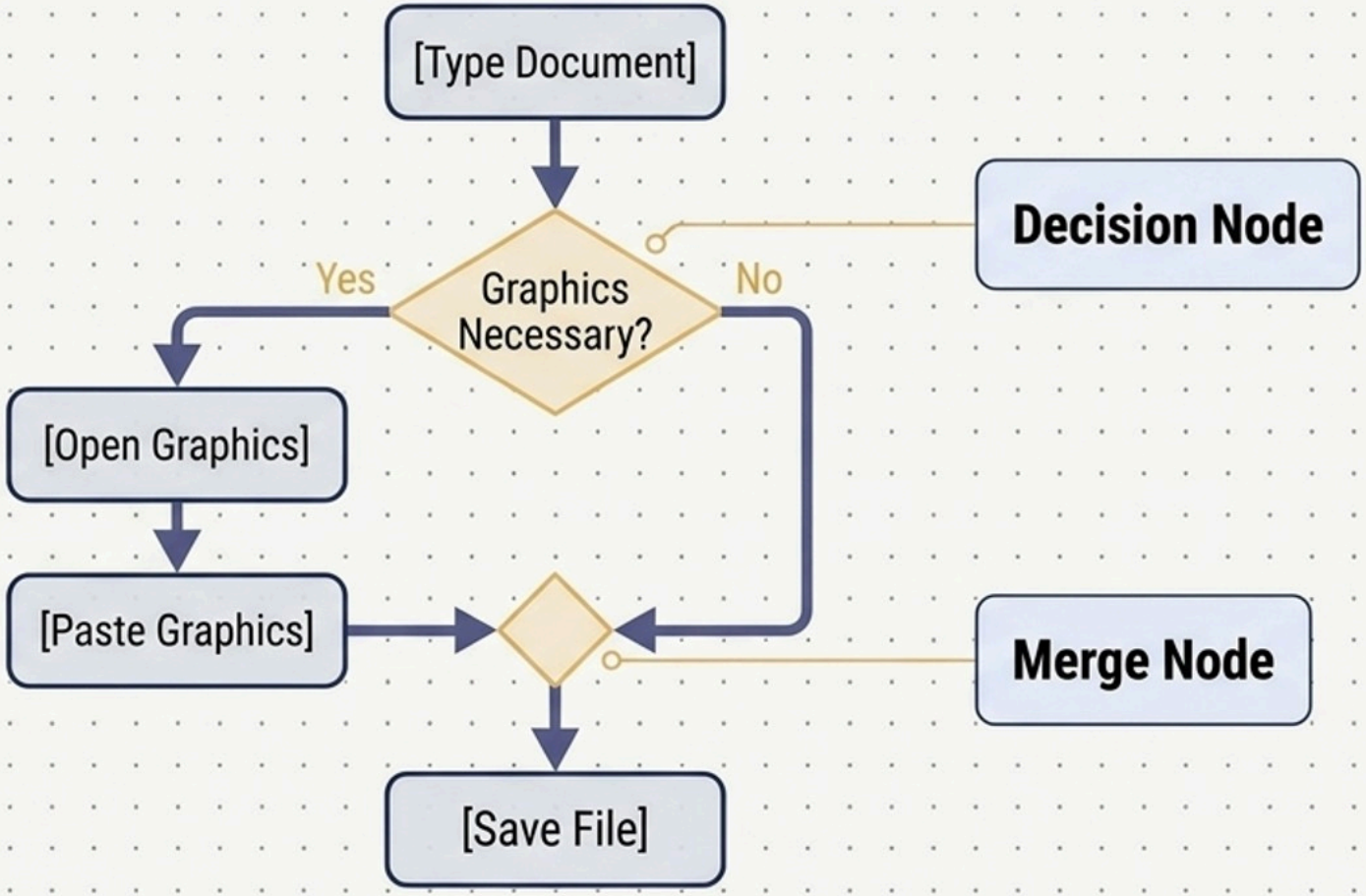


Control Flow
Shows the strict sequence of execution from one node to the next.

Activity Final Node
Stops all control and object flows, terminating the sequence.

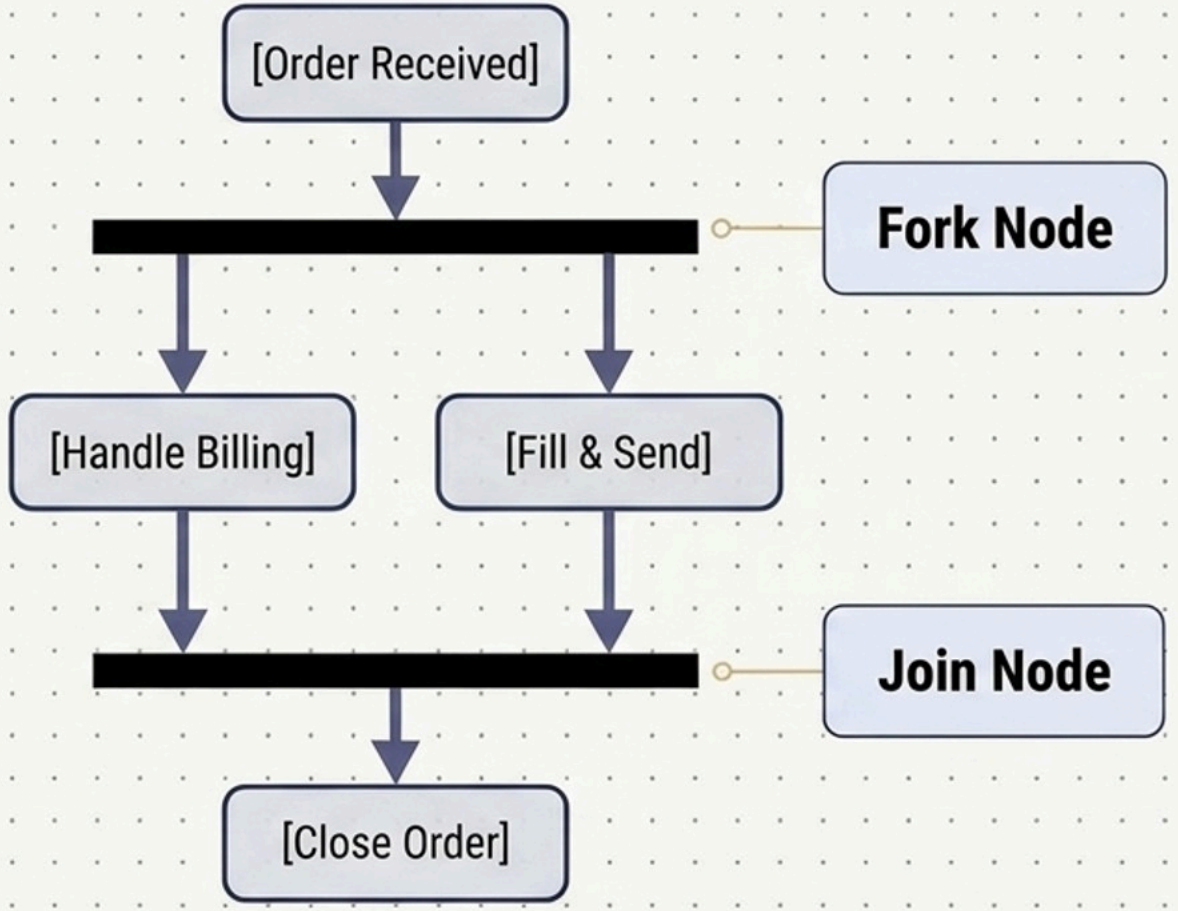
Handling conditional system logic

When a system must evaluate a test condition, a Decision Node ensures the control flow only proceeds down a single valid path. A Merge Node safely brings these disparate decision paths back together.



Orchestrating parallel system actions

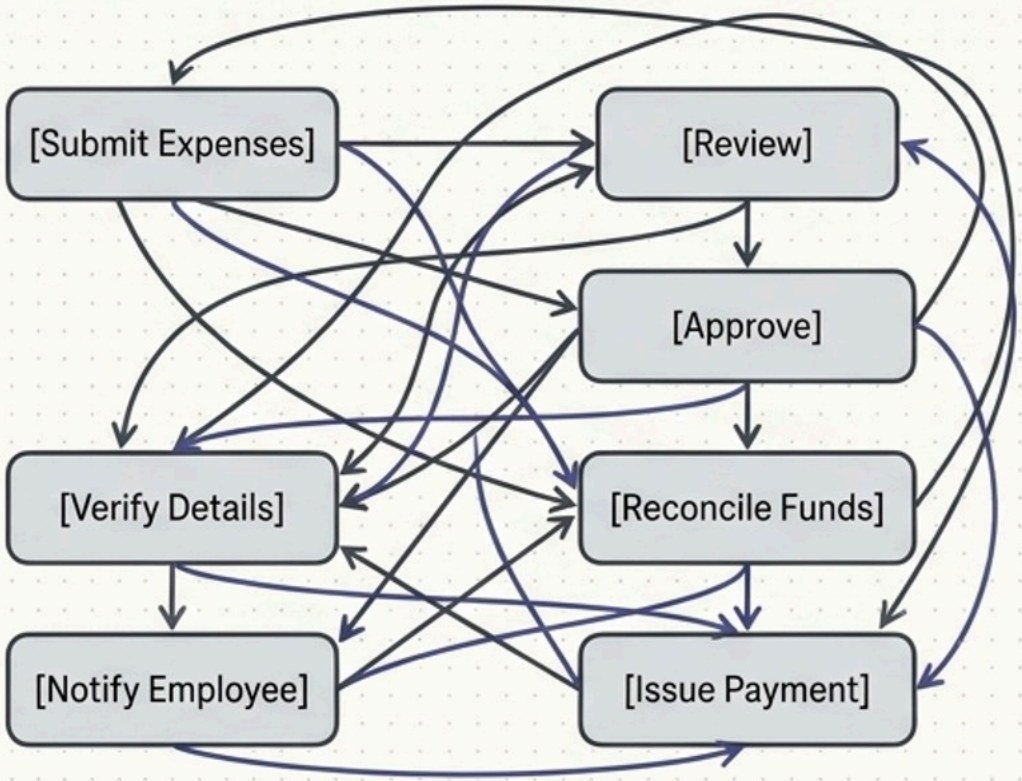
Systems rarely execute one step at a time. The **Fork Node** splits behavior into concurrent actions. The **Join Node** ensures the system waits to bring parallel flows back together before proceeding.



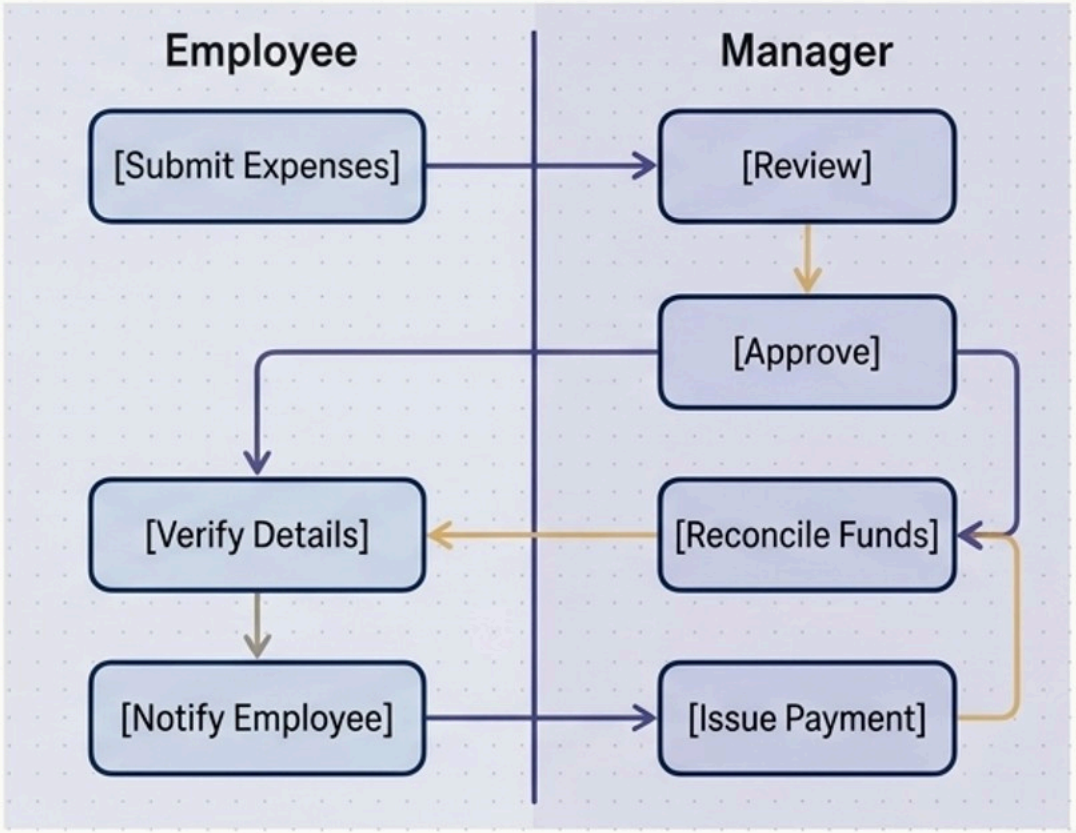
The multi-actor accountability problem

Knowing WHAT happens in a business process is useless if you don't know WHO executes it. Without boundaries, complex workflows degrade into ambiguity.

Without Swimlanes: Ambiguity

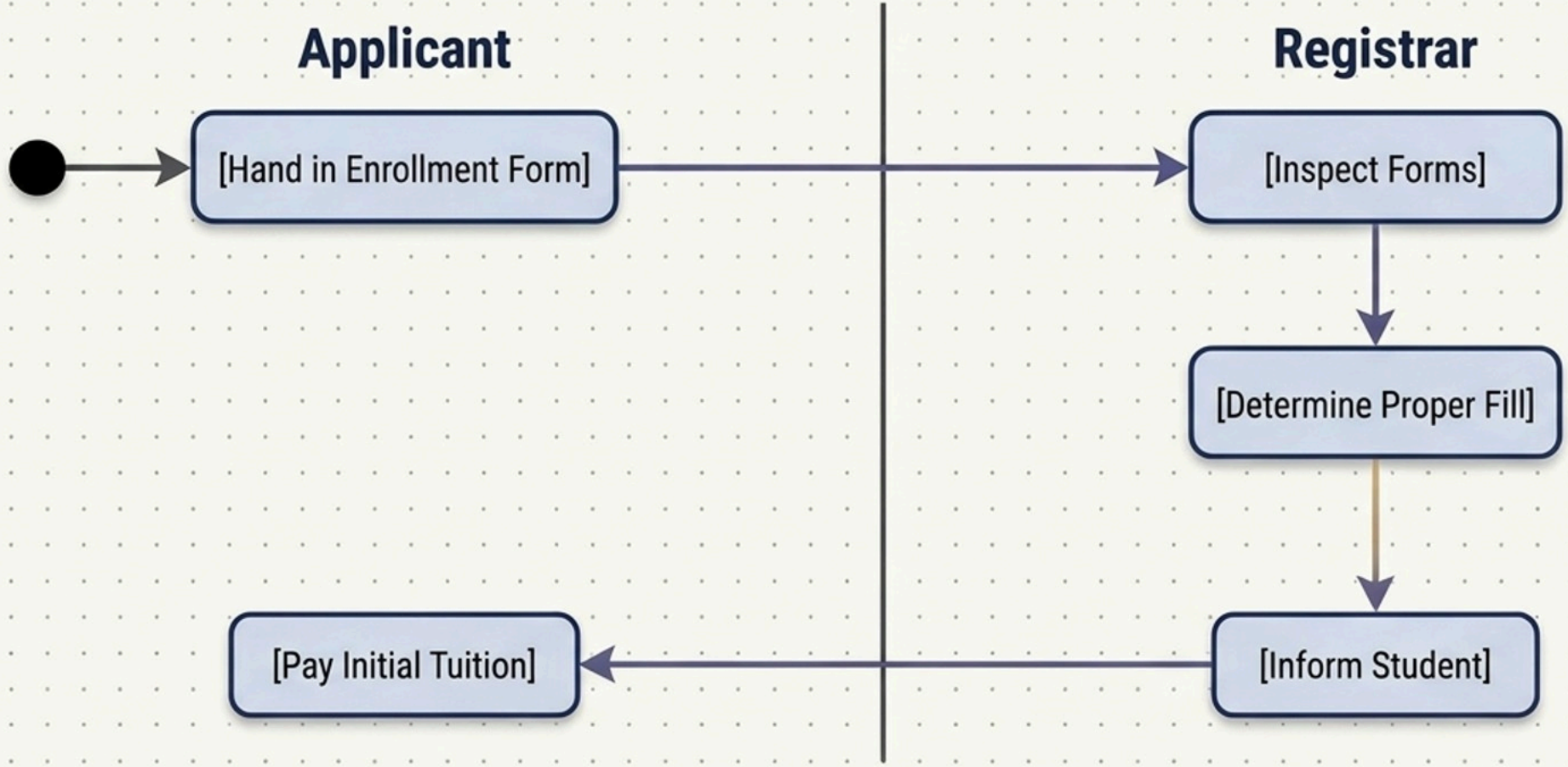


With Swimlanes: Accountability



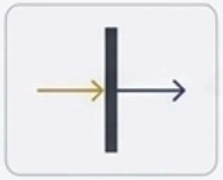
Grouping activities by actor or thread

A Swimlane (or Partition) groups activities performed by the same actor. It visualizes hand-offs seamlessly—such as a student passing paperwork to a registrar and waiting for system validation.



The complete visual taxonomy of dynamic modeling

The Actors



Swimlane / Partition

Groups activities by actor or execution thread.

The Actions

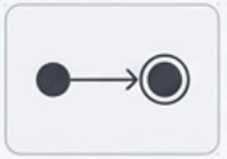
Action Node

A single task to be performed.

Object Node

Represents a connected physical or digital artifact.

The Directors



Initial & Final Nodes

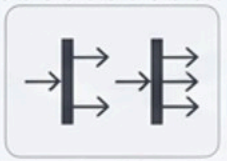
Start and stop all system flows.

Control & Object Flows

Sequence execution and data movement.

Decision & Merge Nodes

Split and return conditional paths.

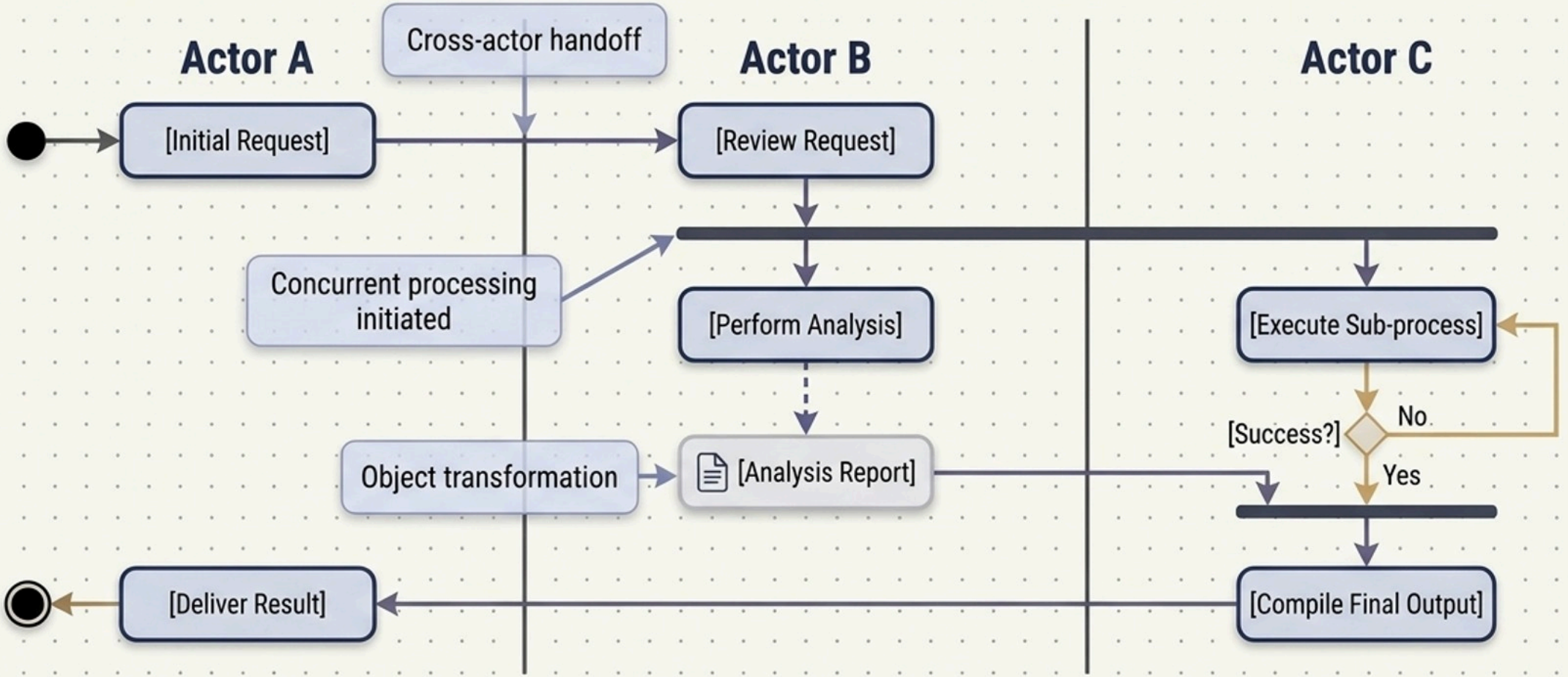


Fork & Join Nodes

Split and return concurrent parallel paths.

The unified workflow model

When Swimlanes, Concurrency, and Conditional logic combine, complex systemic chaos is transformed into a readable, executable blueprint.



Three phases to map your systems

1. Examine the Business Context

Identify candidate use cases by analyzing broader business workflows. Map the precise pre-conditions and post-conditions required.

2. Map the Core Workflows

Build out the activity logic mapping the flow between different use cases, and the sequences contained within them.

3. Zoom In on Complexity

Create detailed, high-level activity diagrams to map out highly complex workflows and intricate operations on specific objects.

AI 輔助指南：活動圖塑模 (Activity Modeling)

- **1. 角色設定與現代語法標準 (Persona & Syntax Standard)：**
  - 指示 AI 扮演「分散式工作流與業務流程架構師 (Workflow Architect)」。
  - **強制要求使用現代語法：** 指定採用 **PlantUML Activity Beta 新版語法** (`start`, `stop`, `if/else`, `fork/fork again/end fork`)，避免過時的舊版標記。
- **2. 泳道當責邊界嚴格分配 (Swimlane Accountability)：**
  - 要求明確定義泳道分區（如 `|顧客|`, `|餐廳廚房|`, `|調度引擎|`, `|外送員|`），動作必須歸屬在對應欄位中。
- **3. 並行與同步工程紀律：**
  - **分岔 (`fork` / `fork again`)：** 明確指示哪些流程必須平行運作（例如廚房做菜與系統配對司機）。
  - **同步結合 (`end fork` / `join`)：** 強制要求在取餐交件前必須同步，防止產生單獨無人外送的競爭條件 (Race Condition)。
- **4. 架構師人工審查清單 (Verification Checklist)：**
  - [ ] 每個 `fork` 是否都有對應的 `end fork` 結合屏障？
  - [ ] 跨泳道的控制流轉移是否合理且具備最小依賴？
  - [ ] 是否存在流程在結合棒無限等待、造成業務死結的缺陷？

AI 提示詞範例：外送平台廚房與外送並行活動模型

1. 角色設定與任務指令：

> 你是一位分散式工作流程架構師。
> 請根據以下外送履約流程，使用 `PlantUML activity beta` 語法產生具備泳道與並行的活動圖。

2. 塑模約束條件：

- 建立 4 個泳道：
  - `|顧客|`，`|餐廳廚房|`，
  - `|調度引擎|`，`|外送員|`
- 訂單付款後立即插入 `fork` 並行分岔：
  - 分支一：廚房接單、備餐並包裝餐點。
  - 分支二：調度系統規劃路徑、媒合司機、司機開車抵達店家。
- 使用 `end fork` (結合) 同步「包裝完成」與「司機抵達」。
- 結合後進行交餐配送，驗證 OTP 後 `stop`。

3. 輸入需求敘述 (Requirements Statement)：

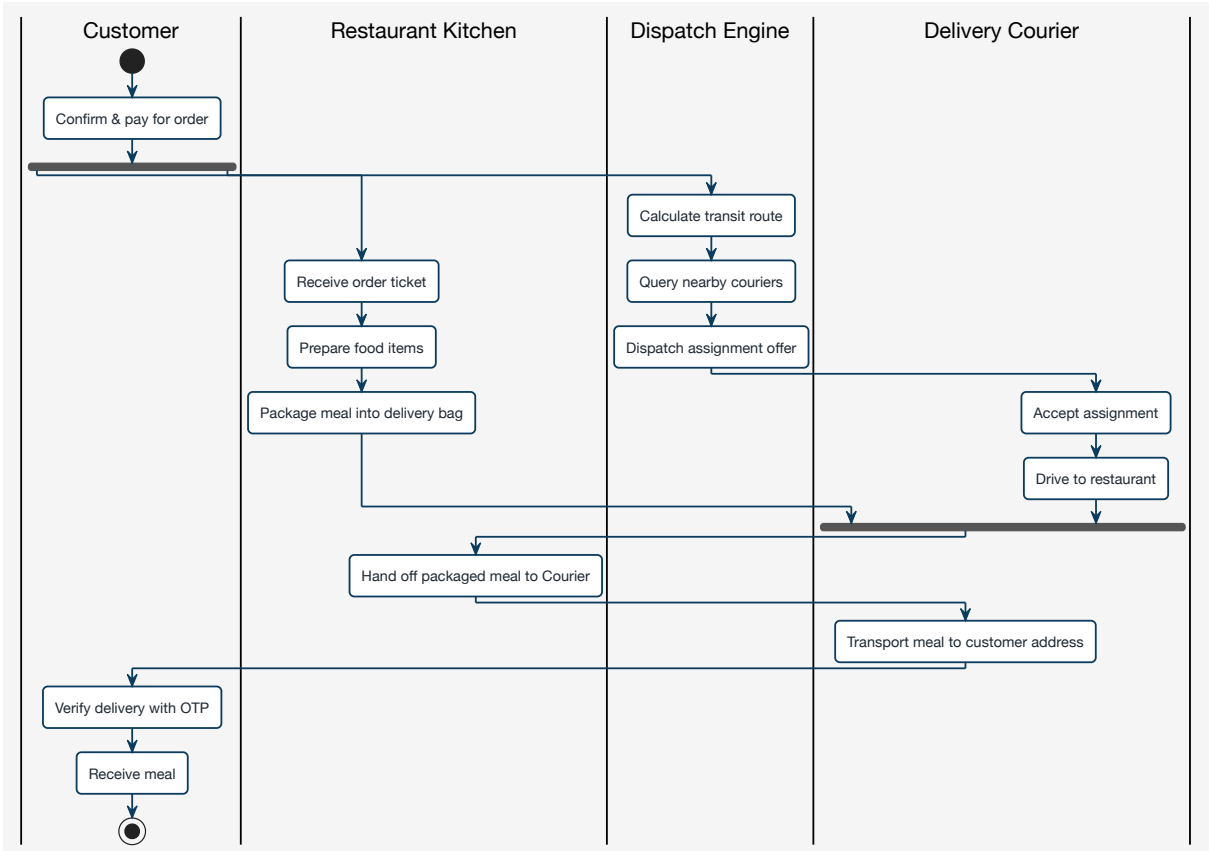
> *「顧客在 |**顧客**| 泳道送出訂單並完成信用卡付款。*
> *付款確認後，系統立即啟動兩條平行並行工作流：*
>
> - *在 |**餐廳廚房**| 泳道：廚房列印工單、廚師開始烹調餐點、完成後將食物裝盒並貼上封條。*
> - *同時間，在 |**調度引擎**| 泳道：系統計算配送里程、搜尋附近空閒外送員並派送接單推播；在 |**外送員**| 泳道，外送員點擊接單並騎車前往餐廳。*
>
> ***同步里程碑：** 外送員抵達櫃檯時，必須等待『餐點包裝完成』且『外送員已經抵店』兩項條件皆達成，才能進行實體交餐。*
> *交餐後，外送員騎車前往顧客地址，顧客輸入 OTP 驗證簽收，流程結束。」*

AI 生成之 PlantUML 與架構圖：外送平台活動流程模型

AI 生成之 PlantUML 規格代碼：

```
@startuml
|Customer|
start
:Confirm & pay for order;

fork
  |Restaurant Kitchen|
  :Receive order ticket;
  :Prepare food items;
  :Package meal into delivery bag;
fork again
  |Dispatch Engine|
  :Calculate transit route;
  :Query nearby couriers;
  :Dispatch assignment offer;
  |Delivery Courier|
  :Accept assignment;
  :Drive to restaurant;
end fork

|Restaurant Kitchen|
:Hand off packaged meal to Courier;

|Delivery Courier|
:Transport meal to customer address;

|Customer|
```

渲染產出之架構圖：



課堂活動：使用 AI 演進活動工作流 (雙人動手實作)

動手實作實驗：變更流程邏輯並審核 AI 產出 (4 分鐘)

- **1. 在活動圖提示詞中加入一條流程規則：**
  - *外送員重新派單：* 若原配對之外送員拒絕接單，系統自動重新搜尋媒合下一位司機。
  - *品管出餐檢驗：* 廚房品檢若發現餐點焦掉，必須重新下鍋製作後方能打包。
  - *顧客自取模式：* 若顧客選擇自取，完全略過外送員調度與派送泳道。
- **2. 要求 AI 重新生成：** 提示 LLM 更新對應的 PlantUML 活動圖代碼。
- **3. 渲染並嚴格審查：** 貼入 PlantText 渲染並審查：
  - 各項活動步驟是否皆正確坐落於所屬角色的**職責泳道**？
  - **平行分岔 (Fork) 與匯聚 (Join)** 是否成對匹配，杜絕死結 (Deadlock)？
  - **決策菱形 (Decision Diamonds)** 是否標註了完整互斥的轉移防護條件？

觀念檢核測驗 6 (CCQ 6)

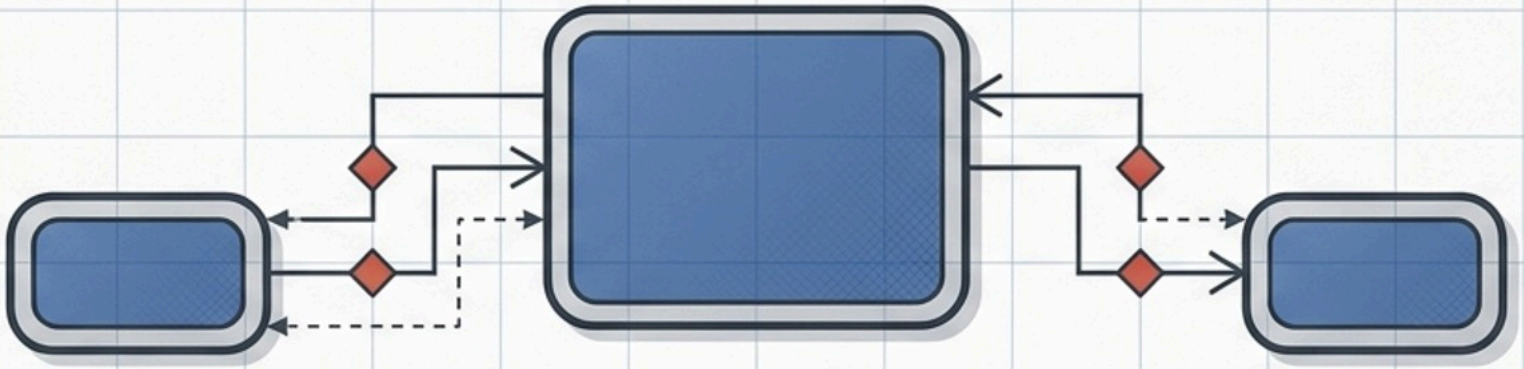
在 UML 活動圖中，**分岔/結合 (Fork/Join 同步條)** 與 **決策/合併 (Decision/Merge 菱形)** 的核心差異為何？

- **A.** Fork/Join 用於將單一流程切分為多條「並行同時執行」的執行緒；Decision/Merge 則依據布林條件由多條路徑中「嚴格選擇其一 (互斥)」執行。
- **B.** Fork/Join 用於類別繼承；Decision/Merge 用於物件實體化。
- **C.** Fork/Join 只能處理資料流；Decision/Merge 只能處理錯誤例外。
- **D.** Fork/Join 必須等待人工確認；Decision/Merge 由系統計時器自動觸發。

4.7 行為與狀態機模型

> *"軟體系統在動態執行中的行為，取決於它當下所處的狀態，以及觸發狀態轉移的事件。"*

The Anatomy of State-Dependent Behavior

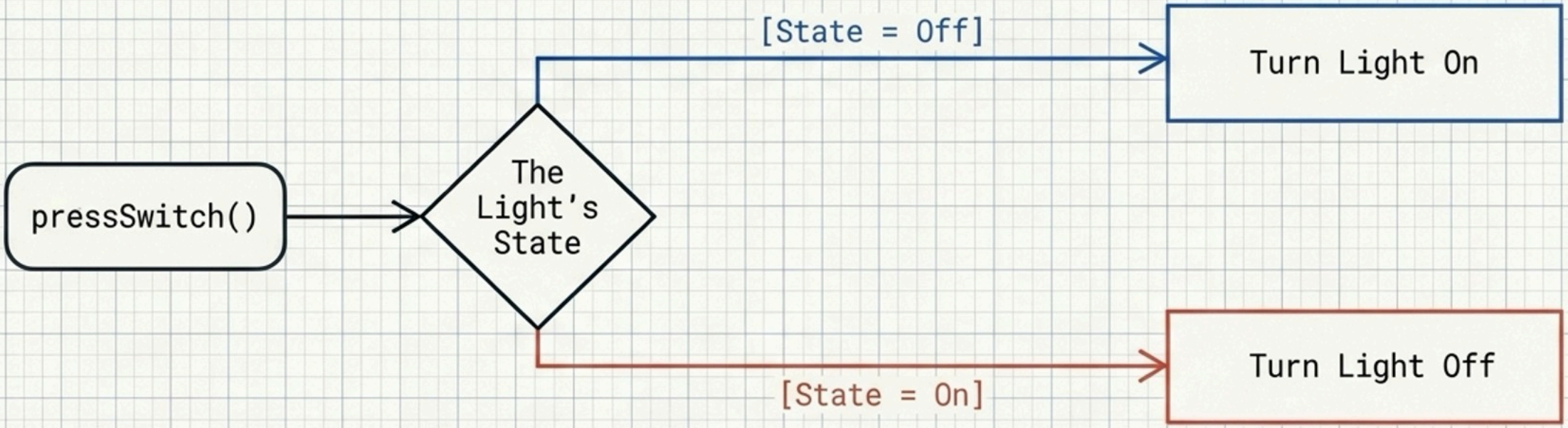


A definitive framework for modeling, deconstructing, and designing UML State Machine Diagrams.

v. Unified Modeling Language | Blueprint Series

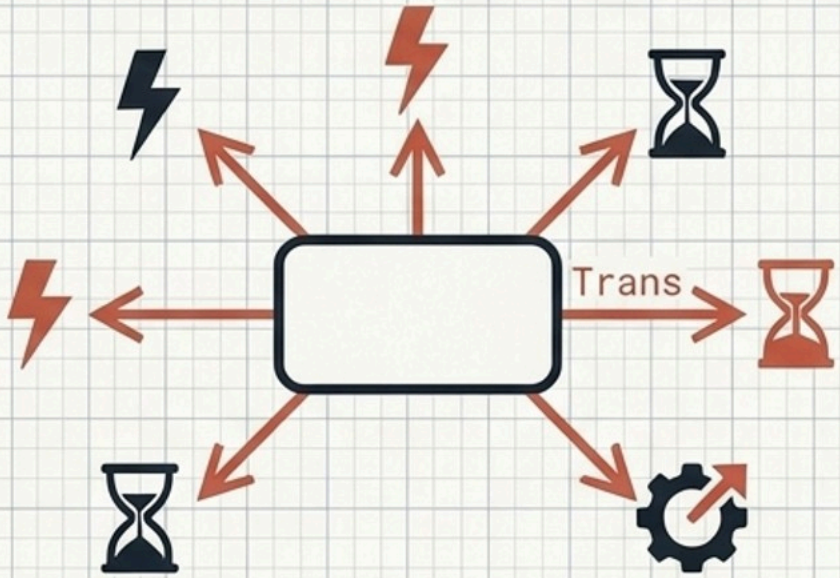
The Same Event Yields Different Results Based on State

A system's behavior is not just a consequence of its current input.
It is entirely dependent on its preceding state and past history.



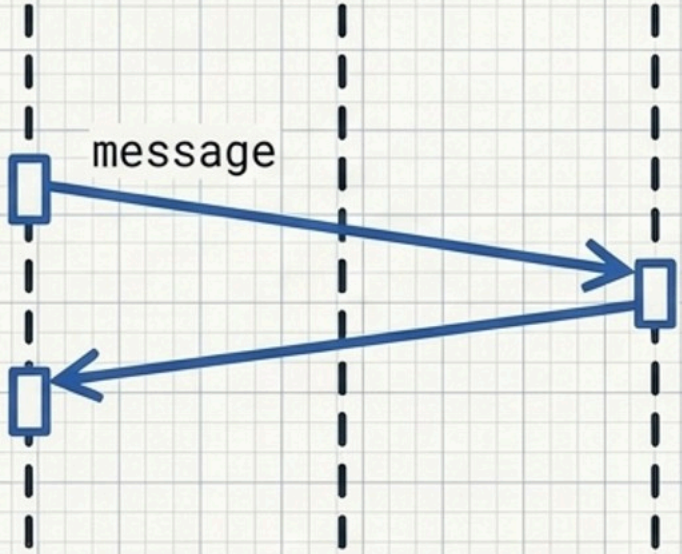
Positioning the Tool Within the UML Ecosystem

State Machine Diagram



Maps all possible events and states for a single isolated object.

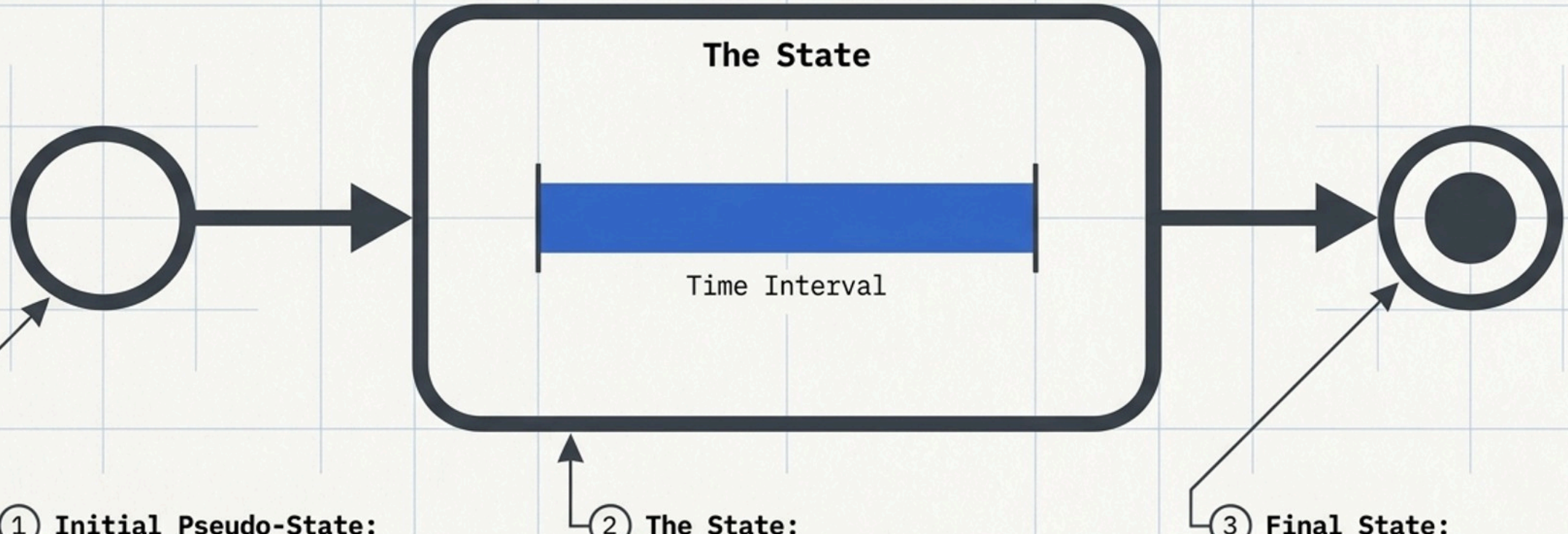
Sequence Diagram



Maps a single linear interaction across multiple connected objects.

A State is a Defined Interval in an Object's Lifecycle

A state represents an abstraction of attribute values where a specific constraint holds true, an activity is executed, or the system waits for an event.



1. **Initial Pseudo-State:** The starting line.
2. **The State:** The active timeframe where conditions are met.
3. **Final State:** The termination point (omitted in closed-loop systems).

The Four Triggers That Initiate State Transitions

Events are the incidents—internal or external—that pass information and force a system to move from one state to another.

Signal Event

Arrival of an asynchronous message or signal.

Call Event

Arrival of a procedural call to an operation.

Time Event

Occurs strictly after a specified time has elapsed.

Change Event

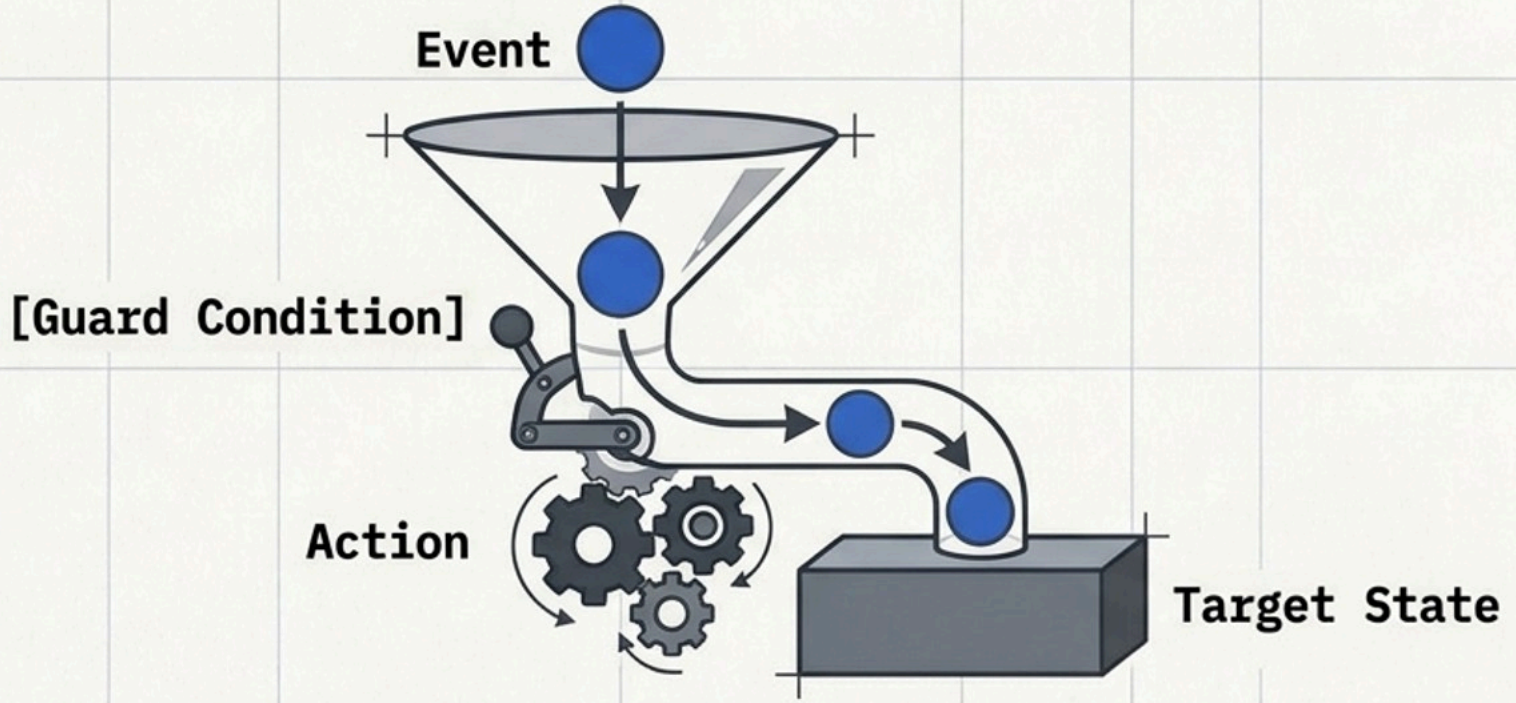
Fires whenever a specified condition is mathematically met.

The Mechanics of a Transition

A transition is the actual movement between states. It requires a source, an event trigger, and an executable action. Automatic transitions skip the event and action entirely.

Transition Equation

[Source State] + (Event) + [Action] = [Target State]



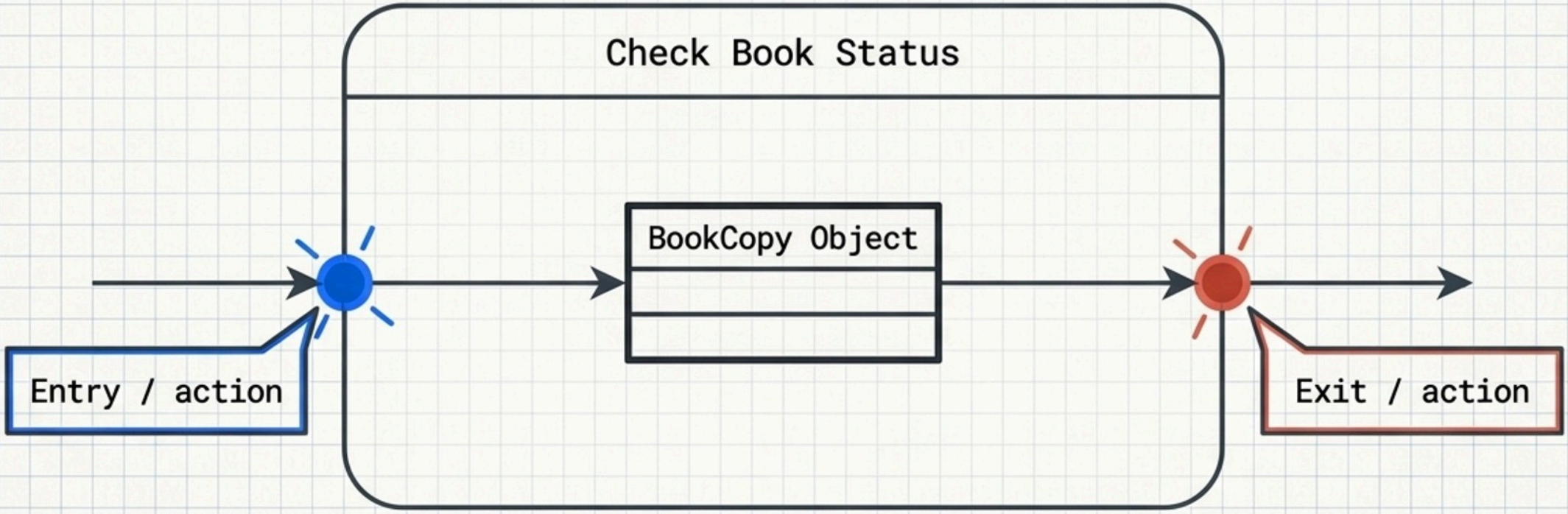
Execution Engines: Actions vs. Activities

States aren't just passive waiting rooms; they trigger computations. Differentiating between instant actions and ongoing activities is critical for system accuracy.

Feature	Action	Activity
Nature	Atomic Computation (Operation call, destroy object)	Non-atomic computation
Duration	Instantaneous	Ongoing (Runs to completion or indefinitely)
Interruptibility	NEVER interruptible	YES, can be interrupted by events
Triggers	On Entry, Do, On Event, On Exit	Sustained state presence

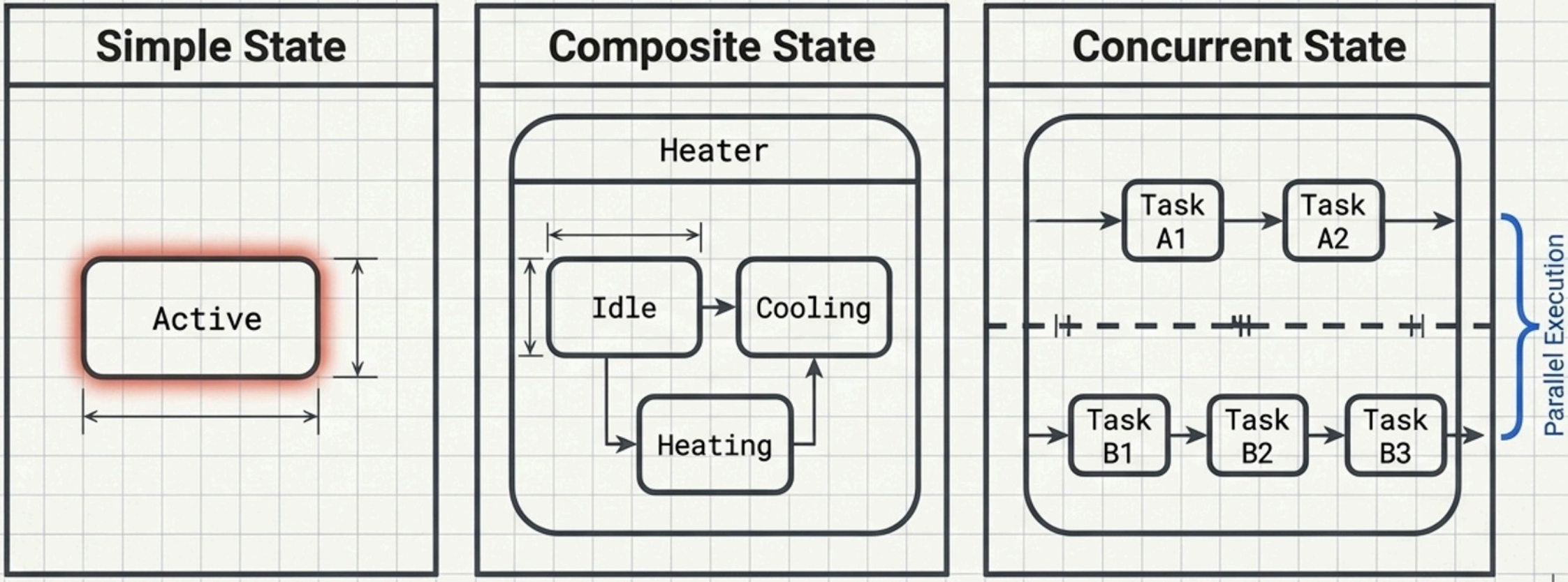
Boundary Executions: Entry and Exit Actions

Actions can be hardwired to the precise millisecond a state is entered or exited. These executions must hold true for every entry or exit occurrence; otherwise, the action belongs on the specific transition arc instead.



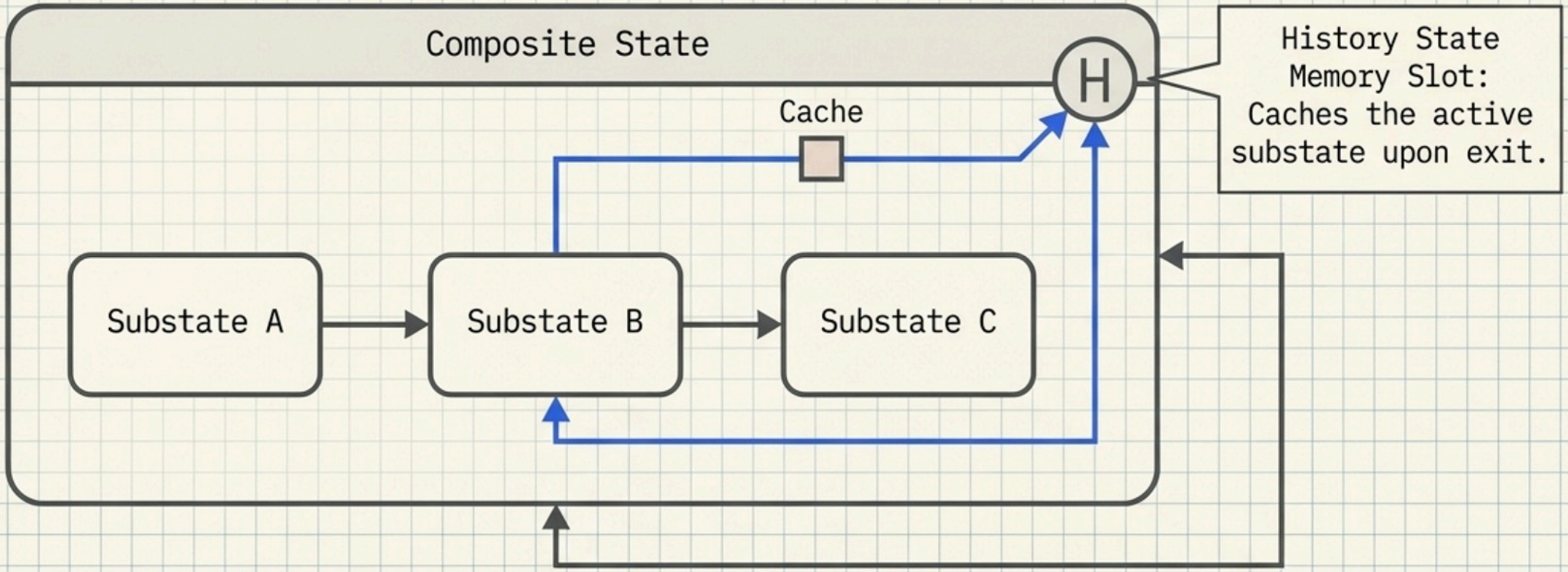
Scaling Architecture: The Complexity Matrix

As system behavior becomes multifaceted, flat state machines are insufficient. Contextual conditions dictate the nesting or parallel execution of states.



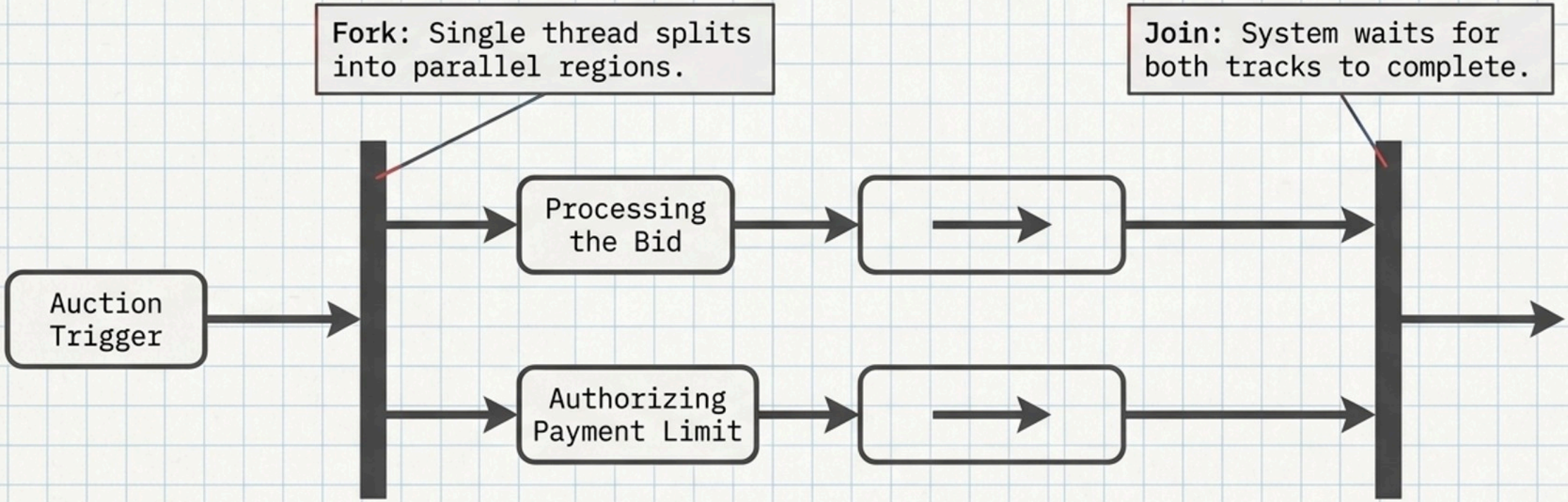
Composite States and System Memory

By default, entering a composite state restarts the nested machine at its initial state. History states act as a cache, allowing the system to remember and re-enter the exact substate that was active prior to leaving.



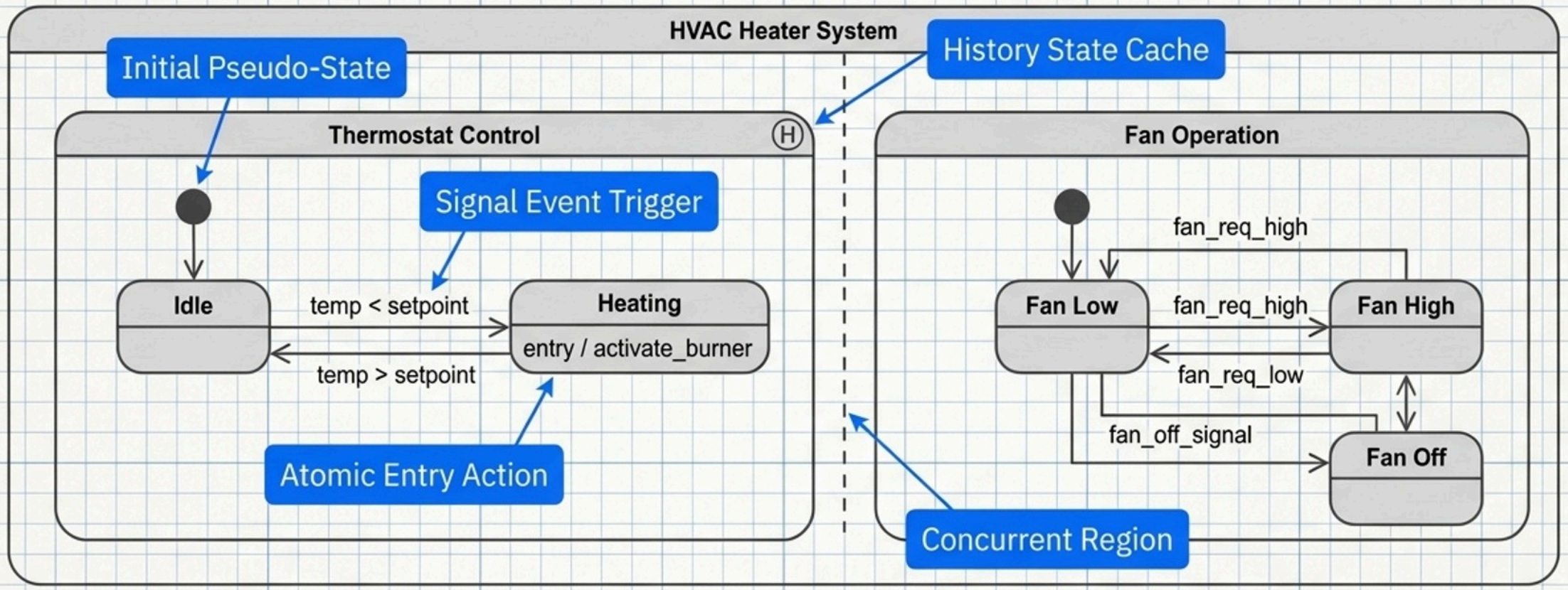
Concurrency: Forking and Joining Execution Threads

For activities requiring concurrent sub-activities, states are nested in parallel regions. The system cannot exit the composite state until all independent substates have successfully terminated.



The Unified Blueprint

By deconstructing the structural variables, complex behavioral models transform from chaotic webs into precise, readable system architectures.



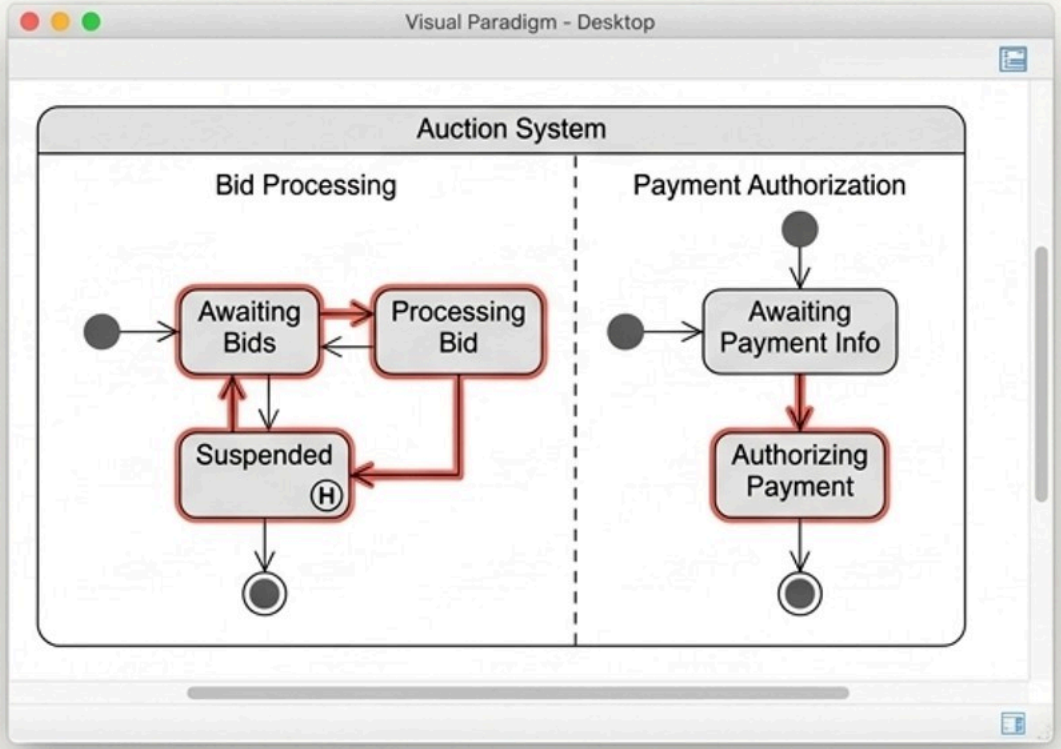
Operationalizing Behavioral Logic with AI

Modern architecture replaces manual drafting with AI-enabled logic generation. Tools like Visual Paradigm transform plain-text behavioral rules into precise, structural UML State Charts in seconds.

The Prompt

Create an auction system with parallel bid processing and payment authorization. Include a history state for suspended auctions.

The Output



[VP Desktop] Integrated AI for real-time model refinement.

[Smart Discovery] Automated identification of nested states and guard conditions.

[Deployment] From system requirement to deployable logic framework instantly.

AI 輔助指南：狀態機塑模 (State Machine Modeling)

- **1. 角色設定與形式化檢驗紀律 (Persona & Formal FSM Discipline)：**
  - 指示 AI 扮演「形式化系統驗證與有限狀態機 (FSM) 專家」。
  - **狀態命名嚴格紀律：** 狀態名稱必須代表物件的「客觀穩態條件」，一律採用形容詞或過去分詞（如 `Placed`, `Preparing`, `Delivered`），嚴禁使用動詞當作狀態名稱！
- **2. 轉換語法與防禦性不變量約束：**
  - **法定語法公式：** 強制要求轉換連線必須遵循 `事件名稱 [守衛條件] / 執行動作`。
  - **進入/離開動作：** 對於具備資源配置或通知的關鍵狀態，明確定義 `entry /` 與 `exit /`。
  - **嚴格封鎖非法路徑：** 明確宣告哪些狀態之間「絕對禁止跳轉」（例如配送中嚴禁直接點擊取消）。
- **3. 架構師人工審查清單 (Verification Checklist)：**
  - [ ] 是否存在無法抵達的「孤島狀態」或進得去出不來的「死結陷阱」？
  - [ ] 守衛條件 `[guard]` 是否互斥且覆蓋所有可能？
  - [ ] 終止狀態 (`[*]`) 是否正確對應至業務的終結？

AI 提示詞範例：外送平台訂單生命週期狀態機模型

1. 角色設定與任務指令：

> 你是一位形式化驗證工程師。
> 請根據以下訂單生命週期規則，產生標準合規的
> PlantUML 狀態機圖 (State Machine Diagram)。

2. 塑模約束條件：

- 狀態清單：`Placed`, `Accepted`, `Preparing`, `ReadyForPickup`, `OutForDelivery`, `Delivered`, `Cancelled`。
- 轉換標籤必須遵循 `Event [Guard] / Action`。
- `OutForDelivery` 狀態必須包含 `entry / startGPSTracking()`。
- 僅允許在 `Placed` 與 `Accepted` 狀態下取消 `[cancellationWindow <= 120s]`；進入 `Preparing` 後嚴禁取消。

3. 輸入需求敘述 (Requirements Statement)：

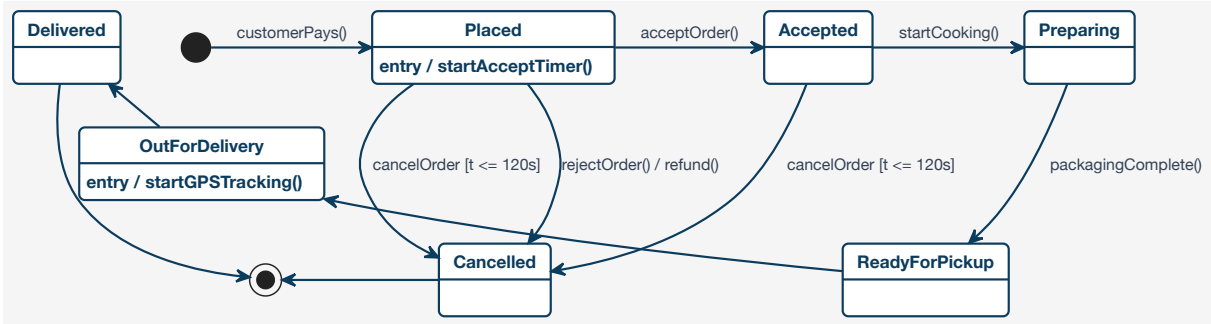
> 「**Order** 生命週期從 `[*]` 開始，顧客完成扣款後進入 **Placed**（已下單）。
> 在 **Placed** 狀態下，餐廳點擊接受則觸發 `acceptOrder` 進入 **Accepted**；若餐廳拒絕則觸發 `rejectOrder` 進入 **Cancelled** `/ issueFullRefund()`。
> 顧客在 **Placed** 或 **Accepted** 狀態下可主動觸發 `cancelOrder`，但必須滿足守衛條件 `[timeElapsed <= 120s]` 才能轉入 **Cancelled**。
> 當廚房點擊開始做菜，觸發 `startCooking` 進入 **Preparing**；此時取消路徑完全鎖死。
> 餐點打包完成後觸發 `packagingComplete` 進入 **ReadyForPickup**。
> 外送員抵達掃描條碼後觸發 `courierPickedUp` 進入 **OutForDelivery**，同時執行 `entry / startGPSTracking()`。
> 顧客輸入 OTP 簽收後觸發 `confirmDropoff` 進入終止狀態 **Delivered**，最後轉入 `[*]`。」

AI 生成之 PlantUML 與架構圖：外送平台狀態機模型

AI 生成之 PlantUML 規格代碼：

```
@startuml
[*] -right-> Placed : customerPays()

state Placed {
  Placed : entry / startAcceptTimer()
}
state OutForDelivery {
  OutForDelivery : entry / startGPSTracking()
}

Placed -right-> Accepted : acceptOrder()
Accepted -right-> Preparing : startCooking()
Preparing -down-> ReadyForPickup : packagingComplete()
ReadyForPickup -left-> OutForDelivery : courierPickedUp()
OutForDelivery -left-> Delivered : confirmDropoff()
Delivered -down-> [*]

Placed -down-> Cancelled : rejectOrder() / refund()
Placed -down-> Cancelled : cancelOrder [t <= 120s]
Accepted -down-> Cancelled : cancelOrder [t <= 120s]
Cancelled -left-> [*]
@enduml
```

渲染產出之架構圖：



課堂活動：使用 AI 演進狀態機生命週期 (雙人動手實作)

動手實作實驗：變更生命週期並審核 AI 產出 (4 分鐘)

- **1. 在狀態機提示詞中加入一個生命週期不變量 (Invariant)：**
  - *司機途中棄單：* 若外送員中途車輛故障棄單，訂單狀態退回 `ReadyForPickup` 重新派單。
  - *顧客延期配送：* 允許在 `Preparing` 備餐中申請延後外送指定時段。
  - *外送失敗處理：* 抵達門前若聯絡不上顧客超過 10 分鐘，轉移至 `FailedDelivery` 並原車退回。
- **2. 要求 AI 重新生成：** 提示 LLM 更新對應的 PlantUML 狀態機代碼。
- **3. 渲染並嚴格審查：** 貼入 PlantText 渲染並審查：
  - AI 是否維護**狀態命名規範**（使用被動式/形容詞，不使用動詞）？
  - **非法狀態轉移**（如餐點在路上卻被隨意取消）是否被嚴格防護？
  - **進入/離開動作**與轉移事件之邏輯前後是否完全一致？

觀念檢核測驗 7 (CCQ 7)

在 UML 狀態機圖中，**動作 (Action，如 transition 上的 `/ refund()` 或 `entry /`) 與 活動 (Activity，如 `do /`)** 最根本的執行語義差異為何？

- **A.** Action 是瞬間完成、不可中斷的原子性操作；Activity 是耗時進行的持續運算，且可被外來事件隨時中斷。
- **B.** Action 只能寫 Python；Activity 只能寫 SQL。
- **C.** Action 只能在進入狀態時執行；Activity 只能在離開狀態時執行。
- **D.** Action 專門處理例外失敗；Activity 專門處理成功路徑。

4.8 PlantUML 宣告式文字塑模

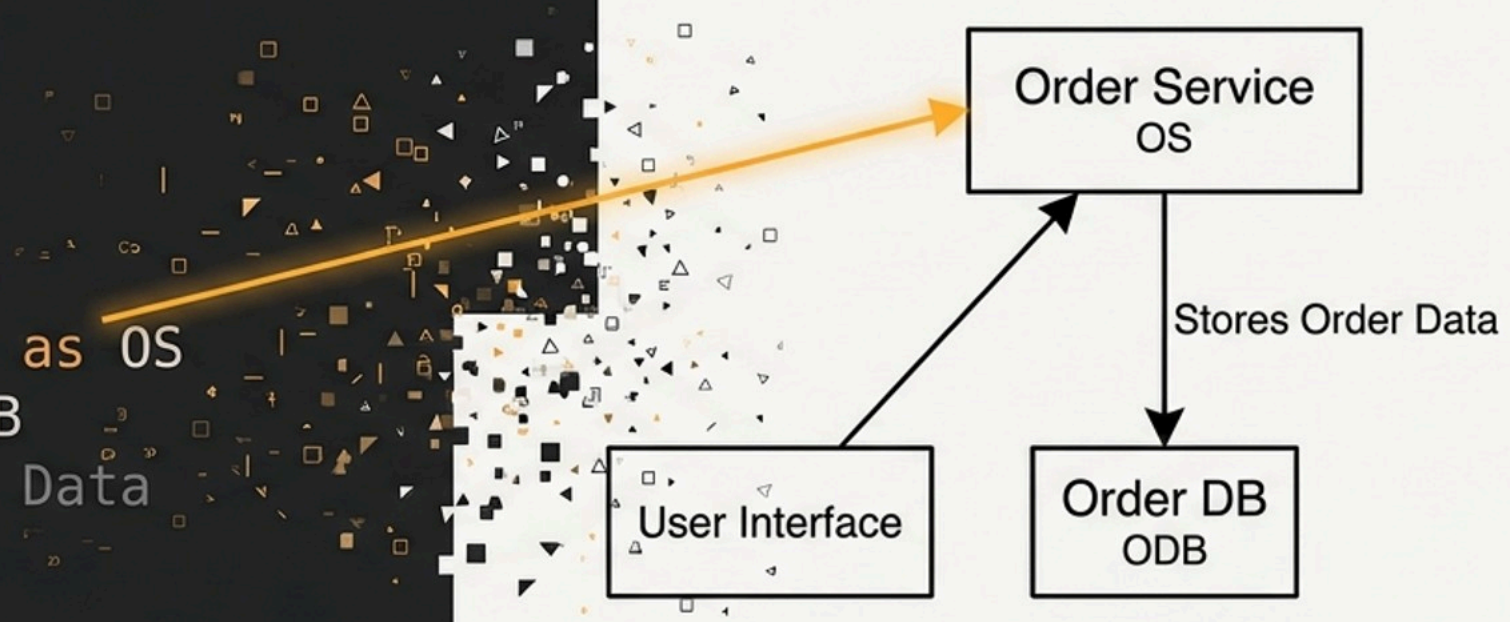
> *"告別手動拖拉圖框的痛苦。架構即代碼，讓軟體模型與程式碼在 Git 倉儲中一同進化。"*

線上編輯器 PlantText 線上實時沙盒 (planttext.com) →

UML Modeling with PlantUML

A visual guide to translating software design from text to architecture.

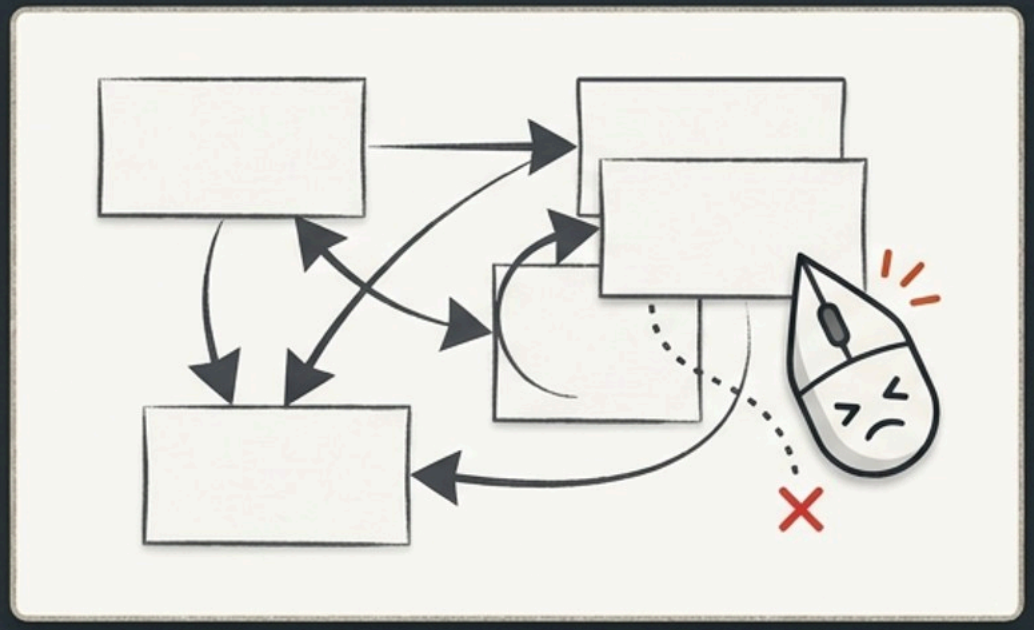
```
@startuml
!theme crt-amber
skinparam monochrome true

component "Order Service" as OS
database "Order DB" as ODB
OS --> ODB : Stores Order Data
@enduml
```



Based on the comprehensive guide by Mohammad Reza Bayat.

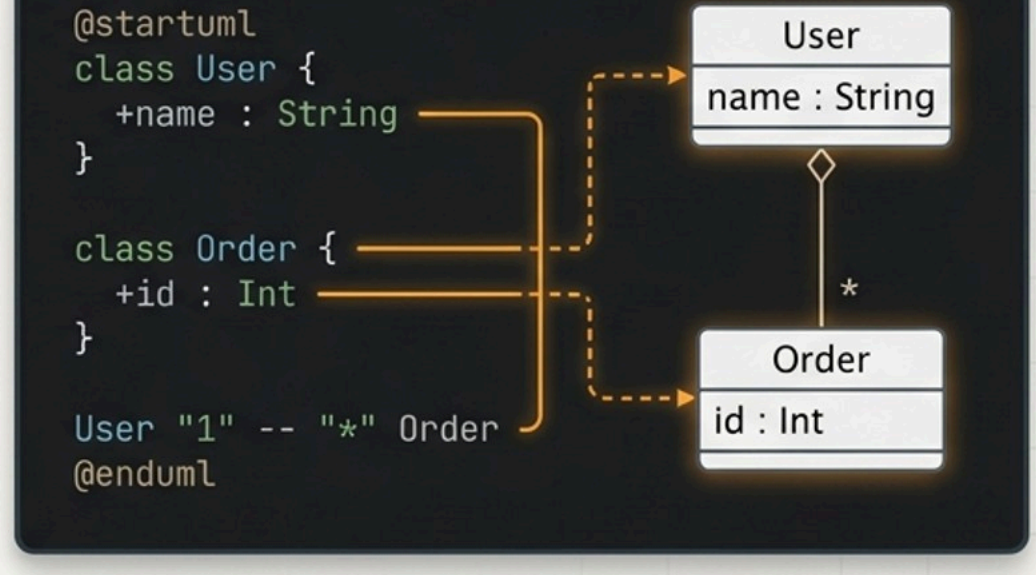
Stop Dragging Boxes. Start Writing Architecture.

The Unified Modeling Language (UML) remains the industry standard for communicating system structure and behavior. PlantUML removes the friction from creating these models. By treating diagrams as simple text syntax, engineers generate complex visual architecture without ever touching a mouse.



Slow. Manual. Disconnected.

```
@startuml
class User {
  +name : String
}

class Order {
  +id : Int
}

User "1" -- "*" Order
@enduml
```



Instant. Version-Controlled. Standardized.

The Frictionless Setup

1. Install

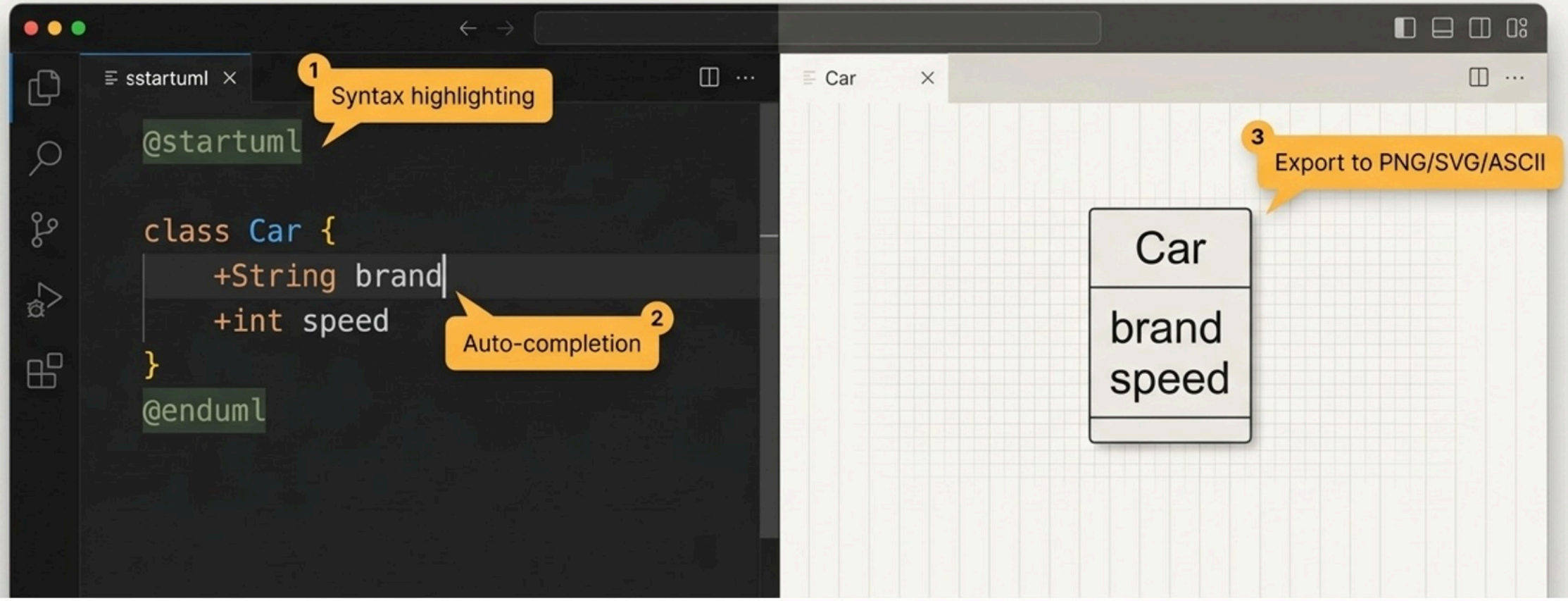
Add the PlantUML Extension in the VS Code marketplace.

2. Write

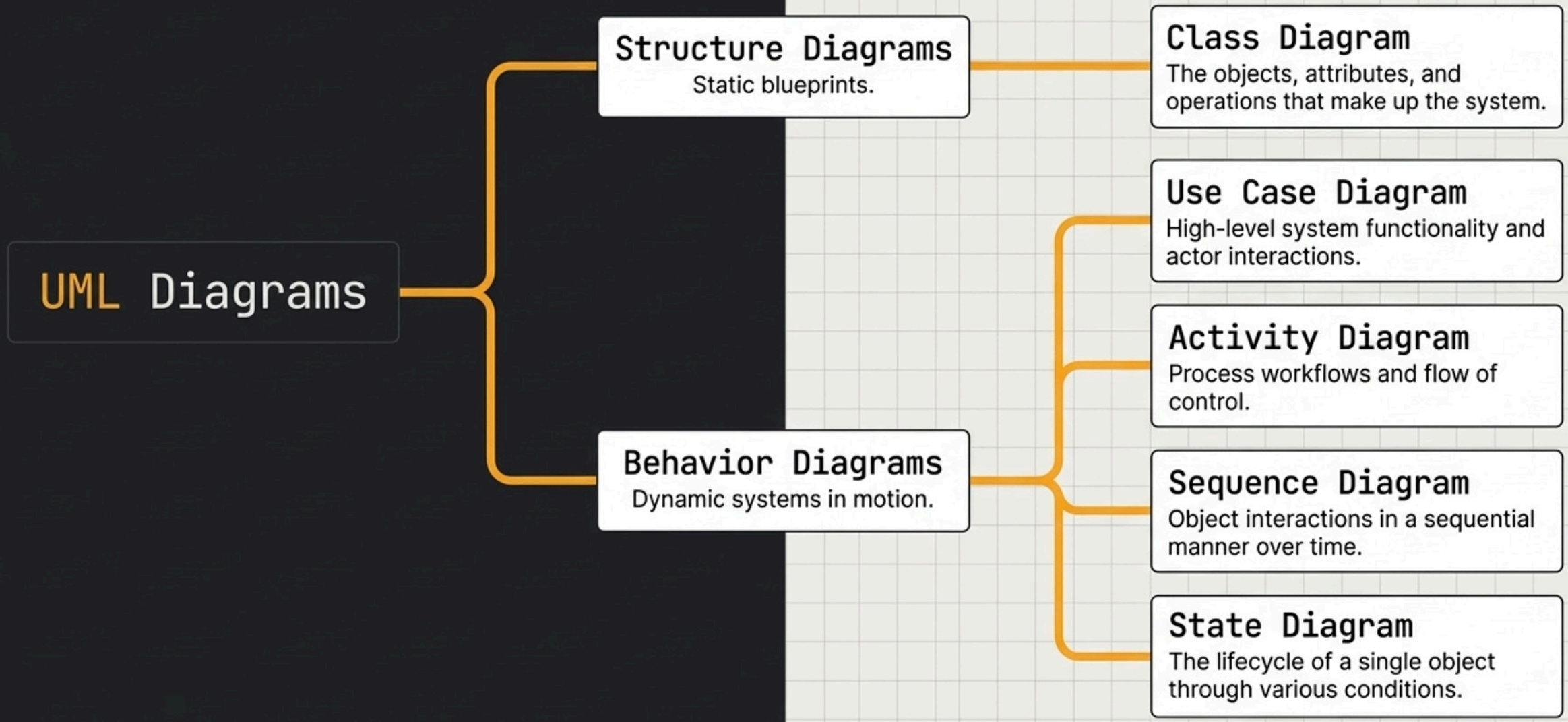
Create a new file with the **.puml** extension.

3. Render

Press **Alt + D** to trigger the live visual preview.



The Architect's Blueprint: 5 Essential Lenses



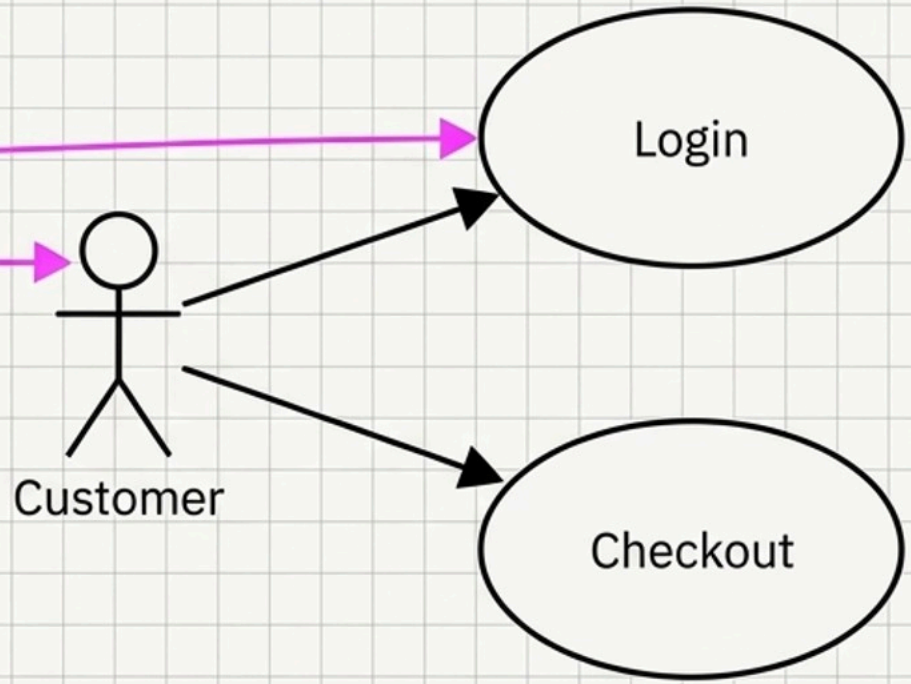
Use Case Diagrams: Mapping System Functionality

Highly effective for Agile methodologies to capture high-level requirements before writing any logic.

```
actor Customer

usecase Login
usecase Checkout

Customer --> Login
Customer --> Checkout
```

External Actors

System Functionalities



Class Diagrams: The Static System Structure

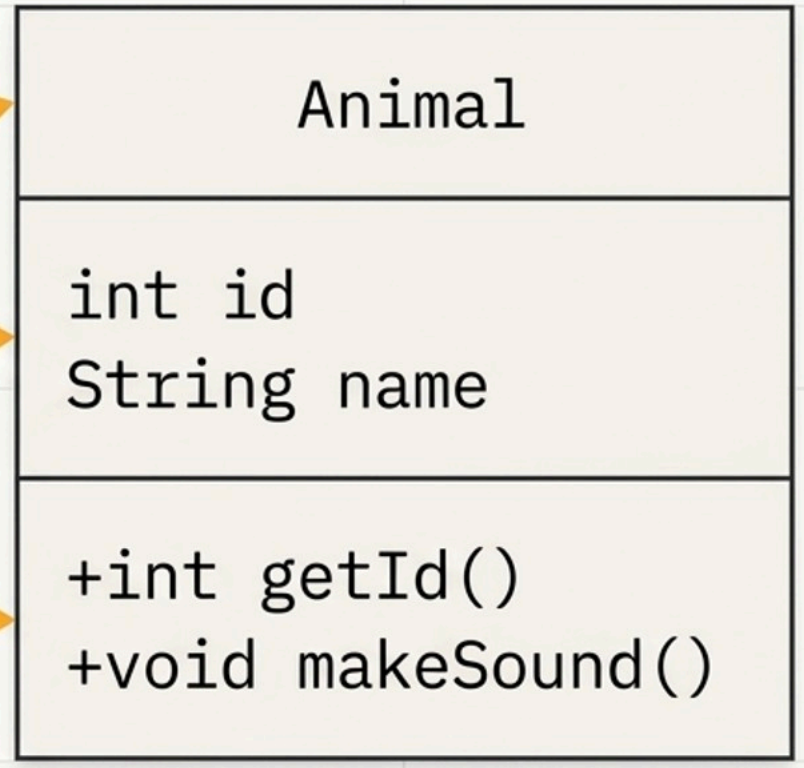
Key Insight: Visibiiity symbols define access:
+ (public), - (private), and # (protected).

```
class Animal {
  int id
  String name
  +int getId()
  +void makeSound()
}
```

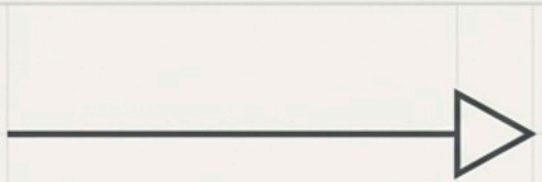
Class Name (The Blueprint)

Attributes

Operations/Methods



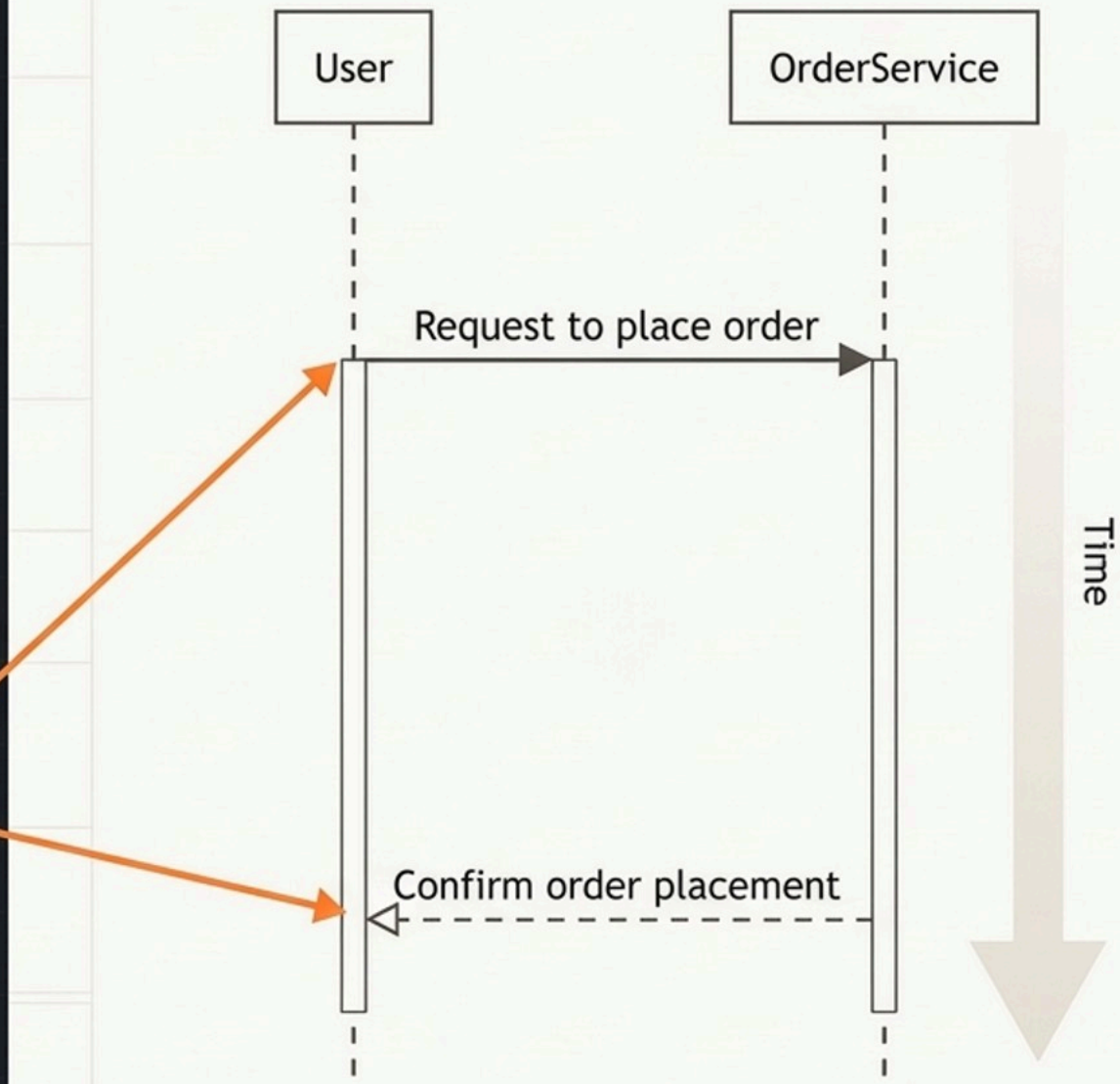
The Class Relationship Matrix

Relationship Type	Definition	PlantUML Syntax	Visual Output
Inheritance	Subclass inherits attributes/operations of a superclass.	`<|--`	
Association	One class uses the services of another.	`--`	
Aggregation	A whole/part relationship where the part can exist independently.	`o--`	
Composition	Strong ownership; the part cannot exist without the whole.	`*--`	

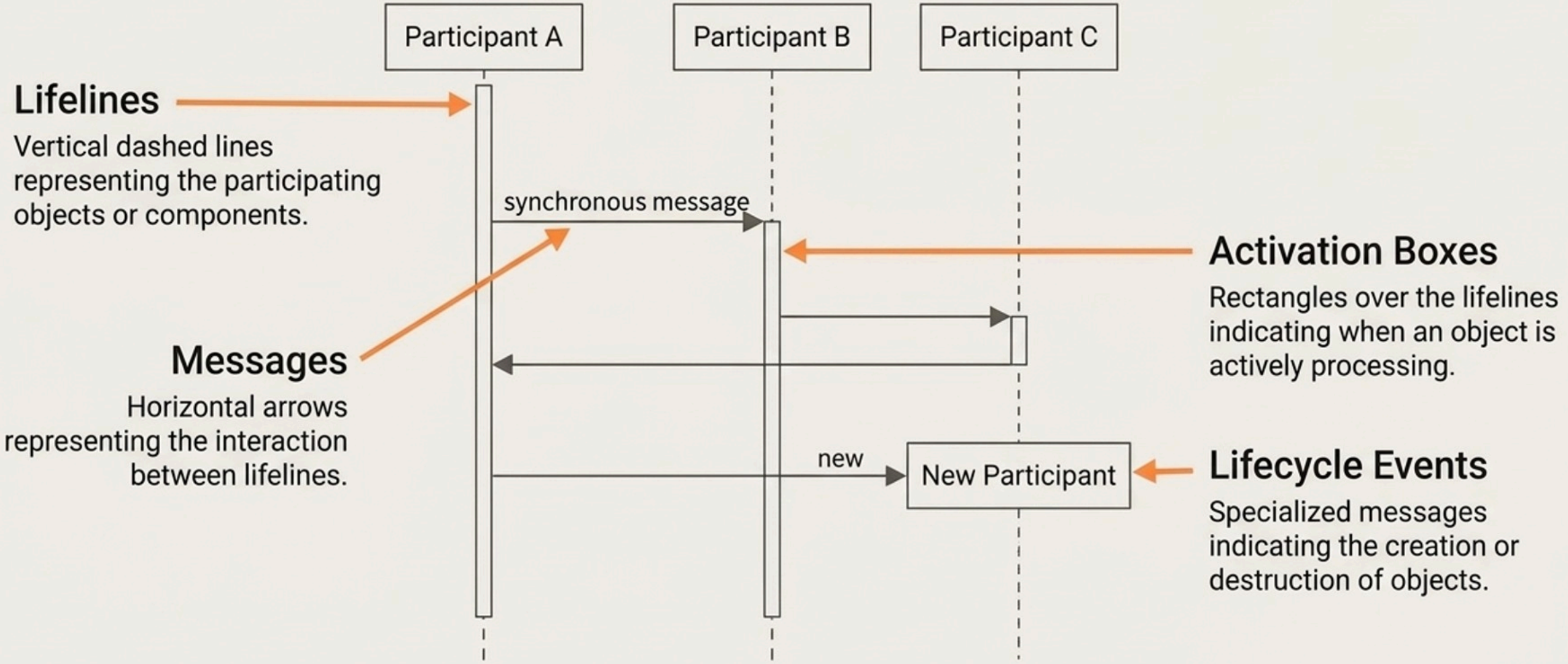
Sequence Diagrams: Interaction Over Time

Essential for modeling dynamic behavior, API calls, and synchronous/asynchronous message flows. The vertical axis represents the flow of time, moving from top to bottom.

```
participant User
participant OrderService

User -> OrderService : Request to place order
OrderService -> User : Confirm order placement
```



Anatomy of a Sequence Diagram



Lifelines

Vertical dashed lines representing the participating objects or components.

Messages

Horizontal arrows representing the interaction between lifelines.

Activation Boxes

Rectangles over the lifelines indicating when an object is actively processing.

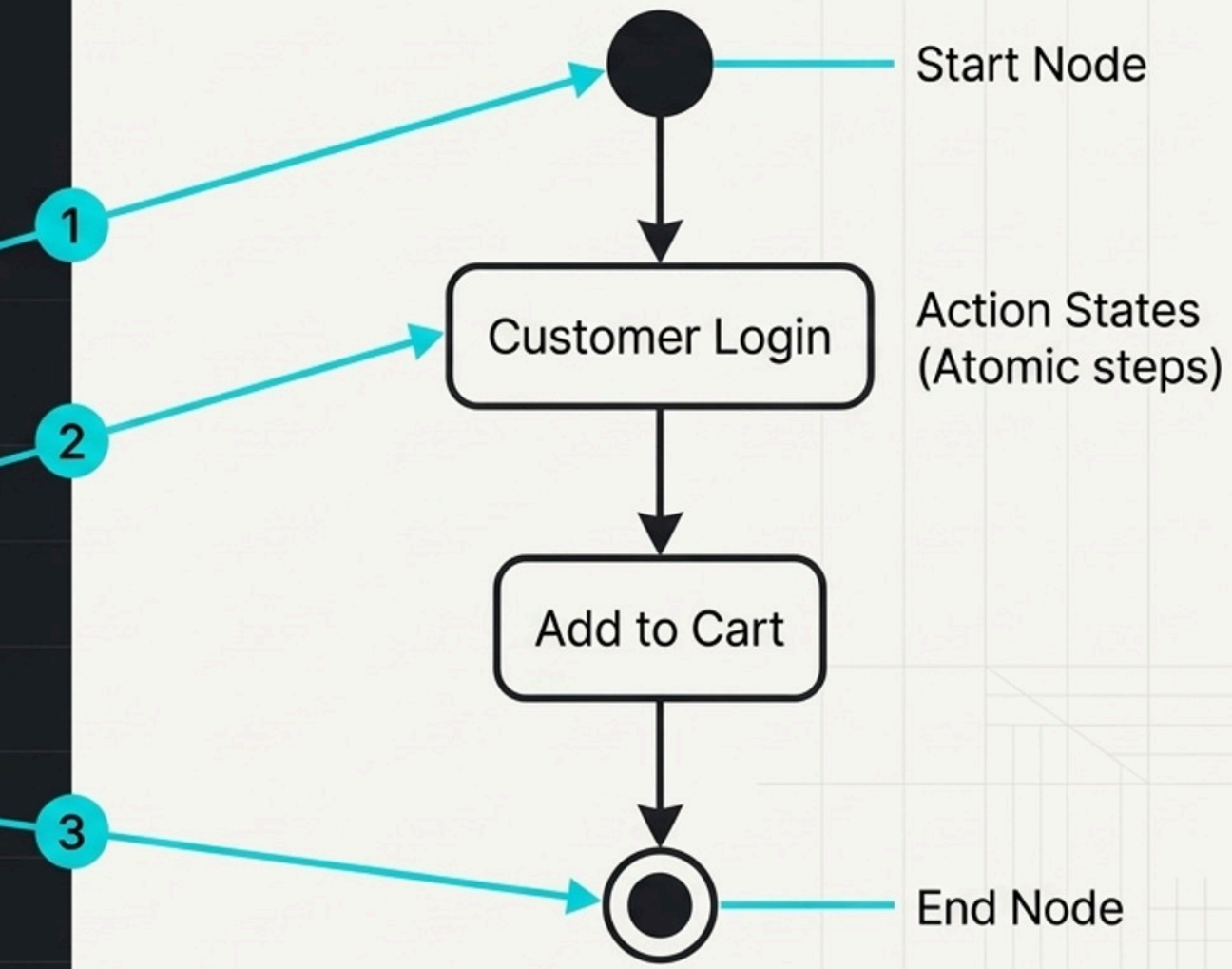
Lifecycle Events

Specialized messages indicating the creation or destruction of objects.

Activity Diagrams: Process Workflows

Powerful for modeling parallel processing via horizontal divisions called swim lanes, and conditional logic via diamond decision nodes.

```
mermaid
start
    :Customer Login;
    :Add to Cart;
stop
```

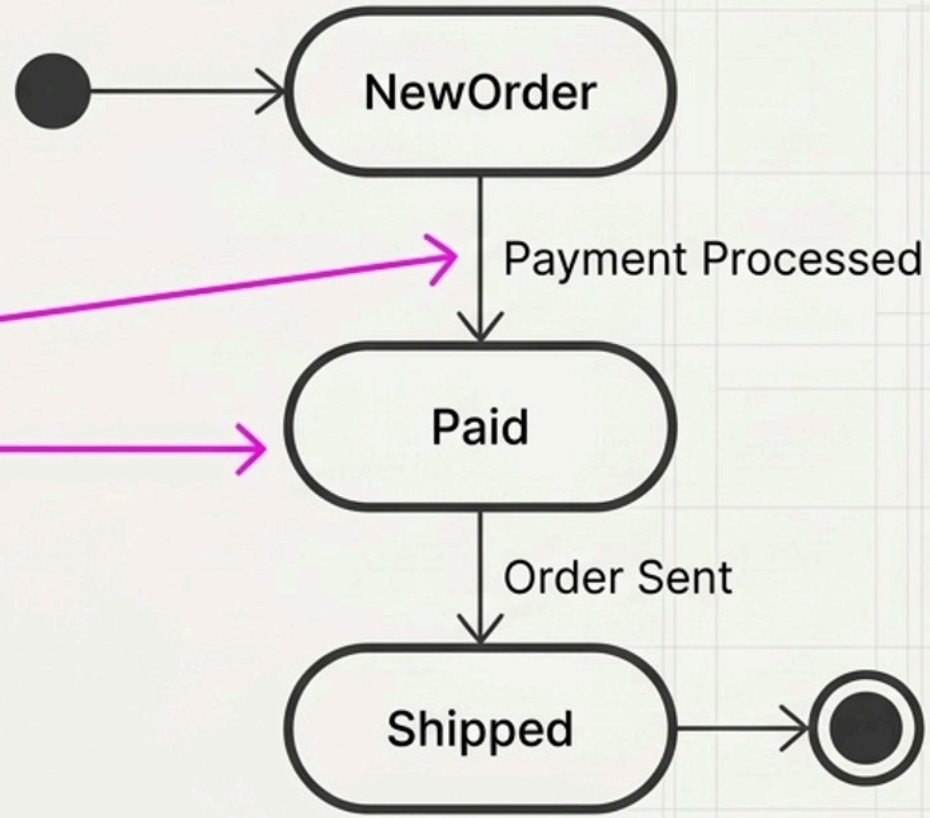


State Diagrams: Object Lifecycles

While Activity diagrams show multi-actor processes, State diagrams focus strictly on the lifecycle of a single object in response to external events.

```
1  [*] --> NewOrder
2  NewOrder --> Paid : Payment Processed
3  Paid --> Shipped : Order Sent
4  Shipped --> [*]
5
```

Format: `StateA --> StateB : Event String`

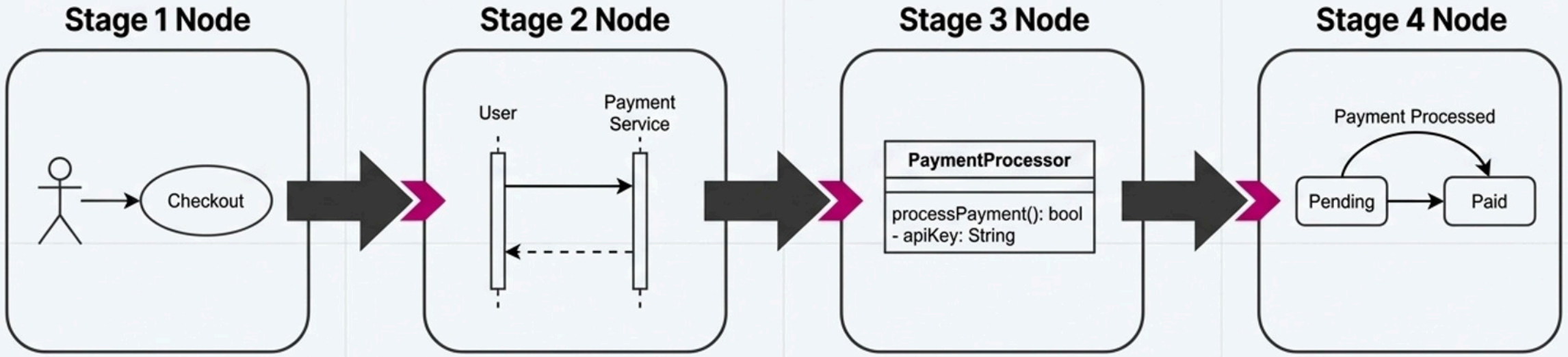


The Architect's Diagnostic: Choosing Your Model

Diagram
Use Case
Class
Sequence
Activity
State

Paradigm	Core Focus	When to Use It
Use Case	Identification centrol	Use case contains/or slopping, retreats and changes.
Class	Identify fioce class	Use class er developers irrvvesz of customers and models.
Sequence	Example-after sequence	Use the ei tercrpated to larg-en:stiums and conticipants in diagram.
Activity	Provenional activity	Activity spectureations arupified its complon placemenz after activity.
State	Model aroinotisnt	State from detorning the targeer maricipation of activity.

The Holistic System View



1. A Use Case (Customer Checkout) dictates the requirement...

2. ...which requires a Sequence (User talks to Payment Service)...

3. ...which requires a specific Class (PaymentProcessor) to be structurally built...

4. ...which ultimately changes the State of an Order (from Pending to Paid).

Insight: UML diagrams are not separate documents; they are overlapping viewpoints of the exact same system.

觀念檢核測驗 8 (CCQ 8)

在軟體架構工程實踐中，相較於傳統以滑鼠手動拖拉圖框的二進位繪圖軟體（如 Visio 或特定商業 CASE 工具），採用 **PlantUML 宣告式文字塑模 (Code-as-Architecture)** 的最核心工程優勢為何？

- **A.** 能保證自動生成 100% 無 Bug 的 Java 實作程式碼，無需單元測試。
- **B.** 可以在純文字中編寫，輕鬆納入 Git 進行精確的行級版本控制、Pull Request 審查與 CI/CD 自動編譯。
- **C.** 完全廢除物件導向原則，讓系統架構不再需要考慮類別或介面。
- **D.** 限制只能產出循序圖，防止工程師建立過多無意義的圖表。

4.9 概念複習與統整

> *"模型是軟體工程的世界語；精通模型，方能洞見架構的真實靈魂。"*

核心觀念統整：互動填空小測驗

測試你對本章核心架構概念的掌握度：

1. 於 Rational Software 攜手催生 UML 標準化的「UML 三巨頭」為 Grady Booch、Jim Rumbaugh 與 ＿＿。
2. 系統塑模的四大基礎核心視角為外部視角、＿＿ 視角、結構視角與行為視角。
3. 在使用案例模型中，多個使用案例強制共用的必要子程序應使用 ＿＿ 關聯。
4. 徹底分離使用者介面、業務邏輯與資料庫持久實體的架構模式稱為 ＿＿ 模式。
5. 在 UML 類別圖中，實心黑色菱形 (◆) 代表 ＿＿ 關聯，部分與整體生命週期緊密綁定連帶銷毀。
6. 在 UML 類別圖中，空心菱形 (◇) 代表 ＿＿ 關聯，部分具備獨立生存的生命週期。
7. UML 狀態機圖中，轉換標籤的國際標準語法為：事件觸發 [＿＿] / 執行動作。

經典文獻與延伸閱讀 (References)

- **權威經典教科書與國際標準：**
  - Sommerville, I. (2016). *Software Engineering* (10th ed.). Chapter 5: System Modeling. Pearson.
  - Booch, G., Rumbaugh, J., & Jacobson, I. (2005). *The Unified Modeling Language User Guide* (2nd ed.). Addison-Wesley.
  - Fowler, M. (2003). *UML Distilled: A Brief Guide to the Standard Object Modeling Language* (3rd ed.). Addison-Wesley.
  - Cockburn, A. (2000). *Writing Effective Use Cases*. Addison-Wesley.
  - Object Management Group (OMG). (2017). *OMG Unified Modeling Language (OMG UML) Specification*, Version 2.5.1.
- **現代宣告式文字塑模資源：**
  - PlantUML 官方標準網站：plantuml.com
  - Mermaid.js 宣告式圖表手冊：mermaid.js.org